News-Jump Regressions:
Weak Identification and Robust Inferences*

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Abstract

News-jumps, by which we mean sharp asset-price changes triggered by public news announcements, are increasingly attractive to empiricists due to their narrative interpretability and clean discontinuity-based identification. However, small news-jumps raise weak-identification concerns as their signal-to-noise ratio is low, an issue highlighted as the first-stage "power problem" in Nakamura and Steinsson (2018). We introduce a notion of weak jumps, whose magnitudes are local to zero at the square-root rate of the event-window length, to better approximate empirically small jumps. We show that weak jumps make the "jump regressions" of Li, Todorov, and Tauchen (2017a) inconsistently estimable and distort its regression-based inferences. We develop identification-robust tests inspired by Anderson–Rubin and conditional approaches. In empirical applications, we implement test inversion to generate valid confidence intervals for regression coefficients using high-frequency returns around FOMC, ISM manufacturing, Conference Board CCI announcements.

Keywords: news announcements, small jumps, jump regressions, continuous-time event studies, weak identification, semimartingale, high-frequency data.

JEL codes: C12, C58.

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1 Introduction

Jumps, as evidence of directional consensus of financial markets about news in a very instant, absorb public-level information shocks as a manifestation of the rationale of the semi-strong form of efficient market hypothesis. Examining high-frequency co-movements of related series around major news announcements, i.e., *news-jumps*, therefore enables the study of cross-sectional jump dependence through exogeneous variation triggered by the events, to tease out causal macro-effects of interest in a discontinuity-based identification scheme.

Despite large-sample discrete-time event study regressions broadly acknowledged in the high-frequency identification literature[1], the same "shock-response" root mechanism is studied within a continuous-time modeling setup by Li, Todorov, and Tauchen (2017a) (henceforth, LTT) who developed jump regressions in a statistically valid two-stage procedure and established corresponding infill-asymptotic central limit theory. For two related high-frequency time series, the first stage detects jumps using the standard truncation method[2], and the second stage conducts least-square regressions on the filtered, large and rare *statistical jumps*, and draws inference on the regression coefficient, a.k.a. in the context of jump regressions, the *jump beta*.

However, when we study news-jumps in high-frequency event-study designs to facilitate economic interpretation[3], it is empirically found high-frequency log-returns around news announcements are often so small that they fall below the truncation threshold and thus fail to be stastically classified as jumps. Such a return may reflect either no jump[4] or a genuine but undetected small jump given the finite sample. Consequently, regression analysis

[1]See Maheu and McCurdy (2004), Pettenuzzo, Sabbatucci, and Timmermann (2020), Nakamura and Steinsson (2018), Känzig (2021) for representative empirical work respectively on firm earnings, cash flow, monetary policy, and oil supply news using high-frequency data. Ramey (2016) provides a comprehensive survey on this subject.

[2]In high-frequency financial econometrics literature for detecting jumps, the truncation threshold is chosen to shrink at a rate that asymptotically separates continuous diffusion increments from fixed-size jumps; see Mancini (2001) and Chapter 9 of Jacod and Protter (2012) for more details.

[3]See Section 3.2 of LTT where inference for jump regressions in the case of news-jumps is discussed, but has not been fully analyzed.

[4]Kandel and Pearson (1995) gives theoretical modeling and empirical evidence suggesting that asset prices do not necessarily jump in response to news announcements.

based on these small log-returns can be heavily contaminated by the continuous diffusion component, yielding misleading inferences. This “small-jump” or “small-shock” issue is well realized in the literature of monetary policy shocks identification, as the “power problem” emphasized by Nakamura and Steinsson (2018).[5]

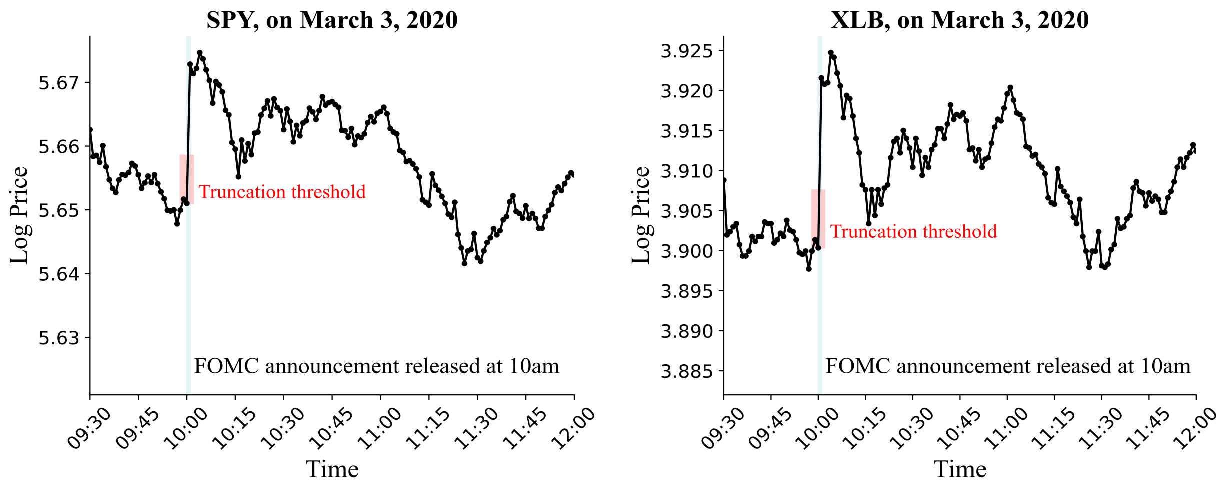
To instructively illustrate the small-jump problem, we examine SPY and XLB prices around Federal Open Market Committee (FOMC) announcements, a canonical macro-news setting. Figure 1 contrasts two representative cases: the unscheduled 50-basis-point rate cut on March 3, 2020, which generated announcement-minute returns in both ETFs exceeding the truncation threshold, and the largely anticipated no-change decision on May 1, 2024, for which both returns remained below the threshold. Figure 2 collects all pairs high-frequency returns around news in scatterplots and shows the weak first stage is pervasive: in the left panel, 121 of 165 available release-minute observations from 2002–2024 fall below truncations; in the right panel, after aggregating returns over the five-minute FOMC window $[t-2, t+2]$, 126 of 156 available observations remain below the threshold. Thus, even around major macro announcements, high-frequency price responses are often too small to be classified as statistical jumps, motivating us to develop valid inferences robust to small or no jumps.

In this paper, we address the small-jump phonomena as the weak identification problem due to its low signal-to-noise ratio nature[6]. To do this, we model the jump smallness by letting jumps shrink at the rate inverse to the square root of sample size, which is one typical way to model weak identification by considering a drifting sequence, and we refer to such vanishing jumps as *weak jumps* in analogy to the weak instrument literature (Staiger and Stock (1997)). For contrast, regular non-shrinking jumps are called strong jumps or strong identification cases. We clarify that our theory is designed for mildly high-frequency data, e.g. sampled at one- or five-minute(s), to avoid market microstructure noise problem, and the one-half shrinkage rate is deliberately chosen in line with the truncation method to let jumps be comparable in magnitude to the diffusion component.

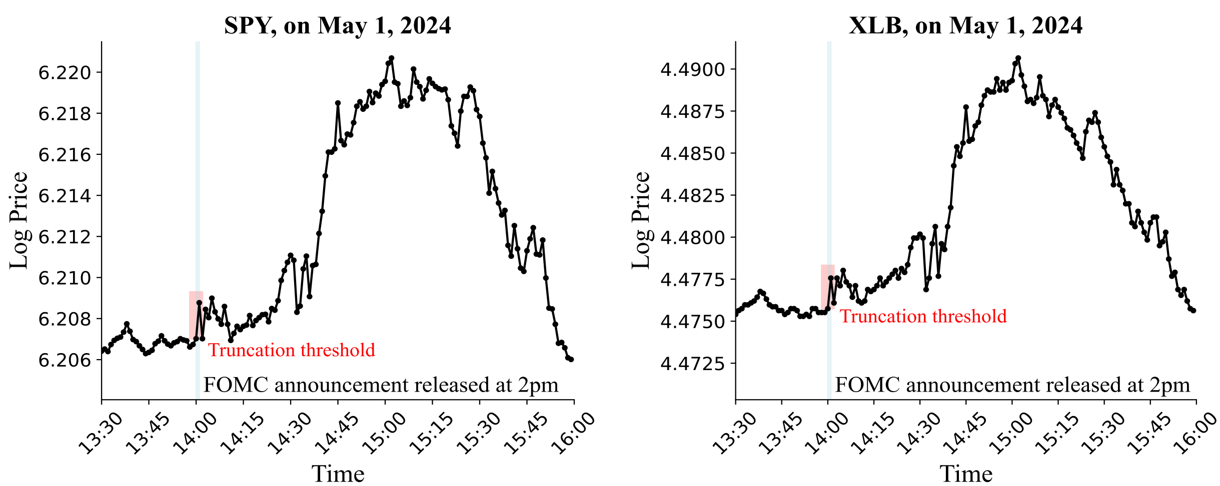
[5]In addition to FOMC announcements, Swanson (2023) and Swanson and Jayawickrema (2024) proposed to study market reaction around the Fed Chair’s speeches as they are stronger instruments to enhance the identification power for monetary policy shocks.

[6]In our setting, each log-return contains three components: drift, diffusion and jump, due to continuous-time modelling of high-frequency data. Signals are jumps and noises correspond to drift and diffusion.

Figure 1: The one-min log-price of indexes around the FOMC announcements



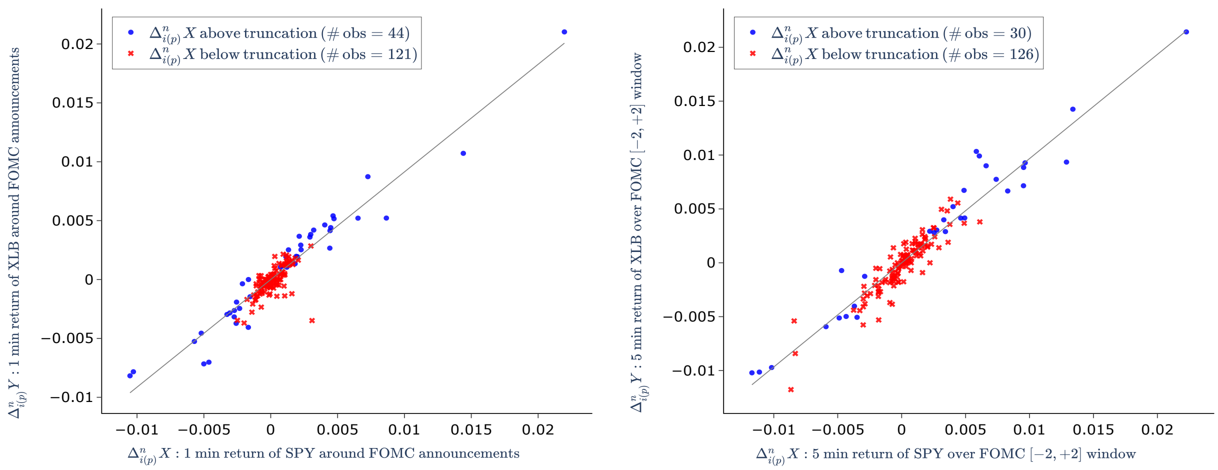
Top panel: Large jumps, the strong identification case.



Bottom panel: Small jumps, the weak identification case.

Notes: The figure plots one-minute log prices of SPY and XLB around two FOMC announcements on March 3, 2020 (top panel) and May 1, 2024 (bottom panel). The light blue band marks the one-minute interval of the announcement release. The red band indicates the truncation threshold for jump detection, given by equation (41), with the shrinkage exponent set to 0.49. In panel (a), returns exceed the thresholds, so both are classified as containing jumps. In panel (b), neither return crosses its threshold, suggesting possibly small jumps or no jumps at all.

Figure 2: The scatter plot of high frequency return pairs around FOMC announcements.



Notes: The figure plots XLB versus SPY log-return pairs around FOMC announcements from 2002 to 2024, using one-minute release-window returns in the left panel and five-minute $[t-2, t+2]$ returns in the right panel. The five-minute panel has nine fewer observations because sparse trading in XLB during 2003–2005 leaves some early announcement windows without valid five-minute returns. Blue dots (red crosses) indicate observations for which the absolute SPY return is above (below) its truncation threshold. The fitted line is a zero-intercept regression based on the blue dots.

We make three contributions. First, we show both in theory and simulations that under weak jumps, LTT's jump regression estimation is no longer consistent and regression-based tests are size-distorted. Second, we provide a identification-robust and asymptotically similar Anderson–Rubin test (Anderson and Rubin (1949)) allowing for price-volatility co-jumps[7]. To fix ideas, let the $i(p)$-th pair of high-frequency log-returns around p-th announcement be denoted by $(\Delta^n_{i(p)}Y, \Delta^n_{i(p)}X)$, and satisfy the reduced-form system

$$\begin{cases} \Delta^n_{i(p)}X = J^X_p + \varepsilon^X_{i(p)}, \\ \Delta^n_{i(p)}Y = J^X_p \beta + \varepsilon^Y_{i(p)} \end{cases} \tag{1}$$

where β denotes our parameter of interest, the jump beta, J^X_p represents the X-jump induced by the p-th announcement, and disturbance terms $(\varepsilon^X_{i(p)}, \varepsilon^Y_{i(p)})$ capture the continuous return part, including both drift and diffusion. The Anderson–Rubin test is constructed as $\Delta^n_{i(p)}Y - \beta_0\Delta^n_{i(p)}X$. When β_0 coincides with the truth β, the jump components cancel out, thereby

[7]Unlike the weak instrumental variable regressions, an additional complication of price-volatility co-jumps is evident in high-frequency data, see Jacod and Todorov (2010), which means the diffusion noise level accompanying the jump signal is allowed to simultaneously jump.

ensuring robustness with respect to jump smallness. Moreover, the problem is inherently heterogeneous, as $\mathrm{Var}(\varepsilon^X_{i(p)}, \varepsilon^Y_{i(p)})$ varies across p. Third, with an additional assumption that no price-volatility co-jumps, we find a sufficient statistic for the jumps so that conditional on it enables us to develop the conditional inferences in the Gaussian limit experiment, i.e. the lagrangian multiplier, conditional likelihood ratio, and conditional linear combination tests as in the weak instrument literature, see Kleibergen (2002), Moreira (2003), and Andrews (2016). Moreover, we provide a conservative method to relax the additional assumption, and justify it by simulations. Overall, this paper econometrically provides constructive solutions on weak identification problem of high-frequency event studies in continuous-time models.[8]

To further unravel the fundamental connection between weak jump regression and the conventional weak instrument problem, recall a linear instrumental variables (IV) model with a single endogenous regressor, written in reduced form,

$$\begin{cases} X_i = Z_i\pi + V_{i1}, \\ Y_i = Z_i\pi\beta + V_{i2} \end{cases} \tag{2}$$

The similarity between (1) and (2) orignates from the same root structure that news shocks, served as intruments, provide exogeneous variations in X and Y that helps identify causal or structural parameters. Our problem (1) is nonstandard in two respects. First, the jump is latent, it can be consistently estimated in strong identification cases[9], but no longer for weak jumps, unlike in weak IV the intruments Z is always observed. Second, the infill asymptotics entails increasingly dense sampling over a fixed time span, implying that the number of announcements is fixed, while the weak IV or typical event study regressions require the number of announcements in the sample to be large.

This paper relates to three strands of literatures: jump dependence, weak identification, and high-frequency identification of shocks. The literature on the dependence structure of co-jumps in financial markets has expanded rapidly. Todorov and Bollerslev (2010) first justified that jumps should be priced differently from continuous risk, thereby introducing

[8]The continuous-time methods have been applied in empirical work. For examples, Piazzesi (2005) models FOMC targets as a pure-jump process and studies resulting response of bond yield, Bollerslev, Li, and Xue (2018) use jump regressions to study volume-volatility dynamics involving disagreement measures.

[9]If the jump J^X_p is strong, its consistent estimator is $\Delta^n_{i(p)}X$ itself.

the concept of jump beta. Building on this, LTT established regression analysis for jumps, including a specification test for linearity, the $\sqrt{n}$-consistency of least-squares estimation, and a semiparametric efficiency result. And jump regressions are generalized by Li, Todorov, and Tauchen (2017b), Li, Todorov, Tauchen, and Chen (2017), Bollerslev, Li, and Chaves (2021). The close interrelation between news and jumps in different levels has been well documented, for examples, by Maheu and McCurdy (2004), Lahaye, Laurent, and Neely (2011), Caporin, Kolokolov, and Renò (2017), Pettenuzzo, Sabbatucci, and Timmermann (2020), Erdemlioglu and Yang (2023), Aleti and Bollerslev (2025), Christensen, Timmermann, and Veliyev (2025) among many others. However, despite this extensive body of findings, few attempts have been devoted to econometric theory for small jumps surrounding news, which presents its own challenges for inference and identification.

Moreover, jump regressions bears a high-level resemblance to the regression discontinuity design (Lee and Lemieux (2010)), to the Wald estimator in IV regressions (Wald (1940)). Our theory therefore shares intellectual parallels with the weak identification literature developed in these settings, and absorbs the methodological insights from Anderson and Rubin (1949), Staiger and Stock (1997), Kleibergen (2002, 2005), Moreira (2003), Andrews, Moreira, and Stock (2006), Feir, Lemieux, and Marmer (2016), Andrews (2016). The notion of weak jumps, to the best of our knowledge, is first analyzed by Li (2013) who adopted shrinkage modelling of jumps and developed a local-to-continuity theory for pre-averaging estimators addressing weak identification of jumps in the presence of microstructure noise.

The empirical literature on high-frequency identification of monetary policy effects is closely related to the empirical purpose of this paper, see Kuttner (2001), Cochrane and Piazzesi (2002), Faust, Rogers, Swanson, and Wright (2003), Gürkaynak, Sack, and Swanson (2005), Bernanke and Kuttner (2005), Nakamura and Steinsson (2018), Bauer and Swanson (2023), Lewis (2025) among others. These studies typically adopt discrete-time asymptotics within ordinary least square (OLS) regressions or vector autoregressions (VARs). Stock and Watson (2012) interpret high-frequency monetary shocks as "external instruments" that facilitate identification in structural VARs and found weak instrument problem. Casini and McCloskey (2025) has similar findings that "narrow size of the window is not sufficient for identification" in high-frequency event-study regressions as in Nakamura and Steinsson

(2018), which in this paper we interpret the concern as weak identification.

In what follows, Section 2 introduces the model setup and assumptions. Section 3 studies weak-jump asymptotics and shows how the limiting distribution departs from LTT's strongly identified theory. Section 4 develops Anderson–Rubin inference that remains valid under weak identification. Section 5 studies conditional inferences. Sections 6 and 7 report simulation evidence and empirical applications, and Section 8 concludes. All proofs are given in the supplment materials.

The following notation is used throughout. The set of real numbers is denoted by $\mathbb{R}$, define $\mathbb{R}_* \equiv \mathbb{R} \setminus \{0\}$ and $\mathbb{R}_+ \equiv [0, \infty)$; the transpose of A is denoted by $A^\intercal$; we write $x_n \asymp y_n$ if there exists a constant $c \geq 1$, such that $x_n/c \leq y_n \leq cx_n$ for all n; for a generic process $(X_t)_{t\geq 0}$, define its left limit as $X_{t-} \equiv \lim_{s<t, s\to t} X_s$ and jump as $\Delta X_t \equiv X_t - X_{t-}$; let $\|\cdot\|$ denote the Euclidean distance, $\mathcal{M}_2$ denote the set of all 2×2 positive semi-definite matrices; for a set B, let $|B|$ denote its cardinality; let $\mathcal{N}$ be the symbol for Gaussian distribution; let $\perp\!\!\!\perp$ denote "independent of"; let $\xrightarrow{\mathcal{L}}$, $\xrightarrow{\mathcal{L}\text{-}s}$, $\xrightarrow{\mathbb{P}}$ denote convergence in law, and stable convergence in law, convergence in probability respectively.

2 Setup and Assumptions

Let $Z_t = (X_t, Y_t)^\intercal$ be a bivariate log-price process, and we model Z_t by an Itô semimartingale model defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$, specified as

$$Z_t = Z_0 + \int_0^t b_s \mathrm{d}s + \int_0^t \sigma_s \mathrm{d}W_s + J_t \tag{3}$$

Here, the drift process $b \in \mathbb{R}^2$ and the stochastic volatility process $\sigma \in \mathcal{M}_2$ are both càdlàg adapted, and we let $c_t = \sigma_t \sigma_t^\intercal$ denote the spot covariance matrix. The driving process $W = (W_X, W_Y)^\intercal$ is a standard two-dimensional Brownian motion. The jump component $J = (\Delta X_t, \Delta Y_t)^\intercal$ is a purely discontinuous local martingale, constructed as the sum of its large- and small-jump parts, both driven by a homogeneous Poisson random measure on "temporal-spatial" region $\mathbb{R}_+ \times \mathbb{R}$. Since we only cares about news-jumps, the jump activity index is saved for simplicity. The semimartingale specification for asset prices is the most general consequence of the no-arbitrage principle (Delbaen and Schachermayer (1994)), and the qualifier "Itô" endows the semimartingale with favorable properties, such as a Brownian-type diffusion and Poisson-type jumps, while preserving a high degree of generality. For systematic expositions on application of semimartingale models for high-frequency data, we refer to monographs Jacod and Protter (2012) and Aït-Sahalia and Jacod (2014).

Suppose an economist is interested in the effect of a type of news announcements, labeled by $\mathfrak{a}$, which occur at pre-scheduled times $\mathfrak{a}_1, \mathfrak{a}_2, \mathfrak{a}_3, \cdots$ and are likely to trigger jumps in Z. We let $J^{\mathfrak{a}}$ denote the news-based jump related to $\mathfrak{a}$, i.e.,

$$J_t^{\mathfrak{a}} = \sum_{p \geq 1} (\Delta X_{\mathfrak{a}_p}, \Delta Y_{\mathfrak{a}_p})^{\intercal} \cdot 1_{\{\mathfrak{a}_p \leq t\}}.$$

The predictable jump process $J_t^{\mathfrak{a}}$ related to pre-scheduled announcements $\mathfrak{a}$ is assumed to be finitely active, which means the number of announcement jumps is finite on any finite time interval. Note that it is possible that $\Delta X_{\mathfrak{a}_p} = 0$ if the information contained in the p-th announcement is fully expected by the market; otherwise, news-based jumps presumably exist.

We study the jump beta using high-frequency log-returns surrounding news announcements in event-study fashion. Following LTT, suppose B is a Borel subset of $\mathbb{R}_+$ with finite Lebesgue measure, representing a selected time period containing news announcements. Define a set $\mathfrak{a}_B \equiv \{\mathfrak{a}_p : \mathfrak{a}_p \in B\}$ as the collection of all announcement times in B, and let $\mathcal{P}_{\mathfrak{a}_B} = \{p \geq 1 : \mathfrak{a}_p \in \mathfrak{a}_B\}$ denote the ordinal indexes of news-jumps related to $\mathfrak{a}$ in the time domain B. We assume a proportional relationship between the jumps, consistent with the empirical pattern shown in Figure 2, namely

$$\Delta Y_{\mathfrak{a}_p} = \beta \Delta X_{\mathfrak{a}_p}, \quad \text{for } p \in \mathcal{P}_{\mathfrak{a}_B} \tag{4}$$

where β denotes our parameter of interest. Equation (4) imposes an exact zero-intercept proportionality restriction on announcement-induced jumps: whenever $\Delta X_{\mathfrak{a}_p} \neq 0$, the ratio $\Delta Y_{\mathfrak{a}_p} / \Delta X_{\mathfrak{a}_p}$ equals β, while $\Delta X_{\mathfrak{a}_p} = 0$ implies $\Delta Y_{\mathfrak{a}_p} = 0$. Our testing problem is

$$H_0 : \beta = \beta_0 \quad \text{against} \quad H_1 : \beta \neq \beta_0, \tag{5}$$

for a candidate jump beta β_0. Meanwhile, the continuous-time process $(Z_t)_{t \in [0,T]}$ is discretely and regularly observed at time points $t = 0, \Delta_n, 2\Delta_n, \ldots, n\Delta_n$, where Δ_n denotes the time gap between consecutive observations, and the sample size $n = \lfloor T/\Delta_n \rfloor$. We adopt the infill asymptotics scheme which means the time mesh $\Delta_n \to 0$ or equivalently $n \to \infty$. The i-th log-return is obtained by a difference operator Δ_i^n defined as follows,

$$\Delta_i^n Z \equiv Z_{i\Delta_n} - Z_{(i-1)\Delta_n}$$

We let the $i(p)$-th sampling interval $((i(p)-1)\Delta_n, i(p)\Delta_n]$ contain the p-th announcement time $\mathfrak{a}_p$, and define

$$\mathcal{I}(\mathfrak{a}_B) = \{i(p) : p \in \mathcal{P}_{\mathfrak{a}_B}\} \tag{6}$$

to collect all the sampling intervals in B that contain news announcements $\mathfrak{a}$. Combining (3) and (4), the news-jump regression is therefore,

$$\begin{aligned}
&\Delta_{i(p)}^n Y = \beta \Delta_{i(p)}^n X + u_{p,n}(\beta), \text{ where} \\
&u_{p,n}(\beta) = (-\beta, 1)\left(\int_{(i(p)-1)\Delta_n}^{i(p)\Delta_n} b_s \, \mathrm{d}s + \int_{(i(p)-1)\Delta_n}^{i(p)\Delta_n} \sigma_s \, \mathrm{d}W_s\right), \quad \text{for } p \in \mathcal{P}_{\mathfrak{a}_B}
\end{aligned} \tag{7}$$

where the error term satisfies $u_{p,n}(\beta) = O_{\mathbb{P}}(\Delta_n^{1/2})$. Moreover, we confine our analysis to a subset of the sample space Ω to ensure that there is at least one X-jump in $\mathfrak{a}_B$, i.e.

$$\Omega(\mathfrak{a}_B) = \{\omega \in \Omega : \Delta X_t \neq 0 \text{ for some } t \in \mathfrak{a}_B\}. \tag{8}$$

On $\Omega(\mathfrak{a}_B)$, condition (4) uniquely identifies β, since

$$\beta = \frac{\sum_{p \in \mathcal{P}_{\mathfrak{a}_B}} \Delta X_{\mathfrak{a}_p} \Delta Y_{\mathfrak{a}_p}}{\sum_{p \in \mathcal{P}_{\mathfrak{a}_B}} (\Delta X_{\mathfrak{a}_p})^2},$$

where the denominator is strictly positive by (8). By default, all the following analysis is restricted to $\Omega(\mathfrak{a}_B)$ if without explicitly mentioning.

To account for potential small jumps that fall below the standard truncation threshold, we impose an asymptotic structure on jump magnitudes, specified as

$$\Delta X_{\mathfrak{a}_p} = O_{\mathbb{P}}(\Delta_n^r) = O_{\mathbb{P}}(n^{-r}), \quad r \geq 0. \tag{9}$$

The rate r governs the strength of identification. When $r = 0$, the model becomes the standard strongly identified case studied in LTT; when $0 < r < 1/2$, identification is semi-strong; and when $r = 1/2$, identification is weak, see Andrews and Cheng (2012). We focus on the weakly identified case, i.e. $r = 1/2$.

Assumption 1 (Weak jumps)**.** For each announcement in $\mathfrak{a}_B$, the X-jump component is *weak* in the sense that

$$\Delta X_{\mathfrak{a}_p} = \xi_{\mathfrak{a}_p} \Delta_n^{1/2}, \quad p \in \mathcal{P}_{\mathfrak{a}_B},$$

where $\xi_{\mathfrak{a}_p}$ is a real-valued $\mathcal{F}_{\mathfrak{a}_p}$-measurable random variable for each $p \in \mathcal{P}_{\mathfrak{a}_B}$. The possibility that $\xi_{\mathfrak{a}_p} = 0$ allows for no X-jump at the p-th announcement.

We have two remarks on Assumption 1. First, the limitation of jump detection is due to a finite sample, which can be inevitable due to data availability or for the purpose to stay away from microstructure noise. Modelling jumps by shrinkage with sample size captures the low-signal difficulty. Second, the choice of the rate 1/2 is intentional to make jumps comparable in magnitude to the Brownian diffusion, noting drift is conventionally treated negligible in theory. Moreover, the shrinkage rate of 1/2 is typical in the weak identification literature.[10]

Assumption 2 (Known announcement times)**.** For any time domain B, the set $\mathfrak{a}_B$ is known.

As discussed in Section 3.2 of LTT, one can rely on pre-scheduled macro announcements to locate jump times by assuming that price immediately adjusts after the announcement in a very liquid market. Assumption 2 is always satisfied in post hoc in-sample analysis. What follows are some regularity conditions.

Assumption 3 (Regularity conditions)**.** (a) The drift process b_t is locally bounded and adapted; (b) the volatility process σ_t is càdlàg adapted and positive semi-definite for any $t \in [0, T]$; (c) no jump occurs on the boundary of B, almost surely; (d) $J^{\mathfrak{a}}$ is finitely active.

Assumption 3(a) renders the drift component negligible relative to the diffusion part, thereby simplifying our problem; Assumption 3(b) allows the existence of price-volatility co-jumps and is necessary to pin down the pre-jump and post-jump volatilities; Assumption 3(c) is easy to satisfy by mildly extending the considered temporal region B; Assumption 3(d) is justified by our focus on finitely many scheduled news announcements.

[10]See, for example, weak instruments (Staiger and Stock (1997)), weak GMM (Stock and Wright (2000)), weak factors (Kleibergen (2009)), and weak fuzzy regression discontinuity designs (Feir, Lemieux, and Marmer (2016)).

3 Weak-Jump Asymptotics

3.1 Strong Identification Case: LTT's Results

Here we briefly reviews LTT's jump regression in strong identification cases. Define

$$\begin{aligned} R_p &\equiv \sqrt{\kappa_p}\,\sigma_{\mathfrak{a}_p-}\Psi_{p-} + \sqrt{1-\kappa_p}\,\sigma_{\mathfrak{a}_p}\Psi_{p+}, \\ \varsigma_p(\beta_0) &\equiv (-\beta_0, 1)R_p. \end{aligned} \tag{10}$$

where $(\kappa_p, \Psi_{p-}, \Psi_{p+})_{p\geq 1}$ is a sequence of triplets of mutually independent random variables, with $(\kappa_p)_{p\geq 1}$ being i.i.d. uniformly distributed on $(0,1]$, and both $(\Psi_{p-})_{p\geq 1}$ and $(\Psi_{p+})_{p\geq 1}$ being i.i.d. bivariate Gaussian vectors with identity covariance matrices. Moreover, consistent estimators for the spot pre-jump and post-jump volatilities are needed. Set

$$k_n = \lfloor c\Delta_n^{-\gamma} \rfloor, \quad \text{for some constant } c \text{ and } 0 < \gamma < 1.$$

in which $k_n \to \infty$ ensures consistency, and $k_n\Delta_n \to 0$ guarantees the estimation targets the spot volatility. At each i-th observation, the large-k estimators are

$$\begin{cases} \hat{c}_{n,i+} = \dfrac{\sum_{j=1}^{k_n} \Delta_{i+j}^n Z \Delta_{i+j}^n Z^\intercal 1_{\{-u_n \leq \Delta_{i+j}^n Z \leq u_n\}}}{\Delta_n \sum_{j=1}^{k_n} 1_{\{-u_n \leq \Delta_{i+j}^n Z \leq u_n\}}}, \\ \hat{c}_{n,i-} = \dfrac{\sum_{j=1}^{k_n} \Delta_{i-j}^n Z \Delta_{i-j}^n Z^\intercal 1_{\{-u_n \leq \Delta_{i-j}^n Z \leq u_n\}}}{\Delta_n \sum_{j=1}^{k_n} 1_{\{-u_n \leq \Delta_{i-j}^n Z \leq u_n\}}} \end{cases} \tag{11}$$

where $u_n \in \mathbb{R}^2$ is truncation level to remove potential jumps, specified as

$$\|u_n\| \asymp \Delta_n^{\varpi}, \quad \text{for some constant } \varpi \in (0, 1/2).$$

Based on Section 3.2 of LTT, the OLS estimator of β in model (7) is therefore

$$\tilde{\beta}_n = \tilde{\beta}_n(\mathfrak{a}_B) = \frac{\sum_{i\in\mathcal{I}(\mathfrak{a}_B)} \Delta_i^n X \Delta_i^n Y}{\sum_{i\in\mathcal{I}(\mathfrak{a}_B)} \Delta_i^n X^2} \tag{12}$$

where $\mathcal{I}(\mathfrak{a}_B)$ is defined in (6). As for the weighted least square (WLS), let the weighting function $w : \mathcal{M}_2 \times \mathcal{M}_2 \times \mathbb{R} \to \mathbb{R}$ be a continuous function, then the WLS estimator is

$$\hat{\beta}_n(\mathfrak{a}_B, w) = \frac{\sum\limits_{i \in \mathcal{I}(\mathfrak{a}_B)} w(\hat{c}_{n,i-}, \hat{c}_{n,i+}, \tilde{\beta}_n) \Delta_i^n X \Delta_i^n Y}{\sum\limits_{i \in \mathcal{I}(\mathfrak{a}_B)} w(\hat{c}_{n,i-}, \hat{c}_{n,i+}, \tilde{\beta}_n) \Delta_i^n X^2} \tag{13}$$

Moreover, the optimal weighting which minimizes asymptotic variance is

$$w^*(\hat{c}_{n,i-}, \hat{c}_{n,i+}, \tilde{\beta}_n) = \frac{2}{(-\tilde{\beta}_n, 1)(\hat{c}_{n,i-} + \hat{c}_{n,i+})(-\tilde{\beta}_n, 1)^\intercal} \tag{14}$$

which is the generalized least square (GLS) that achieves efficiency via transforming heteroskedasticity into homoskedasticity. And LTT shows that

$$\Delta_n^{-1/2}(\hat{\beta}_n(\mathfrak{a}_B, w) - \beta) \xrightarrow{\mathcal{L}\text{-}s} \frac{\sum\limits_{p \in \mathcal{P}_{\mathfrak{a}_B}} w(c_{\mathfrak{a}_p-}, c_{\mathfrak{a}_p}, \beta) \Delta X_{\mathfrak{a}_p} \varsigma_p}{\sum\limits_{p \in \mathcal{P}_{\mathfrak{a}_B}} w(c_{\mathfrak{a}_p-}, c_{\mathfrak{a}_p}, \beta) \Delta X_{\mathfrak{a}_p}^2} \tag{15}$$

the above result clearly subsumes the $\sqrt{n}$-consistency for the OLS and GLS by respectively letting $w = 1$ and $w = w^*$.

3.2 Jump Regressions under Weak-Jump Asymptotics

Given our model setup, we re-examine LTT's two-step jump regression procedure to investigate how weak jumps influence the estimation and inference. In the first step, one might want to apply the truncation method on high-frequency log-returns in $\mathcal{I}(\mathfrak{a}_B)$ to detect jumps. Define

$$\mathcal{I}_n(\mathfrak{a}_B) = \{i \in \mathcal{I}(\mathfrak{a}_B) : |\Delta_i^n X| > u_n\}, \quad u_n \asymp \Delta_n^{\varpi}, \quad \varpi \in (0, 1/2). \tag{16}$$

Clearly, $\mathcal{I}_n(\mathfrak{a}_B) \subseteq \mathcal{I}(\mathfrak{a}_B)$. In the strongly-identified case, we typically apply $\mathcal{I}_n(\mathfrak{a}_B)$ to tease out "strong" jumps in $\mathfrak{a}_B$, as justified by Proposition 1 of LTT. However in the weak jump case, we show in Theorem 1(a) that the truncation method (16) will asymptotically detect nothing. The asymptotic results of news-jump regression under weak jumps are summarized in the following theorem.

Theorem 1 (Weak-jump asymptotics)**.** Under Assumptions 1, 2, and 3, the following hold in restriction to $\Omega(\mathfrak{a}_B)$ which is defined by (8):

(a) $\mathbb{P}(\mathcal{I}_n(\mathfrak{a}_B) = \varnothing) \to 1$ as $n \to \infty$.

(b) For the ordinary least square estimator defined by (12), we have

$$\tilde{\beta}_n - \beta \xrightarrow{\mathcal{L}\text{-}s} \tilde{b} \equiv \frac{\sum_{p \in \mathcal{P}_{\mathfrak{a}_B}} (e_1^\intercal R_p + \xi_{\mathfrak{a}_p}) \varsigma_p(\beta)}{\sum_{p \in \mathcal{P}_{\mathfrak{a}_B}} (e_1^\intercal R_p + \xi_{\mathfrak{a}_p})^2} = O_{\mathbb{P}}(1). \tag{17}$$

where $e_1 = (1,0)^\intercal \in \mathbb{R}^2$, R_p and $\varsigma_p(\beta)$ are defined in (10).

(c) For weighted least square estimator defined by (13), we have

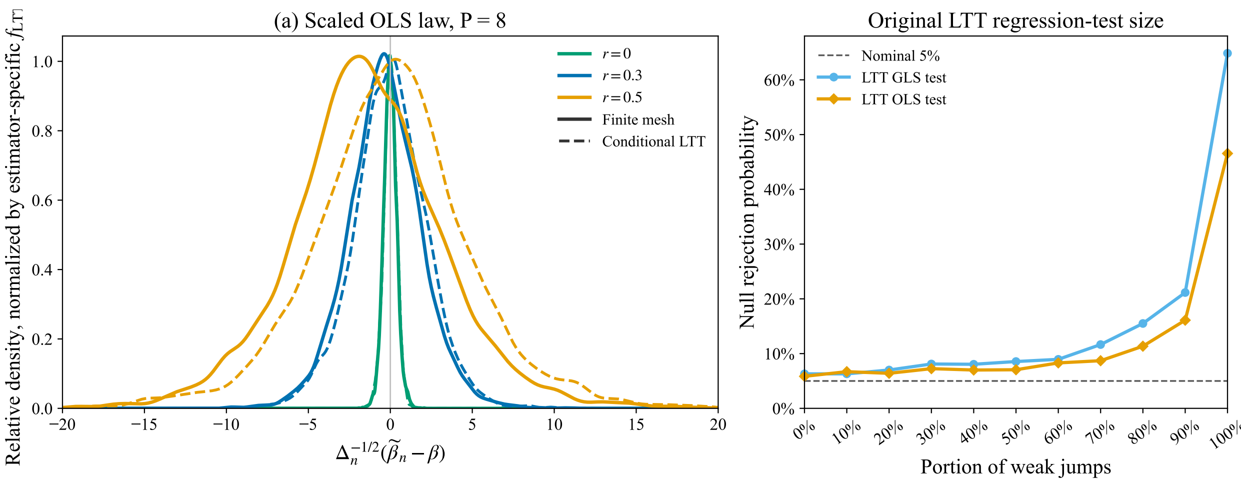
$$\hat{\beta}_n(\mathfrak{a}_B, w) - \beta \xrightarrow{\mathcal{L}\text{-}s} b_w = \frac{\sum_{p \in \mathcal{P}_{\mathfrak{a}_B}} w(c_{\mathfrak{a}_p -}, c_{\mathfrak{a}_p}, \beta + \tilde{b}) \left(e_1^\intercal R_p + \xi_{\mathfrak{a}_p}\right) \varsigma_p(\beta)}{\sum_{p \in \mathcal{P}_{\mathfrak{a}_B}} w(c_{\mathfrak{a}_p -}, c_{\mathfrak{a}_p}, \beta + \tilde{b}) \left(e_1^\intercal R_p + \xi_{\mathfrak{a}_p}\right)^2} \tag{18}$$

Theorem 1(a) indicates the standard truncation method asymptotically fails to detect any weak jumps, which is aligned with the concern over small jumps in real data, i.e., situations where a jump may be visually apparent as an announcement is released, but falls below the truncation threshold and is therefore misclassified as non-jump.

For Theorem 1 (b) and (c), we highlight three implications. First, WLS estimators $\hat{\beta}_n(\mathfrak{a}_B, w)$, which include OLS and GLS as special cases, are inconsistent under weak jumps, reflecting a typical consequence of weak identification that estimation becomes unreliable. Second, the weak-jump limits in Theorem 1 (b) and (c) are substantially heavier-tailed than the strong-identification limit (15), as in weak IV regressions (Phillips (1980, 1989)). For example, in jump regressions with a single jump, the LTT limit is Gaussian-type, whereas (17) is Cauchy-type even without scaling. The left panel of Figure 3 shows that the OLS limit increasingly departs from its strongly-identified counterpart as jumps shrink.[11] The right panel shows the size distortion in mixed samples: while a few weak announcements are negligible, rejection rates based on LTT procedures rise sharply above the nominal level once weak announcements dominate the sample. Third, inference for β with small jumps below the shrinkage trunction is infeasible due to the lack of a consistent estimator for $\xi_{\mathfrak{a}_p}$. These issues necessitate the development of identification-robust inferences.

[11]The GLS admits the severer weakness in the distributional approximation, see Supplement for details.

Figure 3: Scaled OLS law under weak jumps versus the LTT benchmark



Notes: The left panel compares $P = 8$ finite-mesh draws of the scaled OLS statistic, $\Delta_n^{-1/2}(\widetilde{\beta}_n - \beta)$, with pooled sample-specific conditional LTT draws for shrinkage rates $r = 0, 0.3, 0.5$. Each OLS pair is divided by its conditional-LTT density at zero, so the dashed curves have height one at zero. Solid curves are finite-mesh distributions and dashed curves are conditional LTT reference laws. The right panel retains the original mixed-sample OLS/GLS size comparison: null rejection probabilities from the LTT OLS- and GLS-based tests as the portion of weak jumps in a region increases.

4 The Anderson–Rubin Approach

In this section, we design the Anderson–Rubin test which is valid for any jump size by invoking the null hypothesis, inspired by Anderson and Rubin (1949). When the null is exploited and the test controls size uniformly over the parameter space, the test inversion method will generate the identification-robust confidence interval, see Andrews, Stock, and Sun (2019).

4.1 The AR Test and Critical Values

Here we construct a test which is robust to identification strength allowing price-volatility co-jumps. To begin with, we consider the inference for the jump beta β solely at single announcement times. Recalling the news-jump regression (7) and testing problem (5):

$$H_0 : \beta = \beta_0 \quad \text{against} \quad H_1 : \beta \neq \beta_0,$$

we propose the following Anderson–Rubin (AR) statistic on the p-th announcement:

$$S_{n,p}(\beta_0) = \Delta_n^{-1/2} \left(\Delta_{i(p)}^n Y - \beta_0 \Delta_{i(p)}^n X \right), \quad p \in \mathcal{P}_{\mathfrak{a}_B}. \tag{19}$$

The AR statistic (19) exploits the linear specification (4) at news announcement times to cancel jumps. Accordingly, after the proper scaling, $\Delta^n_{i(p)}Y - \beta_0\Delta^n_{i(p)}X$ is the empirical residual corresponding to $u_{p,n}(\beta_0)$ in the news-jump regression (7). If β_0 coincides with the true parameter β, then the X-jump and Y-jump components cancel in $S_{n,p}(\beta_0)$ by the linear specification[12], yielding a null distribution free of the jump size.

In Theorem 2(a), we establish that under the null, $S_{n,p}(\beta_0)$ will stably converge in law to $\varsigma_p(\beta_0)$ which is defined in (10). Then we aggregate the single AR tests $\{S_{n,p}(\beta_0)\}_{p\in\mathcal{P}_{\mathfrak{a}_B}}$ in L_2 norms across evenets to accumulate all signals in $\mathfrak{a}_B$. Since its limiting distribution is highly nonstandard, we apply Monte Carlo simulation to approximate critical values. The procedure is outlined in Algorithm 1 below.

Algorithm 1 (AR critical values)**.** For the inference problem (5), the critical values of our tests depend on β_0, and are simulated by the following steps.

(i) Generate one realization from a sequence of random triplets $(\tilde{\kappa}_p, \tilde{\Psi}_{p-}, \tilde{\Psi}_{p+})_{p\in\mathcal{P}_{\mathfrak{a}_B}}$ which is an independent copy of $(\kappa_p, \Psi_{p-}, \Psi_{p+})_{p\in\mathcal{P}_{\mathfrak{a}_B}}$, and calculate for each p

$$\begin{aligned}
\tilde{R}_{n,p} &= \sqrt{\tilde{\kappa}_p}\hat{\sigma}_{n,i(p)-}\tilde{\Psi}_{p-} + \sqrt{1-\tilde{\kappa}_p}\hat{\sigma}_{n,i(p)+}\tilde{\Psi}_{p+} \\
\tilde{\varsigma}_{n,p}(\beta_0) &= (-\beta_0, 1)\tilde{R}_{n,p} \\
\hat{v}_{S,p}(\beta_0) &= \frac{1}{2}(-\beta_0, 1)(\hat{c}_{n,i(p)-} + \hat{c}_{n,i(p)+})(-\beta_0, 1)^\intercal
\end{aligned}$$

where $(\hat{\sigma}_{n,i(p)-}, \hat{\sigma}_{n,i(p)+})$ are Cholesky decompositions of $\left(\hat{c}_{n,i(p)-}, \hat{c}_{n,i(p)+}\right)$ defined in (11), then we store the value $MC = \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \frac{[\tilde{\varsigma}_{n,p}(\beta_0)]^2}{\hat{v}_{S,p}(\beta_0)}$ in this one trial.

(ii) Given a significance level $\alpha \in (0,1)$, independently repeat step (i) for a large number of times and obtain a Monte Carlo sample of MC. Set $cv^\alpha_{n,AR}(\beta_0)$ as the $(1-\alpha)$-quantile of MC in the Monte Carlo sample.

We have the following identification-robust result.

Theorem 2. In restriction to $\Omega(\mathfrak{a}_B)$, suppose Assumptions 1, 2, 3 are satisfied, then

[12]The method can readily be extended to more flexible specifications, albeit at the cost of a more computationally demanding test-inversion procedure.

(a) $S_{n,p}(\beta_0) \xrightarrow{\mathcal{L}\text{-}s} \varsigma_p(\beta_0) + (\beta - \beta_0)\xi_{\mathfrak{a}_p}$, where $\varsigma_p(\beta_0) = (-\beta_0, 1)R_p$ and $R_p = \sqrt{\kappa_p}\sigma_{\mathfrak{a}_p-}\Psi_{p-} + \sqrt{1-\kappa_p}\sigma_{\mathfrak{a}_p}\Psi_{p+}$ for independent uniform random variable κ_p over the unit interval $(0, 1]$ and independent bivariate normals Ψ_{p-} and Ψ_{p+}.

(b) Normalizing then aggregating $\{S_{n,p}(\beta_0)\}_{p\in\mathcal{P}_{\mathfrak{a}_B}}$ in L_2-norm gives the AR test for $\mathfrak{a}_B$,

$$Q_S \equiv \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \left[\frac{S_{n,p}(\beta_0)}{\sqrt{\hat{v}_{S,p}(\beta_0)}}\right]^2$$

where $\hat{v}_{S,p}(\beta_0)$ is a consistent estimator for $v_{S,p}(\beta_0) \equiv \frac{1}{2}(-\beta_0, 1)(c_{\mathfrak{a}_p-} + c_{\mathfrak{a}_p})(-\beta_0, 1)^\intercal$ which is the conditional variance of $\varsigma_p(\beta_0)$ given $(c_{\mathfrak{a}_p-}, c_{\mathfrak{a}_p})$, by plugging in large-$k$ volatility estimators $(\hat{c}_{\mathfrak{a}_p-}, \hat{c}_{\mathfrak{a}_p})$. Under the null $H_0 : \beta = \beta_0$,

$$Q_S \xrightarrow{\mathcal{L}\text{-}s} \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \frac{[\varsigma_p(\beta_0)]^2}{v_{S,p}(\beta_0)}. \tag{20}$$

(c) The test based on (20) have uniform asymptotic size $\alpha \in (0, 1)$ over β_0, in the sense that $\lim_{n\to\infty} \mathbb{P}\left(Q_S \geq cv^{\alpha}_{n,AR}(\beta_0) \mid H_0\right) = \alpha$ for every $\beta_0 \in \mathbb{R}$, and $cv^{\alpha}_{n,AR}(\beta_0)$ is calculated by the procedure described in Algorithm 1.

Theorem 2(a) shows the null distribution of the AR test at p-th announcement is a Gaussian mixture, which means $\varsigma_p(\beta_0)$ is normally distributed conditional on $(\kappa_p, \sigma_{\mathfrak{a}_p-}, \sigma_{\mathfrak{a}_p})$. Given the Poissonian nature of the jump arrivals, the uniform random variable κ_p exactly locates the jump within the $i(p)$-th sampling interval. Under the null, Q_S is free of jumps and therefore robust to weak identification. Under the alternative, $(\beta-\beta_0)\xi_{\mathfrak{a}_p}$ provides power for the test and the power increases monotonically with $|\beta - \beta_0|$, in line with the AR test of weak IV regressions. Under strong or semi-strong identification, the AR test is consistent against a fixed alternative since $\xi_{\mathfrak{a}_p} = O_{\mathbb{P}}(n^{1/2-r})$ going to infinity for $r \in [0, 1/2)$.

The variance normalization in Q_S places the announcement-level AR tests on a common scale; without it, high-volatility announcements would mechanically receive greater weights. Noting Q_S is the L_2-type member of the general L_q tests $\{\sum_p |S_{n,p}(\beta_0)|^q : q = 1, 2, \cdots, \infty\}$, the insight of high-dimensional testing, see Kock and Preinerstorfer (2026), therefore naturally carries over to our setting: L_2 is well suited to dense signals, whereas L_∞ is more responsive to sparse but strong signals. Due to the fixed time span feature of our problem, we adopt the L_2-norm test to collect power across anouncements when the null is false. Our fixed-$|\mathcal{P}_{\mathfrak{a}_B}|$ simulations using empirically-calibrated parameters in Supplementary Figure S2 confirm that L_2-test is robustly powerful and further show that the t-ratio test based on Bakirov and Szekely (2005) exhibits very low power.

Theorem 2(c) allows us to conduct test inversion for β. Accordingly, define the AR test $\varphi_{AR}^{\alpha} \equiv 1\{Q_S > cv_{n,AR}^{\alpha}(\beta_0)\}$ in which 1 means rejection and 0 indicates non-rejection. And the identification-robust confidence interval using the AR test is defined as

$$\mathrm{CI}_{AR}^{\alpha} \equiv \{\beta_0 \in \mathbb{R} : \varphi_{AR}^{\alpha}(\beta_0) = 0\}.$$

In practice, we invert the test over a fine grid of β_0 values spanning a sufficiently large neighborhood of zero. We emphasize that $\mathrm{CI}_{AR}^{\alpha}$ does not supriously recover the continuous beta which is a nuisance parameter keeping equal under the null and the alternative, therefore does not contribute to distributional discrepancy for hypothesis testing. Additionally, without jumps, the AR confidence interval $\mathrm{CI}_{AR}^{\alpha}$ will be wide because each β_0 is rejected only at the nominal rate α, reflecting identification failure.

5 The Conditional Approach

Consistent with the weak identificaiton literature (Kleibergen (2005), Andrews (2016)), the Anderson–Rubin test is inefficient with strong identification in overidentified cases (in our setting, $|\mathcal{P}_{\mathfrak{a}_B}| \geq 2$), since they equally weights the each moments, while the LTT-GLS optimally weighted those strongly identied news-jumps. This necessaite the exploration on the conditional approach.

When there is no price-volatility co-jump at announcement times, i.e., $c_{\mathfrak{a}_p-} = c_{\mathfrak{a}_p}$ for all $p \in \mathcal{P}_{\mathfrak{a}_B}$, the limit experiment simplifies to a Gaussian shift experiment. This subsection develops the AR, LM, and CLR tests in this tractable case, following Moreira (2003) and the weak-instrument framework in Andrews, Moreira, and Stock (2006). We state the limit experiment, establish sufficiency of two statistics, construct the tests, and prove that they control size with analytical critical values.

5.1 Conditional Inference in Gaussian Limit Problem

We impose the following assumption to simplify the limit experiment from Gaussian mixture to Gaussian shift experiment in which a sufficient statistic for jumps can be found due to the availability of closed-form likelihood.

Assumption 4 (No price-volatility co-jump)**.** For all $p \in \mathcal{P}_{\mathfrak{a}_B}$, $c_{\mathfrak{a}_p-} = c_{\mathfrak{a}_p}$. That is, the spot covariance matrix does not jump at the p-th announcement time.

Assumption 4 is also imposed by LTT in the analysis of semiparametric efficiency. We define $g = (1, \beta)^\intercal$ and $h = (-\beta, 1)^\intercal$ for any $\beta \in \mathbb{R}$, and let g_0 and h_0 denote their counterparts with β replaced by β_0. Let $\delta \equiv \beta - \beta_0$ denote the deviation from the null, therefore $g = g_0 + \delta e_2$ where $e_2 = (0, 1)^\intercal$. Under Assumptions 1, 2, 3, and 4, in restriction to $\Omega(\mathfrak{a}_B)$, the limit experiment for $\Delta_n^{-1/2} \left(\Delta_{i(p)}^n X, \Delta_{i(p)}^n Y\right)^\intercal, p \in \mathcal{P}_{\mathfrak{a}_B}$ is Gaussian, namely

$$\frac{\Delta_{i(p)}^n Z}{\Delta_n^{1/2}} \xrightarrow{\mathcal{L}\text{-}s} Z_p \equiv (X_p, Y_p)^\intercal \sim \mathcal{N}\left(\xi_{\mathfrak{a}_p} g, \boldsymbol{c}_{\mathfrak{a}_p}\right), \tag{21}$$

and Z_p's are independent across $p \in \mathcal{P}_{\mathfrak{a}_B}$. Here $\xi_{\mathfrak{a}_p}$ is the nuisance parameter (rescaled jump) and β is the parameter of interest. Throughout, subscripts t and p refer to different layers of the model: t indexes the underlying continuous-time processes, whereas p indexes announcement-level coordinates of the Gaussian limit experiment obtained under infill asymptotics. According to LeCam's framework (Chapter 9 of Van der Vaart (2000), Müller (2011), Andrews (2016)) in the infill asymptotics, conducting asymptotic inference for the discretized observations is equivalent to conducting the corresponding inferential procedure in the Gaussian model, as in Bollerslev, Li, and Ren (2024).

We then define two statistics exploiting the null candidate β_0. The first one is a sufficient statistic for the rescaled jump $\xi_{\mathfrak{a}_p}$, which captures the strength of the jump signal:

$$T_p \equiv \frac{g_0^\intercal c_{\mathfrak{a}_p}^{-1} Z_p}{g_0^\intercal c_{\mathfrak{a}_p}^{-1} g_0}, \quad p \in \mathcal{P}_{\mathfrak{a}_B}. \tag{22}$$

Moreover, T_p is the profile maximum likelihood estimator of $\xi_{\mathfrak{a}_p}$. Inspired by the literature on conditional inference, see Moreira (2003) and Andrews (2016), we condition on the sufficient statistics for the nuisance parameters $(\xi_{\mathfrak{a}_p})_p$. The resulting conditional distribution is free of these nuisance parameters, thereby ensuring identification-robust inference. The second is an ancillary statistic for β, which is the Anderson–Rubin statistic corresponding to $S_{n,p}(\beta_0)$ in the finite sample:

$$S_p \equiv h_0^\intercal Z_p = Y_p - \beta_0 X_p, \quad p \in \mathcal{P}_{\mathfrak{a}_B}. \tag{23}$$

The properties of the two statistics are displayed in the following lemma.

Lemma 1. In the Gaussian limit experiment (21), for each $p \in \mathcal{P}_{\mathfrak{a}_B}$, the following hold:

(a) Under the null $H_0 : \delta = 0$, the statistic T_p defined in (22) is the profile maximum likelihood estimator of $\xi_{\mathfrak{a}_p}$ and sufficient for $\xi_{\mathfrak{a}_p}$.

(b) Under both null and alternative, the two statistics S_p and T_p are independent. Specifically, their joint distribution is as follows.

$$\begin{pmatrix} S_p \\ T_p \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \delta\xi_{\mathfrak{a}_p} \\ (1+\delta\theta_p)\xi_{\mathfrak{a}_p} \end{pmatrix}, \begin{pmatrix} v_{S,p} & 0 \\ 0 & v_{T,p} \end{pmatrix}\right). \tag{24}$$

where $v_{S,p} \equiv h_0^\intercal c_{\mathfrak{a}_p} h_0$, $v_{T,p} \equiv (g_0^\intercal c_{\mathfrak{a}_p}^{-1} g_0)^{-1}$, and $\theta_p \equiv v_{T,p} g_0^\intercal c_{\mathfrak{a}_p}^{-1} e_2$. Consequently, under the null $\delta = 0$,

$$\begin{pmatrix} S_p \\ T_p \end{pmatrix} \Bigg| \; H_0 \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ \xi_{\mathfrak{a}_p} \end{pmatrix}, \begin{pmatrix} v_{S,p} & 0 \\ 0 & v_{T,p} \end{pmatrix}\right).$$

Lemma 1(a) implies that nuisances $(\xi_{\mathfrak{a}_p})_{p\in\mathcal{P}_{\mathfrak{a}_B}}$ cannot affect the null distribution any tests conditional on $(T_p)_{p\in\mathcal{P}_{\mathfrak{a}_B}}$. The independence between S_p and T_p in Lemma 1(b), which reflects Basu's theorem (see Theorem 1.6.21 of Lehmann and Casella (1998)), allows us to simulate conditional critical values. Let $S_p^* \equiv \frac{S_p}{\sqrt{v_{S,p}}}$, $T_p^* \equiv \frac{T_p}{\sqrt{v_{T,p}}}$, $\xi_{\mathfrak{a}_p}^* \equiv \frac{\xi_{\mathfrak{a}_p}}{\sqrt{v_{T,p}}}$ and thereby we have the standardized version of the two statistics:

$$\begin{pmatrix} S_p^* \\ T_p^* \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \delta\lambda_p\xi_{\mathfrak{a}_p}^* \\ (1+\delta\theta_p)\xi_{\mathfrak{a}_p}^* \end{pmatrix}, I_2\right) \tag{25}$$

where $\lambda_p \equiv \sqrt{v_{T,p}/v_{S,p}}$. Furthermore, we stack S_p^* and T_p^* across p, so define $\xi^* \equiv (\xi_{\mathfrak{a}_p}^*)_{p\in\mathcal{P}_{\mathfrak{a}_B}}$ and

$$S^* = (S_p^*)_{p\in\mathcal{P}_{\mathfrak{a}_B}} \sim \mathcal{N}(\delta\Lambda\xi^*, I_{|\mathcal{P}_{\mathfrak{a}_B}|})$$

$$T^* = (T_p^*)_{p\in\mathcal{P}_{\mathfrak{a}_B}} \sim \mathcal{N}((I_{|\mathcal{P}_{\mathfrak{a}_B}|} + \delta\Theta)\xi^*, I_{|\mathcal{P}_{\mathfrak{a}_B}|})$$

where $\Lambda \equiv \operatorname{diag}((\lambda_p)_{p\in\mathcal{P}_{\mathfrak{a}_B}})$, $\theta \equiv (\theta_p)_{p\in\mathcal{P}_{\mathfrak{a}_B}}$, and $\Theta \equiv \operatorname{diag}(\theta)$. Clearly, T^* maintains the sufficient statistics for $(\xi_{\mathfrak{a}_p})_{p\in\mathcal{P}_{\mathfrak{a}_B}}$. Following Andrews, Moreira, and Stock (2006), define

$$Q = \begin{pmatrix} S^* & T^* \end{pmatrix}^\intercal \begin{pmatrix} S^* & T^* \end{pmatrix} = \begin{pmatrix} \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} (S_p^*)^2 & \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} S_p^* T_p^* \\ \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} S_p^* T_p^* & \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} (T_p^*)^2 \end{pmatrix} \equiv \begin{pmatrix} Q_S & Q_{ST} \\ Q_{ST} & Q_T \end{pmatrix} \tag{26}$$

and the 2×2 matrix Q is a maximal invariant for the transformation $G \equiv$, and each invariant test can be written as a function of the maximal invariant (see Theorem 6.2.1 of Lehmann and Romano (2005)). We first derive the Lagrange multiplier (LM) statistic and its distribution in the limit problem.

Theorem 3 (Lagrangian multiplier test)**.** Under the Gaussian limit experiment (21), the Lagrange multiplier statistic for $H_0 : \beta = \beta_0$ is

$$LM = \frac{\left(\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \lambda_p S_p^* T_p^*\right)^2}{\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \lambda_p^2 (T_p^*)^2}. \tag{27}$$

Under H_0, $LM \mid T^* \sim \chi_1^2$, and therefore unconditionally $LM \sim \chi_1^2$. Additionally, if $c_{\mathfrak{a}_p}$'s are equal for all $p \in \mathcal{P}_{\mathfrak{a}_B}$, the LM statistic simplifies to $LM^h = \frac{Q_{ST}^2}{Q_T}$.

The LM test statisic is exactly in the same spirit as the K-statistic (Kleibergen (2002)) and score-type test (Moreira (2001, 2009)), noting our parameter of interest β is scalar. We let $\varphi_{LM}^\alpha \equiv 1\{LM > \chi^2_{1,1-\alpha}\}$ denote the LM test. Under strong identification, it can be shown that φ_{LM}^α is optimal in the sense of being the uniformly most powerful unbiased (UMPU) test, see Proposition S2(a) in supplement material for a proof, but its power is low under weak identification with power ditches as shown in simulated power curve. Essentially, perceptions about the LM test are consistent with the weak IV literature, see Andrews, Moreira, and Stock (2006) and Andrews (2016).

We next derive the conditional likelihood ratio (CLR) test in the standardized experiment generated by $(S_p^*, T_p^*)_{p\in\mathcal{P}_{\mathfrak{a}_B}}$. Unlike the LM test, the CLR test remains powerful under weak identification due to the conditionality which adjusts critical values according to the signal strength reflected by the sufficient statistics T^*.

Theorem 4 (Conditional likelihood ratio test)**.** In the limit experiment (21) for testing $H_0 : \beta = \beta_0$ against $H_1 : \beta \neq \beta_0$, we have the following results:

(a) The likelihood ratio test is

$$LR = \sup_{\delta \in \mathbb{R}} \sum_{p \in \mathcal{P}_{\mathfrak{a}_B}} \frac{\left\{\delta\lambda_p S_p^* + (1+\delta\theta_p)T_p^*\right\}^2}{\delta^2\lambda_p^2 + (1+\delta\theta_p)^2} - Q_T \tag{28}$$

Let $\zeta_p \overset{iid}{\sim} \mathcal{N}(0,1)$ for $p \in \mathcal{P}_{\mathfrak{a}_B}$, the null distribution of LR conditional on T^* is

$$LR \mid T^* \sim F_{T^*} \equiv \sup_{\delta \in \mathbb{R}} \sum_{p \in \mathcal{P}_{\mathfrak{a}_B}} \frac{\left\{\delta\lambda_p \zeta_p + (1+\delta\theta_p)T_p^*\right\}^2}{\delta^2\lambda_p^2 + (1+\delta\theta_p)^2} - T^{*\intercal}T^*.$$

where F_{T^*} is a continuously distributed random variable. The conditional test $\varphi^{\alpha}_{CLR} \equiv 1\{LR \mid T^* > cv^{\alpha}_{CLR}(T^*)\}$ in which $cv^{\alpha}_{CLR}(T^*)$ denotes the $(1-\alpha)$-quantile of F_{T^*} is conditionally similar in the sense that $\mathbb{E}_{H_0,\mathbb{P}}[\varphi^{\alpha}_{CLR} \mid T^*] = \alpha$.

(b) Additionally, if $c_{\mathfrak{a}_p}$'s are equal for all $p \in \mathcal{P}_{\mathfrak{a}_B}$, then the likelihood ratio test admits a closed form as follows.

$$LR^h = \frac{1}{2}\left(Q_S - Q_T + \sqrt{(Q_S+Q_T)^2 - 4(Q_S Q_T - Q_{ST}^2)}\right). \tag{29}$$

The null distribution of LR^h conditional on T^* is

$$LR^h \mid T^* \sim \frac{1}{2}\left(\chi^2_{|\mathcal{P}_{\mathfrak{a}_B}|} - T^{*\intercal}T^* + \sqrt{\left(\chi^2_{|\mathcal{P}_{\mathfrak{a}_B}|} + T^{*\intercal}T^*\right)^2 - 4T^{*\intercal}T^* \cdot \chi^2_{|\mathcal{P}_{\mathfrak{a}_B}|-1}}\right). \tag{30}$$

We define the conditional test $\varphi^{\alpha}_{CLR^h} \equiv 1\{LR^h \mid T^* > cv^{\alpha}_{CLR^h}(T^*)\}$ where $cv^{\alpha}_{CLR^h}(T^*)$ is the $(1-\alpha)$-quantile of the conditional distribution in (30). The homogeneous CLR test $\varphi^{\alpha}_{CLR^h}$ maintains conditional similarity under heterogeneity in the sense that $\mathbb{E}_{H_0,\mathbb{P}}[\varphi^{\alpha}_{CLR^h} \mid T^*] = \alpha$.

The forms of CLR tests given by Theorem 4 are similar to those in the weak-instrument literature under both heterogeneity, see (2.6) of Moreira and Moreira (2019), and homogeneity, see (3) of Moreira (2003) in which the conditionally pivotal test is presented. This is not surprising because of the nature that jumps as signals play similar roles as instruments for teasing out the structural relationships, as revealed by analogy between (1) and (2) in the Introduction. The key difference lies in asymptotic scheme: in weak IV, the limit experiment is univariate Gaussian due to the central limit theorem, while in jump regressions, limit experiment is multivariate Gaussian induced by the semimatringale modelling.

Under strong identification, we show in Proposition S2(a) in supplementary material that the CLR test φ_{CLR} is asymptotically equivalent to the LM test φ_{LM} and therefore a UMPU test, in line with Theorem 6 in Andrews, Moreira, and Stock (2006). The CLR test under homogeneity, φ^h_{CLR}, although is not the CLR test under heterogeneity, still controls size and lies in the class of Quasi-CLR (QCLR) tests proposed by Kleibergen (2005) in weak GMM setting. Accordingly, provided a function $q : \mathbb{R}^{|\mathcal{P}_{\mathfrak{a}_B}|} \to \mathbb{R} \cup \infty$, we define the class of the QCLR tests as follows.

$$QCLR_q = \frac{1}{2}\left[Q_S - q(T^*) + \sqrt{(Q_S + q(T^*))^2 - 4(Q_S - LM) \cdot q(T^*)}\right]. \tag{31}$$

Intuitively, there is efficiency lost of $\varphi^{\alpha}_{CLR^h}$ compared to φ^{α}_{CLR} with strong jumps under the default heterogeneity setting, this is evident by simulation and the following Corollary 1.

Corollary 1. Under Assumptions 1, 2, 3, 4 and conditional on T^*, the conditional tests φ^{α}_{CLR} and $\varphi^{\alpha}_{CLR^h}$ defined in Theorem 4 are admissible for every $\alpha \in (0, 1)$. That being said, in the weakly-identified limit problem, neither φ^{α}_{CLR} nor $\varphi^{\alpha}_{CLR^h}$ can dominate each other.

Given the AR, LM tests and the conditional procedure with sufficient statistic T^*, we thereby develop the conditional linear combination (CLC) test of jump regressions in the spirit of Andrews (2016). It is theoretically precieved (Example I, page 2166 in Andrews (2016)) and observed in simulations, see Figure 4, that the LM test, although optimal in the strong identification case, can have non-monotonic power curve with weak identification. To compensate the power ditch issue of LM test, Andrews (2016) proposes a linear combination of LM and AR test which has favorable stability in power, and the combination weight is adaptively chosen depending on a statistic capturing the identification strength.

To sharpen the sufficient statistic, we apply a bijective linear transformation $T^* \mapsto D \equiv \Lambda T^*$, and use D as the conditioning statistic for the CLC test. We make this adjustment due to $\frac{\partial}{\partial \delta}\mathbb{E}(S^*) \mid_{\delta=0} = \Lambda \mathbb{E}(T^*) \mid_{\delta=0}$, which means D statistic coincides with the score direction of the AR test deviating from the null and therefore exploits identifying signal in its optimal-gradient way.

To fix ideas, let $J \equiv Q_S - LM$, then we have $LM = S^{*\intercal} P_D S^* \sim \chi^2_1$ where $P_D = D(D^\intercal D)^{-1} D^\intercal$ and $J \sim \chi^2_{|\mathcal{P}_{\mathfrak{a}_B}|-1}$, and LM and J are independent conditional on D. Let $\mathcal{A}$ denote the collection of measurable functions $a : \mathcal{D} \to [0,1]$, where $\mathcal{D}$ is the support of D. For any $a \in \mathcal{A}$, the CLC test is defined as

$$\varphi^\alpha_{CLC,a(D)} = 1\{LM + a(D) \cdot J > c_\alpha(a(D))\}$$

where $c_\alpha(a(D))$ is the upper $(1-\alpha)$-quantile of the conditional distribution $\chi^2_1 + a(D) \cdot \chi^2_{|\mathcal{P}_{\mathfrak{a}_B}|-1}$ given D in which the two chi-squares are independent. The class of CLC tests is rich because it is indexed by the entire collection $\mathcal{A}$, allowing the weight placed on J to vary flexibly with the conditioning statistic D. It nests φ^α_{LM} and φ^α_{AR} as the special cases $a \equiv 0$ and $a \equiv 1$, respectively. Moreover, Theorem 4 of Andrews (2016) shows that the class of CLC tests coincides with the class of QCLR tests defined in (31).

Note $D = \Lambda T^*$, let $\mu_D \equiv \mathbb{E}(D) = \Lambda(I + \delta\Theta)\xi^*$ and $\Sigma_D \equiv \mathrm{Var}(D) = \Lambda^2$. We define the concentration parameter as follows.

$$\mathfrak{c} = \mu_D^\intercal \Sigma_D^{-1} \mu_D = \|(I + \delta\Theta)\xi^*\|^2$$

Under the null $\delta = 0$, we have $\mathfrak{c} = \xi^{*\intercal}\xi^*$. Meanwhile, the deviation from the null about jump beta is captured by the expectation of S^*, we let $m \equiv \mathbb{E}(S^*) = \delta\Lambda\xi^*$. Clearly, for a generic value of μ_D, possible values of m, are restrictive. We let $\mathcal{M}_D(\mu_D)$ denote the compatible set of values of m given μ_D, i.e.

$$\mathcal{M}_D(\mu_D) \equiv \left\{ m = \delta\Lambda\xi^* : \mu_D = \Lambda(I_{|\mathcal{P}_{\mathfrak{a}_B}|} + \delta\Theta)\xi^*, \delta \in \mathbb{R}, \xi^* \in \mathbb{R}^{|\mathcal{P}_{\mathfrak{a}_B}|} \right\}. \tag{32}$$

In summary, we have $S^* \sim \mathcal{N}(m, I_{|\mathcal{P}_{\mathfrak{a}_B}|})$, $m \in \mathcal{M}_D(\mu_D)$ and $D \sim \mathcal{N}(\mu_D, \Lambda^2)$ in which $S^* \perp\!\!\!\perp D$. Due to the computational burden of searching weight function $a(\cdot)$ over its space $\mathcal{A}$, we follow Andrews (2016) and consider plug-in minimax regret unconditional (MMRU) test, which means the weight is chosen as a constant based on μ_D in a minmax regret sense. To illustrate, for a constant weight CLC test $\varphi^\alpha_{CLC,a}$, define the power envelope formed by adjusting a as $\beta^u(m, \mu_D) \equiv \sup_{a \in [0,1]} \mathbb{E}_{m,\mu_D}(\varphi^\alpha_{CLC,a})$ on the alternative (m, μ_D), and determine the weight by

$$a_{\mathrm{MMRU}}(\mu_D) \equiv \max \operatorname*{arg\,min}_{a \in [0,1]} \sup_{m \in \mathcal{M}_D(\mu_D)} \left\{ \beta^u(m, \mu_D) - \mathbb{E}_{m,\mu_D}\left[\varphi^\alpha_{CLC,a}\right] \right\}. \tag{33}$$

Note $a_{\text{MMRU}}(\cdot)$ is numerically available accroding to the non-centrality of chi-square distributions. And the plug-in MMRU test φ_{PI}^{α} is defined as

$$\varphi_{PI}^{\alpha} \equiv \varphi_{CLC,a_{\text{MMRU}}(\hat{\mu}_D)}^{\alpha} = 1\{LM + a_{\text{MMRU}}(\hat{\mu}_D) \cdot J > c_{\alpha}(a_{\text{MMRU}}(\hat{\mu}_D))\}$$

for a plausible estimate $\hat{\mu}_D$ of μ_D. A naive esimate of μ_D is D, however plugining in $\hat{\mu}_D = D$ will bias the estimation of non-centrality parameter since $D\Lambda^{-2}D \sim \chi^2_{|\mathcal{P}_{\mathfrak{a}_B}|}(\mathfrak{c})$ implies $\mathbb{E}(D\Lambda^{-2}D) = \mathfrak{c} + |\mathcal{P}_{\mathfrak{a}_B}|$ and affect the optimization (33). Therefore, we follow Andrews (2016) and shrink D to tame quadratic-form bias based on concentration parameter. Define $\hat{\mathfrak{c}} \equiv D^{-1}\Lambda^{-2}D = Q_T$ as the naive estimator for $\mathfrak{c}$, and consider $\hat{\mathfrak{c}}_{\text{MLE}}$, the MLE whose closed form is not available, positive part $\hat{\mathfrak{c}}_{\text{PP}} \equiv \max(Q_T - |\mathcal{P}_{\mathfrak{a}_B}|, 0)$, and an estimator proposed by Kubokawa, Roberts, and Saleh (1993):

$$\hat{\mathfrak{c}}_{\text{KRS}} \equiv \hat{\mathfrak{c}} - k + e^{-\frac{\hat{\mathfrak{c}}}{2}} \left(\sum_{j=0}^{\infty} \left(-\frac{\hat{\mathfrak{c}}}{2} \right)^2 \frac{1}{j!(k+2j)} \right)^{-1}$$

Accordingly, the plug-in MMRU test should apply the estimated mean of D

$$\hat{\mu}_{D,i} \equiv D \cdot \sqrt{\frac{\hat{\mathfrak{c}}_i}{\hat{\mathfrak{c}}}}, \quad i \in \{\text{MLE, PP, KRS}\} \tag{34}$$

Due to its similar but different nature of jump regressions to IV regressions rooted in infill asymptotics, we go throught the details of CLC tests of Andrews (2016) in jump regressions and collect the necessary results in the following theorems.

Theorem 5 (Conditional linear combination test)**.** Under Assumptions, 2, 3, 4, we have the following results for the CLC test.

(a) For any measurable function $a : \mathbb{R}^{|\mathcal{P}_{\mathfrak{a}_B}|} \to [0,1]$, the CLC test $\varphi_{CLC,a(D)}^{\alpha} = 1\{LM + a(D) \cdot J > c_{\alpha}(a(D))\}$ satisfies $\mathbb{E}_{H_0,\mu_D}\left[\varphi_{CLC,a(D)}^{\alpha} \mid D\right] = \alpha$ for every μ_D, outsite a $\mathbb{P}$-null set $\{D = 0\}$. That being said, the CLC test is almost surely conditionally similar.

(b) Let $P \equiv |\mathcal{P}_{\mathfrak{a}_B}| \geq 2$. In the strong or semi-strong identification cases, the MMRU weight defined by (33) satisfies

$$a_{\text{MMRU}}(\mu_D) \to 0. \tag{35}$$

5.2 Conservative Inference in the Mixed Gaussian Limit Experiments

In light of the high-frequency evidence of price–volatility co-jumps and announcement-induced volatility jumps (see Jacod and Todorov (2010); Bollerslev, Li, and Liao (2021)), Assumption 4 appears empirically restrictive. However, allowing price-volatility co-jumps complicates the theoretical analysis, in the sense that the limit experiment become mixed normal and sufficient statistics are no longer available so as to render conditioanl inference infeasible. Without Assumption 4, the shift limit experiment is

$$Z_p \sim \xi_{\mathfrak{a}_p} g + \sqrt{\kappa_p}\sigma_{\mathfrak{a}_p-}\Psi_{p-} + \sqrt{1-\kappa_p}\sigma_{\mathfrak{a}_p}\Psi_{p+}, \quad p \in \mathcal{P}_{\mathfrak{a}_B}. \tag{36}$$

A natural idea is to still consider the conditional inference in the Gaussian limit experiment with the same mean $\xi_{\mathfrak{a}_p}$ but a adjusted variance $\tilde{\sigma}_p$ to control size. Two reasonable choices are $\tilde{\sigma}_p = (\sigma_{\mathfrak{a}_p-} + \sigma_{\mathfrak{a}_p})/2$ or $\tilde{\sigma}_p = \max(\sigma_{\mathfrak{a}_p-}, \sigma_{\mathfrak{a}_p})$. While the former choice is more efficient and the average is exactly the spot variance estimator at $\mathfrak{a}_p$, the latter one is more conservative for using the variance envelope. In simulations, we find using $(\sigma_{\mathfrak{a}_p-} + \sigma_{\mathfrak{a}_p})/2$ is mildly oversized, using $\max(\sigma_{\mathfrak{a}_p-}, \sigma_{\mathfrak{a}_p})$ well controls size, and both method generate more powerful tests than the AR method. In the empirical application, we find that the two choices yield similar results.

Theorem 6. Suppose Assumptions 1, 2, 3 hold. In the limit experiment (36), the conditional tests φ^{α}_{CLR}, $\varphi^{\alpha}_{CLR^h}$, φ^{α}_{CLC} using Gaussian limit experiment with volatilities $\tilde{\sigma}_p = \max(\sigma_{\mathfrak{a}_p-}, \sigma_{\mathfrak{a}_p})$ control size, in the sense that $\mathbb{E}(\varphi^{\alpha}_{j}|H_0) \leq \alpha$ for $j = CLR, CLR^h, CLC$.

6 Simulation Evidence

6.1 The Benchmark Data Generating Processes

Following Bollerslev and Todorov (2011), we adopt a two-factor structure of volatility for our data generating process (DGP) and calibrate the DGP by incorporating key features of high-frequency financial data, including the continuous beta, leverage effect, price-volatility co-jumps, and idiosyncratic jumps. The process is set to span over one year, corresponding to 250 trading days and each day includes 6.5 trading hours. We sample the process at 1-minute frequency to ameliorate the microstructure noise concern while maintaining the role of infill asymptotics. The announcements $\mathfrak{a}$ are pre-scheduled to arrive immediately after 2 p.m. on the 20th, 60th, 80th, 110th, 130th, 160th, 205th, and 230th trading days in a typical year, mimicking the FOMC announcements time. We let β denote the jump beta which is our parameter of interest. The diffusion coefficients (σ_X, σ_Y) of the process Z are governed by two factors V_1 and V_2 which are designed as follows,

$$\begin{aligned}
&\mathrm{d}V_{1,t} = 0.0128(0.4068 - V_{1,t})\mathrm{d}t + 0.0954\sqrt{V_{1,t}}(\rho \mathrm{d}W_{1,t} + \sqrt{1-\rho^2}\mathrm{d}B_{1,t}),\\
&\mathrm{d}V_{2,t} = 0.6930(0.4068 - V_{2,t})\mathrm{d}t + 0.7023\sqrt{V_{2,t}}(\rho \mathrm{d}W_{1,t} + \sqrt{1-\rho^2}\mathrm{d}B_{2,t}),\\
&\rho = -0.7, \quad V_{1,0} = 0.5, \quad V_{2,0} = 0.5
\end{aligned}$$

where W_1, B_1, and B_2 are independent standard Brownian motions, with W_1 capturing the systematic shock, while B_1 and B_2 drive idiosyncratic shocks. The leverage effect is incorporated through $\rho = -0.7$, a negative correlation between price and volatility shocks. The volatility processes σ_X and σ_Y depend on V_1 and V_2 in the following way,

$$\sigma^2_{X,t} = V_{1,t} + V_{2,t} + \sum_{p=1}^{8} J^\sigma_p \cdot 1\{\mathfrak{a}_p \le t\}, \quad J^\sigma_p \overset{iid}{\sim} \exp(0.125) \tag{37}$$

$$\sigma^2_{Y,t} = [2 - 0.75\sin^2(t)](V_{1,t} + V_{2,t}) \tag{38}$$

where $J^\sigma_p, p = 1, \cdots, 8$ are *i.i.d.* exponential random variables with mean equal to eight. We set $Z_0 = 0$, and the dynamics of log-prices are described as follows.

$$\mathrm{d}X_t = \sigma_{X,t}\mathrm{d}W_{1,t} + \sum_{p=1}^{6} \xi_{\mathfrak{a}_p} 1\{\mathfrak{a}_p \le t\} \cdot \Delta^r_n, \quad \xi_{\mathfrak{a}_p} \overset{iid}{\sim} \eta_p \cdot \mathcal{N}(5, 0.25), \tag{39}$$

$$\mathrm{d}Y_t = \beta_c \cdot \sigma_{X,t}\mathrm{d}W_{1,t} + \sigma_{Y,t}\mathrm{d}W_{2,t} + \varphi_{Y,t}\mathrm{d}N_t \cdot 1\{t \notin \mathfrak{a}\} + \beta \sum_{p=1}^{6} \xi_{\mathfrak{a}_p} 1\{\mathfrak{a}_p \le t\} \cdot \Delta^r_n, \tag{40}$$

$$\beta_c = 0.89, \quad \varphi_{Y,t}|\sigma_{Y,t} \sim \mathcal{N}(0, \phi^2\sigma^2_{Y,t}), \quad \phi = 5,$$

N_t is a Poisson process with intensity $\lambda_N = 15$.

where β_c is the continuous beta and η_p's denote independent Rademacher random variable. The shrinkage component Δ^r_n controls the identification strength: when $r = 0$, jumps are strongly identified, whereas $r = 0.5$ corresponds to the weak jump case. To demonstrate the robustness of our methods for the absence of jumps, we assume no jump occurs at the last two announcements, $\mathfrak{a}_7$ and $\mathfrak{a}_8$. Realizations of the DGP are presented in Figure S3 and Figure S4 in the supplementary material for visualization.

Two tuning parameters are required: the bandwidth k_n for the large-k estimators spot volatility, and the truncation threshold u_n for filtering jumps. Here we exploit the balanced window for spot volatility estimation efficiency, specifically, $k_n = 3 \times \lfloor \Delta_n^{-1/2} \rfloor = 57$. As for the truncation u_n, it is necessary to scale the threshold magnitude by the volatility level as emphasized by LTT. To do so, we use the truncation level $u_n = (u_n^X, u_n^Y)^\intercal$ for $M = X, Y$, and on day d,

$$u_n^M = 4 \times \sqrt{\mathrm{BV}_d} \times \Delta_n^{0.49}, \quad \mathrm{BV}_d = \frac{pi}{2} \frac{\lfloor 1/\Delta_n \rfloor}{\lfloor 1/\Delta_n \rfloor - 1} \sum_{i=\lfloor (d-1)/\Delta_n \rfloor + 2}^{\lfloor d/\Delta_n \rfloor} |\Delta_{i-1}^n M| |\Delta_i^n M|, \tag{41}$$

where $pi = 3.1415926$. Here BV_d is the bipower variation estimator for the integrated variance that is robust to jump, see Barndorff-Nielsen and Shephard (2004, 2006). The rate $\varpi = 0.49$ is close to one-half to enhance jump detection at the expense of an increased false discovery rate. Consequently, jumps below u_n are regarded as small.

We let $\beta_0 = 1$, then the testing problem is

$$H_0 : \beta = 1 \qquad H_1 : \beta \neq 1.$$

We generate 10,000 simulated processes from which we calculate the rejection probabilities of the considered tests under the null and alternatives to assess their size control and power.

6.2 On Size Control

To compare size, we set $\beta = 1$. Table 1 summarizes size control for different tests at the nominal level of 5%. The parameter r governs the shrinkage intensity of news-jumps, so that jumps around announcements become smaller as r increases from 0 to 0.7 across the rows. We find that the first two categories, corresponding to OLS- and GLS-based inferences, are substantially oversized as jumps become weak, although they perform well under strong identification ($r = 0$). This occurs because, as jumps diminish, the diffusion component of high-frequency returns around news events dominates. Moreover, LTT's GLS distorts size more severely than LTT's OLS, reflecting the efficiency–robustness trade-off. By contrast, in the last three categories, the AR-statistic-based inferences consistently control size across values of r, despite mild discrepancies between the nominal level and the simulated rejection probabilities, which can be attributed to estimation error in the large-k spot volatility estimation.

6.3 On Power

Figure xx display the power curves of different tests under weak identification in which the left panel showcase the weak identification case while the right panel corresponds to the case that all jumps are strongly identified. In the weak identification case, we find the sup-test is the most powerful among the three which is quite intuitive: the sup-test just lean all the weights to the jump who disagrees most with the candidate β_0 if $\beta \neq \beta_0$, thus increases the probability of rejection comparing to the L_1- and L_2-tests which incorporate the jumps with less confidence to reject. In the strong identification case, we find that the tests based on OLS and GLS control the test size and the GLS-test is the most powerful which reflects the optimality result in LTT. Additionally, the three AR-based tests are consistent in the strong identification case. In the very local alternatives, we find that the sup-test is still better than the other two AR-based tests with the same intuition. Therefore, the simulation verifies that our tests are robust to the identification strength of jumps.

7 Empirical Applications

7.1 FOMC announcements

Our first empirical application focuses on FOMC announcements, a central source of monetary policy news that typically convey changes in the federal funds rate target range, forward guidance, and, since the 2008 financial crisis, large-scale asset purchases (LSAPs). These announcements reshape market expectations and frequently generate asset price jumps. We examine the dependence structure of stock market jumps around FOMC announcements from May 15, 2001, which corresponds to the fifth announcement of that year, through December 18, 2024, which marks the final announcement in 2024, encompassing 190 scheduled

Table 1: Monte Carlo null rejection frequencies at the 5% level across identification strengths and covariance calibrations (percent).

Method	$r = 0$	$r = 0.1$	$r = 0.3$	$r = 0.5$	$r = 0.7$
Panel A: Identification-robust benchmark					
AR/Q_S	6.85	6.32	6.57	6.74	6.37
Panel B: Conditional tests using the average covariance					
LM/K	6.02	6.10	6.06	6.03	6.27
CLR	5.92	6.10	5.96	6.69	6.81
CLR^h	5.57	5.61	5.52	6.01	6.81
CLC–MMRU (MLE)	5.96	6.03	6.01	6.94	6.83
CLC–MMRU (PP)	5.96	6.03	6.01	7.00	6.82
CLC–MMRU (KRS)	5.96	6.03	6.01	6.98	6.96
Panel C: Conditional tests using the larger residual-variance endpoint					
LM/K	4.70	4.68	4.76	4.89	5.12
CLR	4.58	4.59	4.67	4.60	4.55
CLR^h	4.36	4.28	4.26	4.21	4.70
CLC–MMRU (MLE)	4.53	4.48	4.49	4.74	4.43
CLC–MMRU (PP)	4.53	4.48	4.49	4.70	4.39
CLC–MMRU (KRS)	4.53	4.48	4.49	4.74	4.53
Panel D: Strong-identification benchmarks					
LTT–GLS	6.09	6.67	11.66	30.59	39.02
LTT–OLS	5.49	6.17	10.90	26.11	31.85

Notes: Entries are rejection probabilities under $H_0 : \beta = \beta_0 = 1$; r indexes news-jump magnitude $O_{\mathbb{P}}(\Delta_n^r)$. The full one-minute DGP and the large-k estimator with $k_n = 60$ are used in 10,000 Monte Carlo replications. In Panel B, the conditional coordinates and critical values use $\bar{c}_p = (\hat{c}_{p-} + \hat{c}_{p+})/2$. In Panel C, $j_p^\star = \arg\max_{j\in\{-,+\}} h_0^\intercal \widehat{C}_{pj} h_0$ (ties select post) and the full conditional procedure uses $\widehat{C}_{p,j_p^\star}$. For Panel C, the endpoint rule does not by itself establish conditional similarity or uniform conservatism for LM, CLR, CLR^h, or CLC tests. The estimated post endpoint is selected in 55.4%, 55.4%, 55.4%, 55.3%, and 55.3% of announcement–replication cells across the five columns. CLC–MMRU uses the plug-in MLE, PP, and KRS rules with strength-adaptive local alternatives in its minimax-regret grid. LTT retains its original procedure. At a 5% rejection probability, the outer Monte Carlo standard error is approximately 0.22 percentage points.

Table 2: Maximal power shortfalls relative to the tests considered in Gaussian jump-regression experiments (percentage points)

		CLR	CLR^h	Plug-in MMRU			AR/Q_S	LM/K
				$\text{PI}-\hat{\mathfrak{c}}_{\text{MLE}}$	$\text{PI}-\hat{\mathfrak{c}}_{\text{PP}}$	$\text{PI}-\hat{\mathfrak{c}}_{\text{KRS}}$		
Panel A: Andrews-style homogeneous Gaussian experiment								
ρ	$\mathfrak{c}_D$							
0.5	5	3.82	3.82	2.04	1.76	3.50	18.18	59.42
0.5	20	0.36	0.36	1.24	1.20	0.48	26.26	40.44
0.95	5	0.14	0.14	1.02	1.34	0.60	32.12	1.22
0.95	20	0.44	0.44	1.34	1.42	0.38	32.48	6.44
Panel B: Current heterogeneous calibration without volatility jumps								
	$\mathfrak{c}_D$							
	5	3.42	5.74	3.82	3.96	3.66	15.86	63.78
	20	2.90	4.70	4.74	4.62	4.44	24.06	56.78

Notes: We simulate the exact Gaussian experiment $S^* \sim N(\delta\Lambda\xi^*, I_8)$ and $T^* \sim N((I_8 + \delta\Theta)\xi^*, I_8)$ independently, where $\mathfrak{c}_D = \|\xi^*\|^2 \in \{5, 20\}$ and $\delta = h/\sqrt{\mathcal{I}_0}$, $\mathcal{I}_0 = \xi^{*\prime}\Lambda^2\xi^*$. Each entry is the largest percentage-point shortfall from the pointwise power envelope of the seven named tests within that row over $h \in \{-6, -5.7, \ldots, -0.3, 0.3, \ldots, 5.7, 6\}$. The null $h = 0$ is used only for size validation and is excluded. Panel A uses homogeneous coordinates with $\rho \in \{0.5, 0.95\}$ and the exact common $\text{CLR} = \text{CLR}^h$ rejection decision. Panel B uses the current heterogeneous (λ_p, θ_p) calibration with no volatility jumps. The three PI columns apply the MLE, PP, and KRS plug-in rules of Andrews (2016). Results use 5,000 replications, nominal 5% critical values, and no size adjustment.

Table 3: Maximal rejection-power shortfalls relative to the tests considered across identification strengths and covariance calibrations (percentage points).

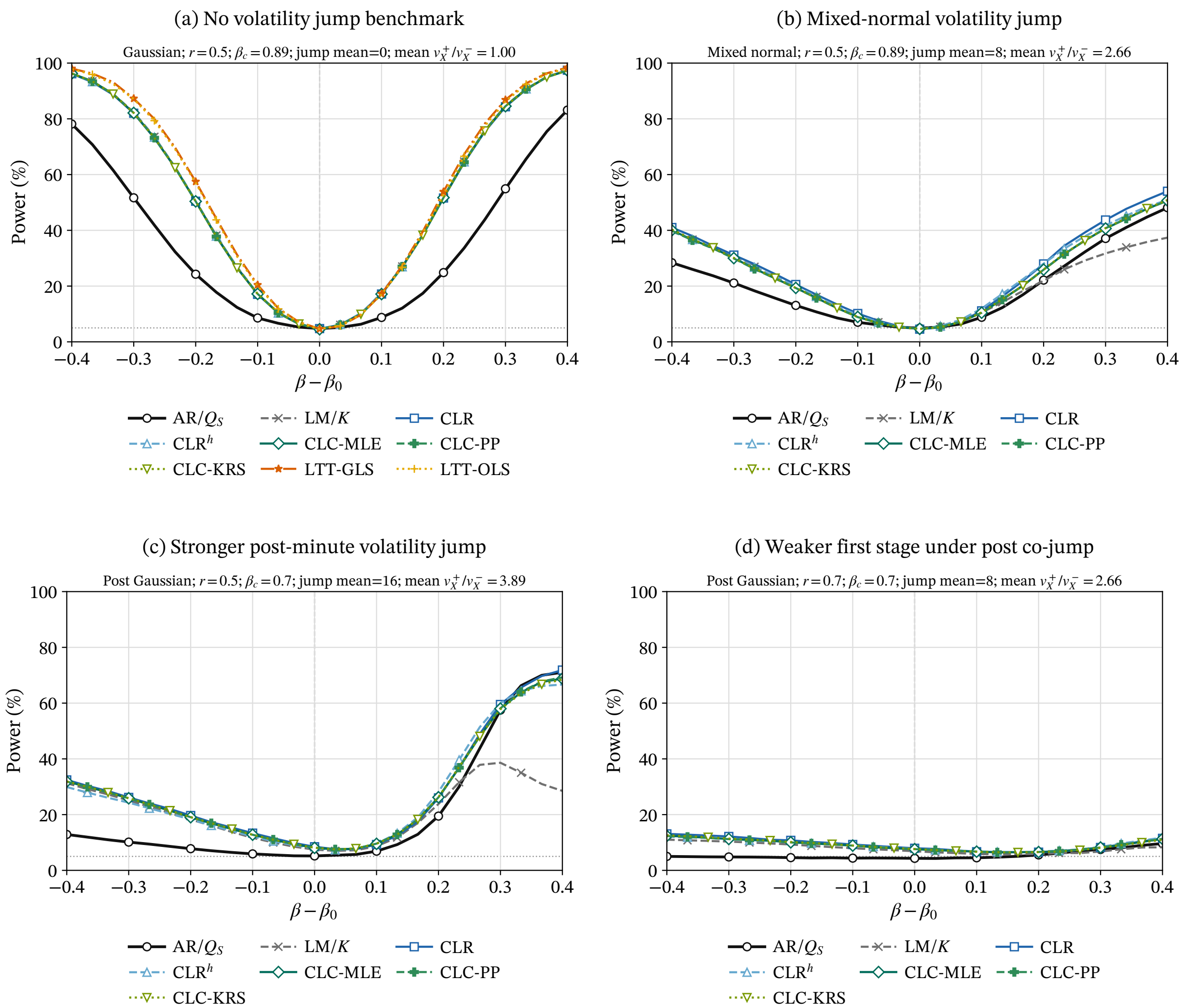
r	CLR	CLR^h	Plug-in MMRU			AR/Q_S	LM/K
			$\text{PI}-\widehat{\mathfrak{c}}_{\text{MLE}}$	$\text{PI}-\widehat{\mathfrak{c}}_{\text{PP}}$	$\text{PI}-\widehat{\mathfrak{c}}_{\text{KRS}}$		
Panel A: Tests using the average covariance							
0.00	0.00	0.06	0.00	0.00	0.00	0.14	0.83
0.10	0.10	1.30	0.08	0.08	0.08	14.15	1.58
0.30	0.00	5.24	0.44	0.44	0.44	29.43	7.87
0.50	0.22	6.51	3.92	3.91	3.83	22.07	22.37
0.70	0.82	1.28	0.97	1.04	0.97	7.23	3.71
Panel B: Tests using the larger residual-variance endpoint							
0.00	0.00	0.09	0.00	0.00	0.00	0.14	1.48
0.10	0.12	3.97	0.14	0.14	0.14	13.44	2.71
0.30	0.25	12.13	1.10	1.10	1.10	24.15	12.97
0.50	2.31	7.10	4.37	4.44	4.40	9.36	20.72
0.70	3.10	3.36	3.44	3.47	3.54	0.52	2.62

Note: For covariance calibration $c \in \{\text{avg}, \text{end}\}$, identification strength r, and method m, each entry is the empirical shortfall

$$100 \max_{\delta \in \mathcal{D}} \left\{ \max_{j \in \mathcal{M}} \widehat{\pi}_{c,r,j}(\delta) - \widehat{\pi}_{c,r,m}(\delta) \right\},$$

where $\mathcal{M}$ is the set of the seven displayed tests and the pointwise envelope is formed separately within each covariance calibration c and r. The alternative grid is $\mathcal{D} = \{\pm 0.04, \pm 0.08, \ldots, \pm 0.40\}$, with true $\beta = 1$ and $\beta_0 = 1 - \delta$; $\delta = 0$ is not evaluated. Rejection decisions use nominal 5% critical values, with no size adjustment. AR/Q_S is common to both panels; only the six conditional procedures change covariance calibration. Results use the same 10,000 saved full one-minute DGP paths and the same critical-value and CLC budgets as Table 1. The three PI columns use the MLE, PP, and KRS plug-in MMRU rules of Andrews (2016).

Figure 4: Power curves under different DGP settings.



meetings. This period is chosen in light of the temporal limitations of the X- and Y-processes under consideration. While some announcements produce strongly identified jumps, a larger share are characterized by weak jumps.

We select both the X- and Y-processes to be exchange-traded funds (ETFs), given their high liquidity and professional management, which ensures that they react promptly to major news events such as FOMC announcements. The X-process is fixed as the ETF tracking the S&P 500 index (ticker: SPY), serving as the proxy for market-wide jumps. For the Y-process, we draw from industry-specific ETFs in order to study their exposures to market jump risk. The candidates, with corresponding tickers in parentheses, are: materials (XLB), communication services (XLC)[13], energy (XLE), financials (XLF), industrials (XLI), technology (XLK), consumer staples (XLP), utilities (XLU), health care (XLV), and consumer discretionary (XLY). The high-frequency intraday data are obtained from Tick Data[14]. Crucially, the exact announcement times of FOMC releases are available on Bloomberg[15], which allows us to precisely anchor the timing of news-based jumps.

The sampling frequency of the series is set to one minute in order to enhance the effectiveness of infill asymptotics while mitigating microstructure frictions such as noise and rounding errors. Figure 5 presents scatter plots of the pairs $(\Delta^n_{i(p)}X, \Delta^n_{i(p)}Y)$, the high-frequency log-price increments around news announcements. The number of observations differs across subplots due to variations in the data availability of the Y-processes. In the plots, blue dots denote cases of strong identification, which can be reliably analyzed using LTT's jump regressions, while red crosses denote weakly identified cases, which LTT's method removes. Notably, the number of weakly identified jumps is nearly three times that of strongly identified ones.

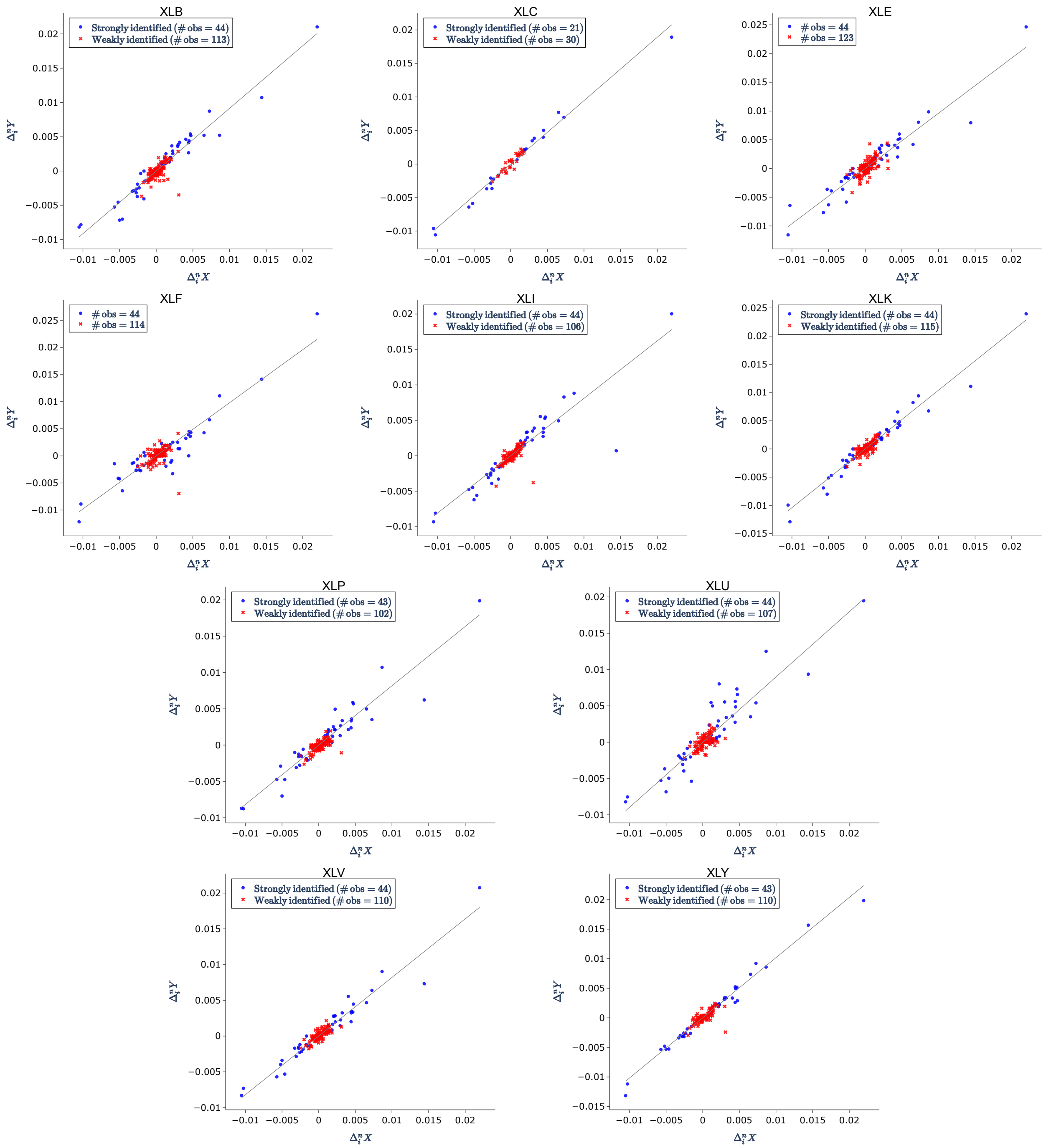
We then perform test inversion to obtain the identification-robust confidence intervals. In

[13]XLC is a relatively recent sector ETF. It was launched on June 18, 2018, around the 2018 GICS reclassification that replaced the former Telecommunication Services sector with the broader Communication Services sector, incorporating selected media, entertainment, and internet-platform firms.

[14]Tick Data website: `https://www.tickdata.com/`.

[15]The FOMC announcement times, accurate to the second, can be found under events such as "FOMC Rate Decision (Upper Bound)" in the Monetary Sector, Economic Releases, Economic Calendars (United States).

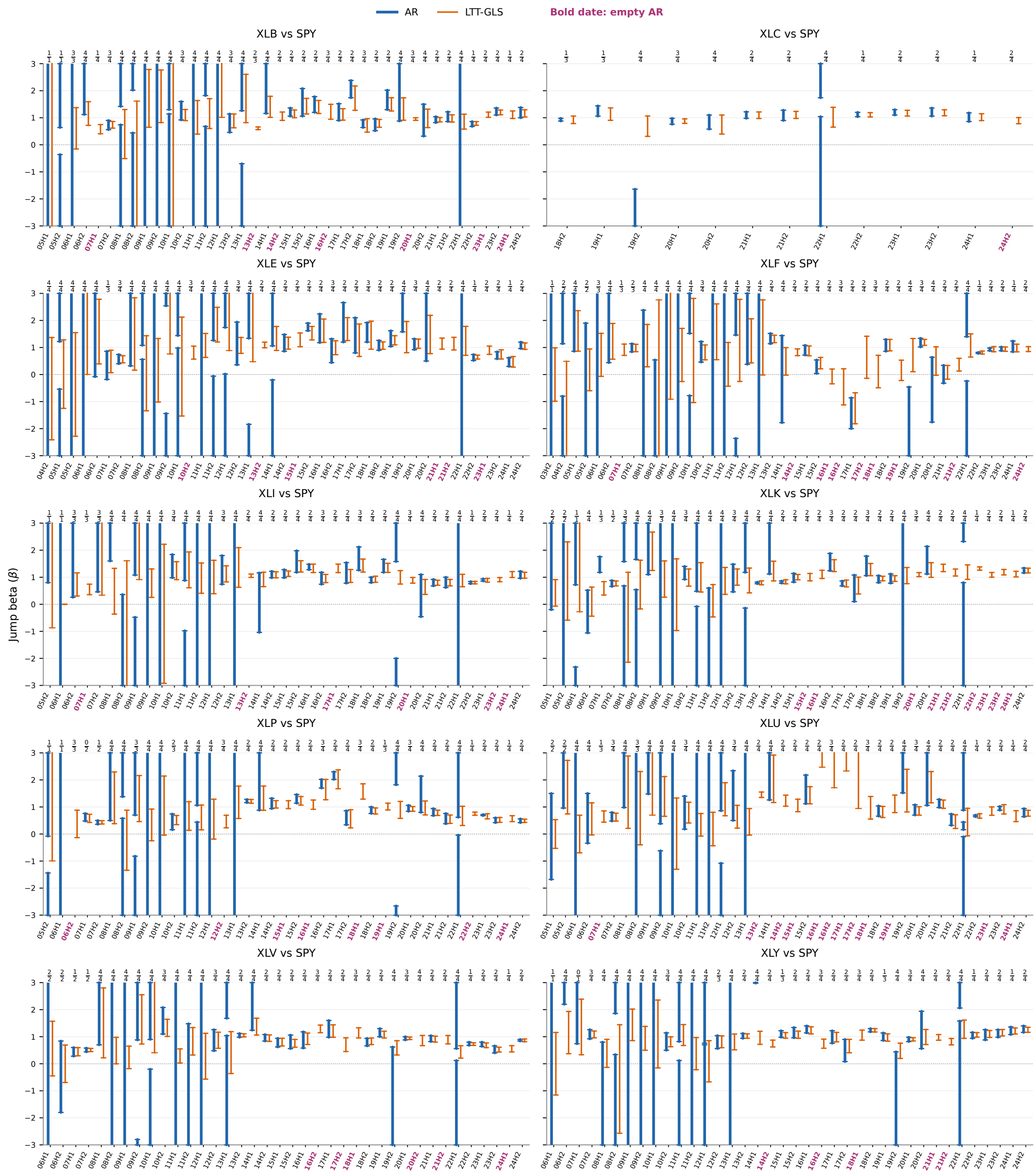
Figure 5: Scatterplots of one-minute log-returns around FOMC announcements for ten ETFs



Notes: Blue dots denote strongly identified cases, where ΔX exceeds the truncation level; red crosses denote weakly identified cases. The straight line is the simple linear fit using strongly identified jumps.

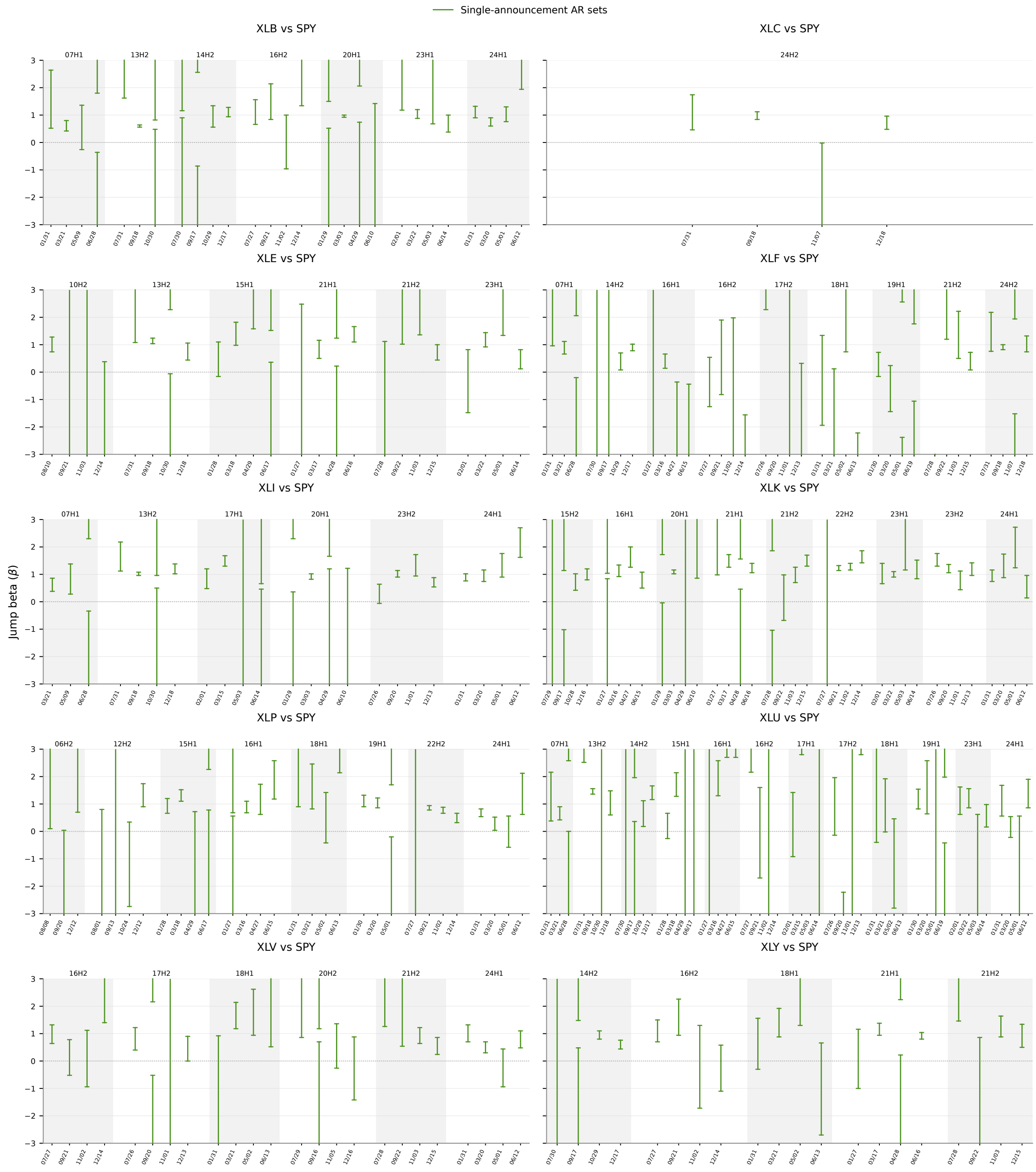
the empirical application, the bandwidth for the pre- and post-announcement spot volatility estimator is set to $k_n = 8$, and the truncation threshold is determined in the same manner as in the simulation study; see (41). The scatter plots of the aforementioned Y-assets against the X-asset (SPY) are presented in Figure 5, where weak identification appears to be more prevalent. The results from the test inversion are reported below.

Figure 6: Half-year FOMC jump-beta confidence sets



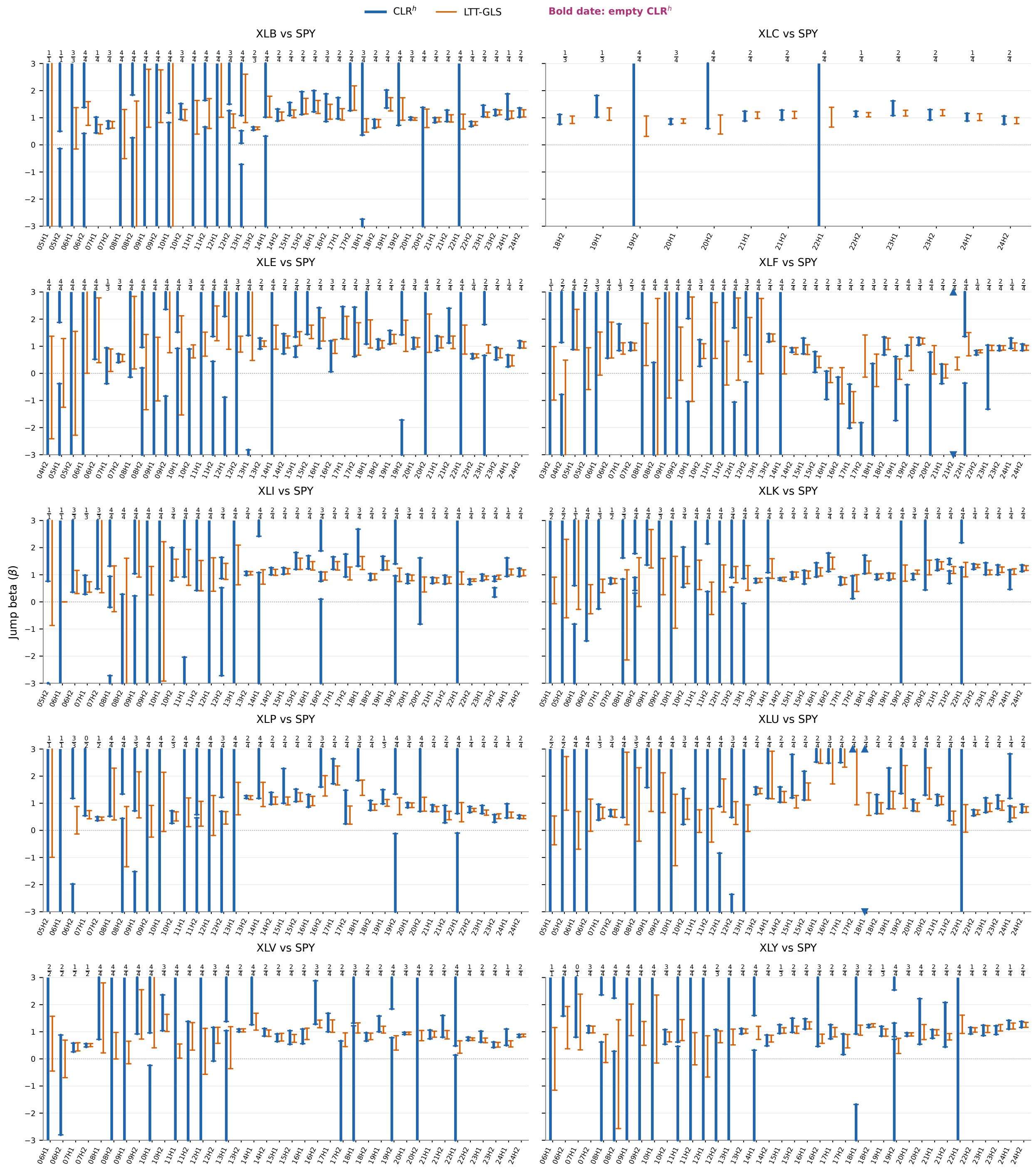
Notes: Blue: pooled half-year AR; orange: LTT–GLS. A short bar marks a finite endpoint; a line reaching a plotting boundary without a bar may continue beyond the displayed range. Bold magenta date labels indicate pooled AR sets that are empty on the maintained parameter space $[-3, 3]$. Fractions report weak/valid announcements.

Figure 7: Single-announcement AR confidence sets within empty half-year cells



Notes: Green lines are pointwise 95% single-announcement AR sets intersected with the maintained parameter space $[-3, 3]$. Exact event days appear on the lower axis; upper labels group them by half-year. Within every displayed group, no beta on the maintained 0.02 grid in that space is accepted by all announcements; this diagnostic is not a simultaneous 95% statement. Interior bars have finite test-determined endpoints; uncapped lines at ± 3 are truncated by the maintained prior.

Figure 8: Half-year FOMC jump-beta confidence sets: CLR^h

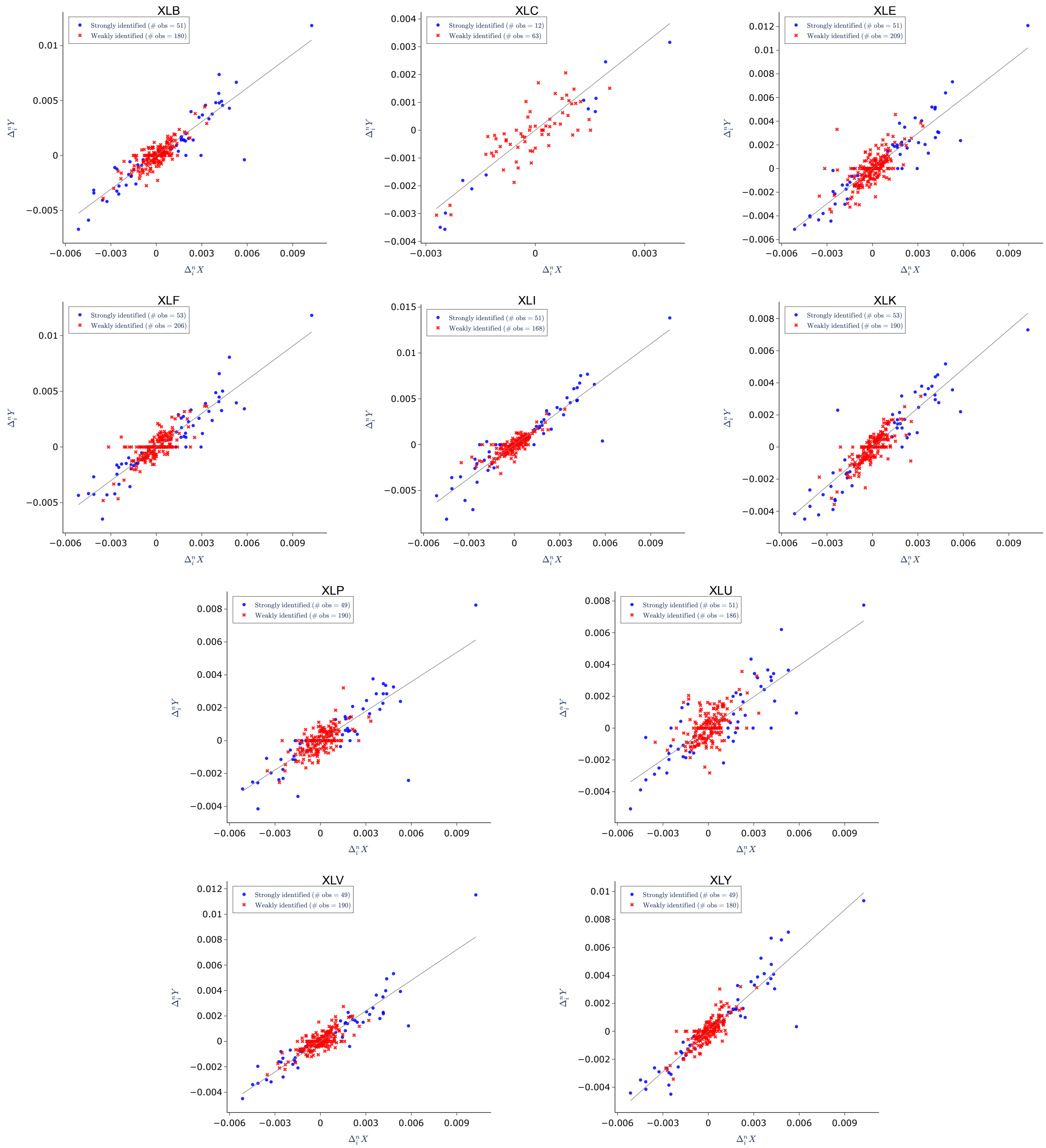


Notes: Blue: CLR^h using the post-announcement truncated Large-K covariance; orange: LTT–GLS. A short bar marks a finite endpoint; a line reaching a plotting boundary without a bar may continue beyond the displayed range. Bold magenta date labels indicate audited empty CLR^h cells. Fractions report weak/valid announcements.

7.2 Other Weaker Macroeconomic Announcements

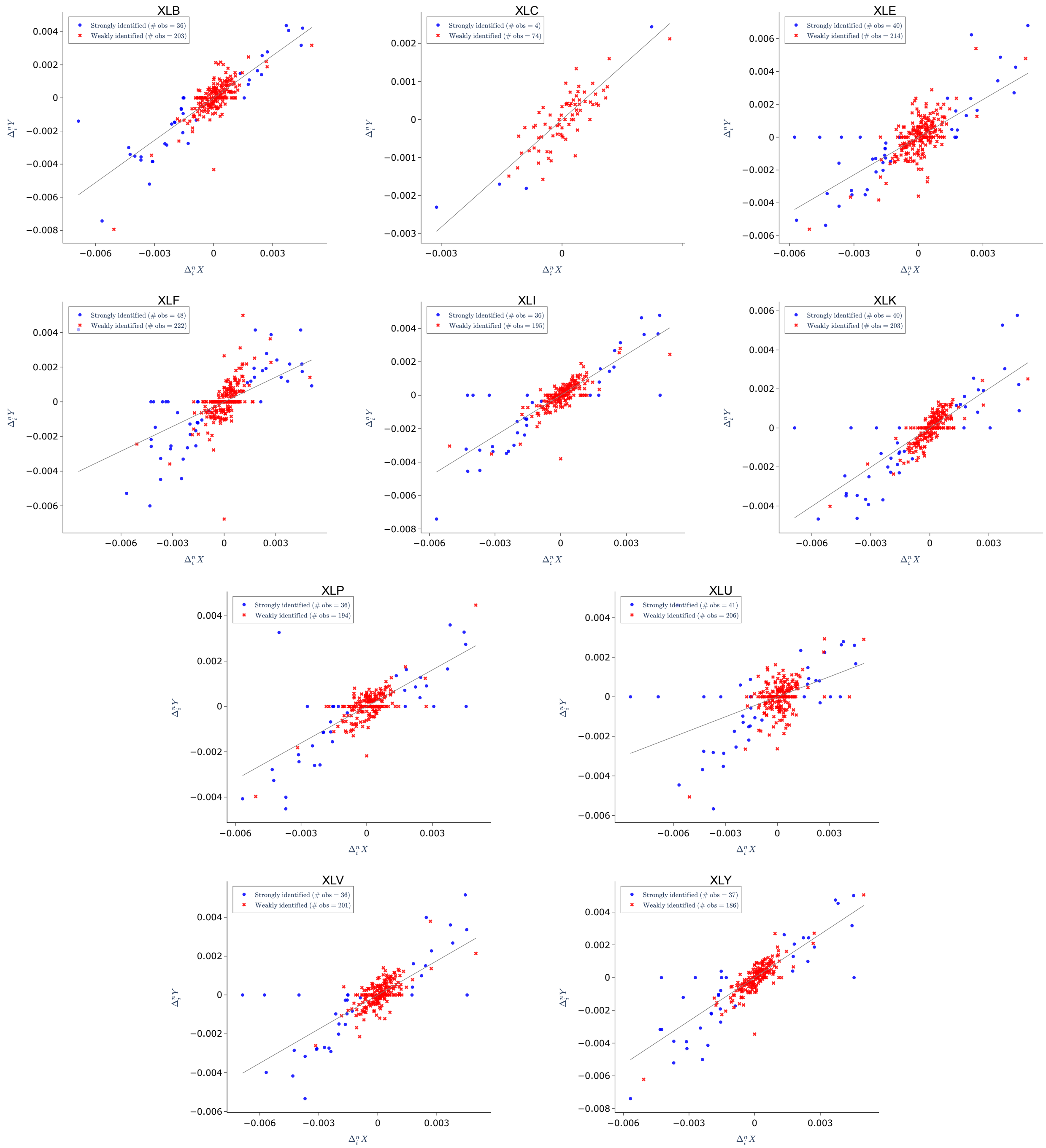
We also examine two weaker macroeconomic announcements, the ISM manufacturing report and the Conference Board consumer confidence release. Figures 9 and 10 present the corresponding sector-ETF scatterplots. Relative to the FOMC sample, both releases contain substantially larger weakly identified groups, making the first-stage weakness especially visible.

Figure 9: Scatterplots around ISM manufacturing announcements for ten ETFs



Notes: Blue dots denote strongly identified cases, for which the absolute one-minute SPY return exceeds its truncation level; red crosses denote weakly identified cases, for which both the SPY and sector-ETF returns are below their respective truncation levels. Observations with a below-threshold SPY return but an above-threshold sector-ETF return are omitted as idiosyncratic price moves. The straight line is the zero-intercept fit using strongly identified observations. Sample sizes differ across ETFs because of data availability.

Figure 10: Scatterplots around Conference Board consumer confidence releases for ten ETFs



Notes: Blue dots denote strongly identified cases, for which the absolute one-minute SPY return exceeds its truncation level; red crosses denote weakly identified cases, for which both the SPY and sector-ETF returns are below their respective truncation levels. Observations with a below-threshold SPY return but an above-threshold sector-ETF return are omitted as idiosyncratic price moves. The straight line is the zero-intercept fit using strongly identified observations. Sample sizes differ across ETFs because of data availability.

8 Conclusion

News-jumps, compared with statistical jumps, offer clear interpretability and rich economic meaning. Quantitative analysis of these news-induced jumps not only helps to understand cross-sectional risk but also reveals how markets assess information shocks. Major macroeconomic announcements often trigger strong reactions in financial markets when the information is surprising, yet in most cases, the responses are mild. Consequently, when an empiricist conducts a high-frequency event study that collects all high-frequency asset price changes around public news announcements over a long sample period for regression analysis, particular caution is warranted if small shocks substantially outnumber large ones. This paper demonstrates that when jump regressions are applied to weakly-identified jumps, the conventional asymptotic approximations under strong identification fail to capture the finite-sample distribution of the jump-beta estimator. To address this weak-identification problem, we propose an Anderson–Rubin-type test that remains robust, albeit at the cost of efficiency under strong identification. Such robustness is especially valuable in the presence of weak announcements. Additionally, one can easily extend the robust inferential methods in this paper to other specifications, such as a linear fit with intercept or quadratic nonlinearity. The development of a power-optimal test remains an open question for future research.

Appendix: Algorithms of Conditional Tests

Algorithm 2 (CLR test procedure)**.** For a given candidate β_0 and significance level α:

(i) At each announcement $p \in \mathcal{P}_{\mathfrak{a}_B}$, compute the spot covariance estimate $\hat{\Sigma}_p = (\hat{c}_{n,i(p)-} + \hat{c}_{n,i(p)+})/2$ via the large-k estimator, and calculate

$$S_{n,p}(\beta_0) = \Delta_n^{-1/2}(\Delta_{i(p)}^n Y - \beta_0 \Delta_{i(p)}^n X),$$
$$\hat{v}_{S,p} = h_0^\intercal \hat{\Sigma}_p h_0, \quad \hat{\xi}_p = \frac{g_0^\intercal \hat{\Sigma}_p^{-1} Z_p}{g_0^\intercal \hat{\Sigma}_p^{-1} g_0}, \quad \hat{v}_{T,p} = \frac{\det(\hat{\Sigma}_p)}{\hat{v}_{S,p}}.$$

(ii) Compute Q_S, Q_T, and Q_{ST}, and the QLR/CLR and LM statistics.

(iii) For $m = 1, \ldots, M$, draw $\tilde{\kappa}_p \sim U(0,1]$, $\tilde{\Psi}_{p-}, \tilde{\Psi}_{p+} \sim \mathcal{N}(0, I_2)$ for each p, form $\tilde{R}_p = \sqrt{\tilde{\kappa}_p}\hat{\sigma}_{n,i(p)-}\tilde{\Psi}_{p-} + \sqrt{1-\tilde{\kappa}_p}\hat{\sigma}_{n,i(p)+}\tilde{\Psi}_{p+}$ (where $\hat{\sigma}_{n,i(p)\pm}$ are Cholesky factors of $\hat{c}_{n,i(p)\pm}$ as in Algorithm 1), $\tilde{S}_p = (-\beta_0, 1)\tilde{R}_p$, and $\tilde{\hat{\xi}}_p = (g_0^\intercal \hat{\Sigma}_p^{-1} \tilde{R}_p)/(g_0^\intercal \hat{\Sigma}_p^{-1} g_0)$ (under H_0, $Z_p = R_p$ with $\xi_{\mathfrak{a}_p} = 0$). Compute $\tilde{Q}_S$, $\tilde{Q}_T$, $\tilde{Q}_{ST}$, and the simulated QLR/CLR and LM statistics. Set cv_{QLR}^{α} and cv_{LM}^{α} as the $(1-\alpha)$-quantiles.

(iv) Reject H_0 if $\text{QLR}(\beta_0) > cv_{\text{QLR}}^{\alpha}$ (or $\text{LM}(\beta_0) > cv_{\text{LM}}^{\alpha}$ for the LM test).

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Supplement to "News-Jump Regressions: Weak Identification and Robust Inferences"

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Abstract

This supplement provides the proofs of the main results, auxiliary theoretical results, additional simulation results, and further empirical findings of Li and Wei (2026), "News-Jump Regressions: Weak Identification and Robust Inferences".

S1. Proofs of Main Results

Proof of Theorem 1

(a) Let X^c and J_X respectively denote the continuous part and the jump part of the X-process. Set $|\mathcal{I}_n(\mathfrak{a}_B)| = q < \infty$ where $\mathcal{I}_n(\mathfrak{a}_B)$ is defined in (16), note q is fixed as $n \to \infty$, then for any strictly positive integer q,

$$
\begin{aligned}
\mathbb{P}(\mathcal{I}_n(\mathfrak{a}_B) \neq \varnothing) &= \mathbb{P}(\exists i, i \in \mathcal{I}_n(\mathfrak{a}_B), \text{ s.t. } |\Delta_i^n X| > u_n) = \mathbb{P}\left(\bigcup_{i \in \mathcal{I}_n(\mathfrak{a}_B)} \{|\Delta_i^n X| > u_n\} \right) \\
&\leq \sum_{i \in \mathcal{I}_n(\mathfrak{a}_B)} \mathbb{P}(|\Delta_i^n X^c + \Delta_i^n J_X| > u_n) \leq \sum_{i \in \mathcal{I}_n(\mathfrak{a}_B)} \mathbb{P}(|\Delta_i^n X^c| + |\Delta_i^n J_X| > u_n) \\
&\leq \sum_{i \in \mathcal{I}_n(\mathfrak{a}_B)} \mathbb{P}(|\Delta_i^n X^c| > u_n/2) + \sum_{i \in \mathcal{I}_n(\mathfrak{a}_B)} \mathbb{P}(|\Delta_i^n J_X| > u_n/2) \\
&\leq \sum_{i \in \mathcal{I}_n(\mathfrak{a}_B)} \mathbb{P}(|\Delta_i^n X^c| > u_n/2) + \sum_{i \in \mathcal{I}_n(\mathfrak{a}_B)} \mathbb{P}(|\Delta_i^n J_X| > u_n/2) \\
&= \sum_{i \in \mathcal{I}_n(\mathfrak{a}_B)} \mathbb{P}(|\Delta_i^n X^c| > u_n/2) + \sum_{i \in \mathcal{I}(\mathfrak{a}_B)} \mathbb{P}(|\Delta_i^n J_X| > u_n/2)
\end{aligned}
$$

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$$
\begin{aligned}
&\overset{(i)}{\leq} K_q \Delta_n^{q/2}/u_n^q + \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \mathbb{P}(|\xi_{\mathfrak{a}_p}| > u_n\Delta_n^{-1/2}/2) \\
&\xrightarrow{(ii)} 0 + 0 = 0
\end{aligned}
$$

where the inequality (i) is due to the Markov's inequality and $\mathbb{E}|\Delta_i^n X^c|^q \leq K_q \Delta_n^{q/2}$ for some constant K_q that depends on q. Note that the convergence (ii) is trivial if $|\mathcal{P}_{\mathfrak{a}_B}| = 0$, so we let $|\mathcal{P}_{\mathfrak{a}_B}| \neq 0$. And $|\mathcal{P}_{\mathfrak{a}_B}| < \infty$ because B is a Borel subset of finite Lebesgue measure and $J^{\mathfrak{a}}$ is finitely active. By Assumption 1, $\forall \varepsilon > 0$, $\exists K_\varepsilon > 0$ s.t. $\mathbb{P}(|\xi_{\mathfrak{a}_p}| > K_\varepsilon) < \frac{\varepsilon}{|\mathcal{P}_{\mathfrak{a}_B}|}$, hence for a large enough n,

$$
\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \mathbb{P}(|\xi_{\mathfrak{a}_p}| > u_n\Delta_n^{-1/2}/2) \leq \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \mathbb{P}(|\xi_{\mathfrak{a}_p}| > K_\varepsilon) < \varepsilon
$$

and letting $\varepsilon \to 0$ gives the convergence (ii).

(b) Since $R_{n,p} = \Delta_n^{-1/2}(\Delta_{i(p)}^n Z - \Delta Z_{\mathfrak{a}_p})$, $p \in \mathcal{P}_{\mathfrak{a}_B}$, we have

$$
\begin{aligned}
&\Delta_n^{-1/2}\left(\Delta_{i(p)}^n Y - \beta\Delta_{i(p)}^n X\right) = (-\beta, 1)R_{n,p} \\
&\frac{\Delta_{i(p)}^n X}{\Delta_n^{1/2}} = \frac{\Delta_{i(p)}^n X - \Delta X_{\mathfrak{a}_p} + \Delta X_{\mathfrak{a}_p}}{\Delta_n^{1/2}} = (1,0)R_{n,p} + \xi_{\mathfrak{a}_p}
\end{aligned}
$$

Therefore,

$$
\begin{aligned}
\tilde{\beta}_n - \beta &= \frac{\sum_{i\in\mathcal{I}(\mathfrak{a}_B)} \Delta_i^n X \Delta_i^n Y}{\sum_{i\in\mathcal{I}(\mathfrak{a}_B)} (\Delta_i^n X)^2} - \beta = \frac{\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \Delta_{i(p)}^n X(\Delta_{i(p)}^n Y - \beta\Delta_{i(p)}^n X)}{\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} (\Delta_{i(p)}^n X)^2} \\
&= \frac{\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} ((1,0)R_{n,p} + \xi_{\mathfrak{a}_p})(-\beta,1)R_{n,p}}{\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} ((1,0)R_{n,p} + \xi_{\mathfrak{a}_p})^2} \\
&\xrightarrow{\mathcal{L}\text{-}s} \frac{\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \left[(1,0)R_p + \xi_{\mathfrak{a}_p}\right] \varsigma_p(\beta)}{\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \left[(1,0)R_p + \xi_{\mathfrak{a}_p}\right]^2} = \tilde{b}
\end{aligned} \tag{S1}
$$

where the convergence in the last step is by the stable convergence in law (S3) in the proof of Theorem 2 and the continuous mapping theorem.

It remains to verify that the limiting ratio in (S1) is well defined and bounded in probability. Write

$$
A_p \equiv e_1^\intercal R_p + \xi_{\mathfrak{a}_p}, \qquad V_p \equiv \varsigma_p(\beta), \qquad D \equiv \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} A_p^2.
$$

We first show that $D > 0$ almost surely. On $\Omega(\mathfrak{a}_B)$, Assumption 1 implies that $\xi_{\mathfrak{a}_p} \neq 0$ for at least one $p \in \mathcal{P}_{\mathfrak{a}_B}$. On the extension supporting the stable limit, conditionally on the original sigma-field and on κ_p,

$$e_1^\intercal R_p \mid \mathcal{F}, \kappa_p \sim \mathcal{N}(0, q_p), \qquad q_p \equiv \kappa_p e_1^\intercal c_{\mathfrak{a}_p -} e_1 + (1-\kappa_p) e_1^\intercal c_{\mathfrak{a}_p} e_1 \geq 0.$$

If $q_p > 0$, this conditional distribution has no atoms, and hence $\mathbb{P}(A_p = 0 \mid \mathcal{F}, \kappa_p) = 0$. If $q_p = 0$, then $e_1^\intercal R_p = 0$ almost surely, so $A_p = \xi_{\mathfrak{a}_p} \neq 0$ whenever $\xi_{\mathfrak{a}_p} \neq 0$. Since $|\mathcal{P}_{\mathfrak{a}_B}| < \infty$, it follows that

$$\mathbb{P}(D = 0, \Omega(\mathfrak{a}_B)) \leq \sum_{p \in \mathcal{P}_{\mathfrak{a}_B}} \mathbb{P}\left(A_p = 0, \xi_{\mathfrak{a}_p} \neq 0\right) = 0.$$

Hence $D > 0$ almost surely on $\Omega(\mathfrak{a}_B)$; all probabilities below are understood under this restriction. This also verifies that the discontinuity set of the ratio map used in the continuous mapping theorem has probability zero.

Finally, let $A = (A_p)_{p \in \mathcal{P}_{\mathfrak{a}_B}}$ and $V = (V_p)_{p \in \mathcal{P}_{\mathfrak{a}_B}}$. The number of announcements is finite and every component of A and V is finite almost surely. By the Cauchy–Schwarz inequality,

$$|\tilde{b}| = \frac{|A^\intercal V|}{\|A\|^2} \leq \frac{\|V\|}{\|A\|}.$$

For any $\varepsilon > 0$, $\mathbb{P}(\|A\| = 0) = 0$ allows us to choose $\delta > 0$ such that $\mathbb{P}(\|A\| \leq \delta) < \varepsilon/2$, while the almost sure finiteness of $\|V\|$ allows us to choose $K < \infty$ such that $\mathbb{P}(\|V\| > K) < \varepsilon/2$. Therefore,

$$\mathbb{P}\left(|\tilde{b}| > K/\delta\right) \leq \mathbb{P}(\|A\| \leq \delta) + \mathbb{P}(\|V\| > K) < \varepsilon.$$

Thus $\tilde{b}$ is a proper random variable and $\tilde{b} = O_{\mathbb{P}}(1)$. Moreover, since stable convergence implies convergence in law to a proper limit, (S1) also implies $\tilde{\beta}_n - \beta = O_{\mathbb{P}}(1)$.

(c) For the weighted least square, we note

$$\begin{aligned}\hat{\beta}_n(\mathfrak{a}_B, w) - \beta &= \frac{\sum_{p \in \mathcal{P}_{\mathfrak{a}_B}} w(\hat{\boldsymbol{c}}_{n,i-}, \hat{\boldsymbol{c}}_{n,i+}, \tilde{\beta}_n) \Delta_{i(p)}^n X \left(\Delta_{i(p)}^n Y - \beta \Delta_{i(p)}^n X\right)}{\sum_{p \in \mathcal{P}_{\mathfrak{a}_B}} w(\hat{\boldsymbol{c}}_{n,i-}, \hat{\boldsymbol{c}}_{n,i+}, \tilde{\beta}_n) \Delta_{i(p)}^n X^2} \\ &\xrightarrow{\mathcal{L}\text{-}s} \frac{\sum_{p \in \mathcal{P}_{\mathfrak{a}_B}} w(\boldsymbol{c}_{\mathfrak{a}_p -}, \boldsymbol{c}_{\mathfrak{a}_p}, \beta + \tilde{b}) \left[(1,0) R_p + \xi_{\mathfrak{a}_p}\right] \varsigma_p(\beta)}{\sum_{p \in \mathcal{P}_{\mathfrak{a}_B}} w(\boldsymbol{c}_{\mathfrak{a}_p -}, \boldsymbol{c}_{\mathfrak{a}_p}, \beta + \tilde{b}) \left[(1,0) R_p + \xi_{\mathfrak{a}_p}\right]^2} = b_w\end{aligned}$$

where the stable convergence in law is due to (S4), (S3), (S1), w being a continuous function, and the continuous mapping theorem.

Proof of Theorem 2

(a) Note $\Delta Y_{\mathfrak{a}_p} = \beta \Delta X_{\mathfrak{a}_p}$, thus

$$\Delta Y_{\mathfrak{a}_p} - \Delta_{i(p)}^n Y + \Delta_{i(p)}^n Y = \beta(\Delta X_{\mathfrak{a}_p} - \Delta_{i(p)}^n X + \Delta_{i(p)}^n X). \tag{S2}$$

Define $R_{n,p} = \Delta_n^{-1/2}(\Delta_{i(p)}^n Z - \Delta Z_{\mathfrak{a}_p})$, then plug $R_{n,p}$ into (S2) and rearrange the equality,

$$\Delta_{i(p)}^n Y = \beta \Delta_{i(p)}^n X + (-\beta, 1)\Delta_n^{1/2} R_{n,p}$$

Then subtract $\beta_0 \Delta_{i(p)}^n X$ from both sides and then divide both sides by $\Delta_n^{1/2}$, we obtain

$$\begin{aligned}
S_{n,p}(\beta_0) = \Delta_n^{-1/2}(\Delta_{i(p)}^n Y - \beta_0 \Delta_{i(p)}^n X) &= (\beta - \beta_0)\frac{\Delta_{i(p)}^n X}{\Delta_n^{1/2}} + (-\beta, 1)R_{n,p} \\
&= (\beta - \beta_0)\frac{\Delta_{i(p)}^n X - \Delta X_{\mathfrak{a}_p}}{\Delta_n^{1/2}} + (\beta - \beta_0)\frac{\Delta X_{\mathfrak{a}_p}}{\Delta_n^{1/2}} + (-\beta, 1)R_{n,p} \\
&\overset{(i)}{=} (\beta - \beta_0)(1, 0)R_{n,p} + (\beta - \beta_0)\xi_{\mathfrak{a}_p} + (-\beta, 1)R_{n,p} \\
&= (-\beta_0, 1)R_{n,p} + (\beta - \beta_0)\xi_{\mathfrak{a}_p}
\end{aligned}$$

where equality (i) is due to Assumption 1 and the definition of $R_{n,p}$. By Proposition 4.4.10 of Jacod and Protter (2012), we have the following joint convergence,

$$(R_{n,p})_{p\geq 1} \xrightarrow{\mathcal{L}\text{-}s} (R_p)_{p\geq 1} \tag{S3}$$

where R_p is defined in (10). And it implies $(R_{n,p}, \xi_{\mathfrak{a}_p}) \xrightarrow{\mathcal{L}\text{-}s} (R_p, \xi_{\mathfrak{a}_p})$ due to the property of stable convergence. According to the continuous mapping theorem,

$$S_{n,p}(\beta_0) \xrightarrow{\mathcal{L}\text{-}s} (-\beta_0, 1)R_p + (\beta - \beta_0)\xi_{\mathfrak{a}_p} = \varsigma_p(\beta_0) + (\beta - \beta_0)\xi_{\mathfrak{a}_p}$$

(b) We note $\mathbb{E}(R_p | \sigma_{\mathfrak{a}_p -}, \sigma_{\mathfrak{a}_p}) = 0$ and

$$\begin{aligned}
&\mathrm{Var}(R_p | \sigma_{\mathfrak{a}_p -}, \sigma_{\mathfrak{a}_p}) = \mathbb{E}(R_p R_p^\intercal | \sigma_{\mathfrak{a}_p -}, \sigma_{\mathfrak{a}_p}) \\
&= \mathbb{E}\left\{\mathbb{E}\left[\kappa \sigma_{\mathfrak{a}_p -}\Psi_{p-}\Psi_{p-}^\intercal \sigma_{\mathfrak{a}_p -}^\intercal + (1-\kappa)\sigma_{\mathfrak{a}_p}\Psi_{p+}\Psi_{p+}^\intercal \sigma_{\mathfrak{a}_p}^\intercal | \kappa, \sigma_{\mathfrak{a}_p -}, \sigma_{\mathfrak{a}_p}\right] \middle| \sigma_{\mathfrak{a}_p -}, \sigma_{\mathfrak{a}_p}\right\} \\
&= \mathbb{E}\left[\kappa \sigma_{\mathfrak{a}_p -}\sigma_{\mathfrak{a}_p -}^\intercal + (1-\kappa)\sigma_{\mathfrak{a}_p}\sigma_{\mathfrak{a}_p}^\intercal \middle| \sigma_{\mathfrak{a}_p -}, \sigma_{\mathfrak{a}_p}\right] \\
&= \frac{c_{\mathfrak{a}_p -} + c_{\mathfrak{a}_p}}{2}
\end{aligned}$$

Hence $\mathrm{Var}(\varsigma_p(\beta_0)|c_{\mathfrak{a}_p-}, c_{\mathfrak{a}_p}) = \frac{1}{2}(-\beta_0, 1)(c_{\mathfrak{a}_p-} + c_{\mathfrak{a}_p})(-\beta_0, 1)^\intercal = v_{S,p}(\beta_0)$. By Theorem 9.3.2 in Jacod and Protter (2012) we have

$$\hat{c}_{n,i(p)+} \xrightarrow{\mathbb{P}} c_{\mathfrak{a}_p} \qquad \hat{c}_{n,i(p)-} \xrightarrow{\mathbb{P}} c_{\mathfrak{a}_p-} \tag{S4}$$

and the continuous mapping theorem implies $\hat{v}_{S,p}(\beta_0) \xrightarrow{\mathbb{P}} v_{S,p}(\beta_0)$. By the stable convergence in law under the null in Theorem 2(a), we have

$$Q_S \xrightarrow{\mathcal{L}\text{-}s} \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \frac{[\varsigma_p(\beta_0)]^2}{v_{S,p}(\beta_0)}.$$

(c) Let $cv_1^\alpha(\beta_0)$ denote the $(1-\alpha)$ quantile of the distribution of $\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}}[\varsigma_p(\beta_0)]^2/v_{S,p}(\beta_0)$, clearly $cv_1^\alpha(\beta_0) > 0$ for $\alpha \in (0,1)$. For the large-k estimators in (S4), let $\mathbb{N}_1$ be an arbitrary subsequence of $\mathbb{N}$, hence $(\hat{c}_{n,i(p)+}, \hat{c}_{n,i(p)-}) \xrightarrow{\mathbb{P}} (c_{\mathfrak{a}_p}, c_{\mathfrak{a}_p-})$ along $\mathbb{N}_1$, so there is a sub-subsequence $\mathbb{N}_2$ of $\mathbb{N}_1$, s.t. $(\hat{c}_{n,i(p)+}, \hat{c}_{n,i(p)-}) \xrightarrow{a.s.} (c_{\mathfrak{a}_p}, c_{\mathfrak{a}_p-})$ along $\mathbb{N}_2$, i.e., if we define

$$\tilde{\Omega} = \{\omega \in \Omega : (\hat{c}_{n,i(p)+}(\omega), \hat{c}_{n,i(p)-}(\omega)) \to (c_{\mathfrak{a}_p}(\omega), c_{\mathfrak{a}_p-}(\omega)) \text{ along } \mathbb{N}_2.\}$$

we have $\mathbb{P}(\tilde{\Omega}) = 1$. Fix $\omega \in \tilde{\Omega}$, we have

$$\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} |\tilde{\varsigma}_{n,p}(\beta_0)| \xrightarrow{\mathcal{L}} \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} |\varsigma_p(\beta_0)|, \text{ as } n\to\infty.$$

where $\tilde{\varsigma}_{n,p}(\beta_0)$ is defined in Algorithm 1. Notice that the distribution $\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}}|\varsigma_p(\beta_0)|$ has no atoms over $[0,\infty)$, so by Lemma 21.2 of Van der Vaart (2000) and $cv_{n,1}^\alpha(\beta_0)$ being the $(1-\alpha)$-quantile of $\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}}|\tilde{\varsigma}_{n,p}(\beta_0)|$, we have $cv_{n,1}^\alpha(\beta_0,\omega) \longrightarrow cv_1^\alpha(\beta_0)$ as $n\to\infty$. Hence, for an arbitrary subsequence $\mathbb{N}_1$, there exists a subsubsequence $\mathbb{N}_2$ along which $cv_{n,1}^\alpha(\beta_0) \xrightarrow{a.s.} cv_1^\alpha(\beta_0)$. By the subsequence characterization of convergence in probability, we have $cv_{n,1}^\alpha(\beta_0) \xrightarrow{\mathbb{P}} cv_1^\alpha(\beta_0)$ along $\mathbb{N}$, therefore under the null,

$$\big((|S_{n,p}(\beta_0)|)_{p\geq 1}, cv_{n,1}^\alpha(\beta_0)\big) \xrightarrow{\mathcal{L}\text{-s}} ((|\varsigma_p(\beta_0)|)_{p\geq 1}, cv_1^\alpha(\beta_0))$$

hence by continuous mapping theorem, $Q_S - cv_{n,1}^\alpha(\beta_0) \xrightarrow{\mathcal{L}\text{-s}} \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}}[\varsigma_p(\beta_0)]^2/v_{S,p}(\beta_0) - cv_1^\alpha(\beta_0)$, which implies $\mathbb{P}(Q_S - cv_{n,1}^\alpha(\beta_0) > 0) \to \mathbb{P}\left(\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}}[\varsigma_p(\beta_0)]^2/v_{S,p}(\beta_0) - cv_1^\alpha(\beta_0) > 0\right)$, that is, $\mathbb{P}(Q_S > cv_{n,AR}^\alpha(\beta_0)) \to \mathbb{P}(\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}}[\varsigma_p(\beta_0)]^2/v_{S,p}(\beta_0) > cv_{AR}^\alpha(\beta_0)) = \alpha$.

Proof of Lemma 1

(a) Under $H_0 : \beta = \beta_0$, we have $Z_p \sim \mathcal{N}(g_0 \xi_{\mathfrak{a}_p}, c_{\mathfrak{a}_p})$, so the log-likelihood of Z_p in terms of $\xi_{\mathfrak{a}_p}$ is, up to constants,

$$\begin{aligned}\ell(\xi_{\mathfrak{a}_p}) &= -\frac{1}{2}(Z_p - g_0\xi_{\mathfrak{a}_p})^\intercal c_{\mathfrak{a}_p}^{-1}(Z_p - g_0\xi_{\mathfrak{a}_p}) \\ &= -\frac{1}{2}Z_p^\intercal c_{\mathfrak{a}_p}^{-1} Z_p + \xi_{\mathfrak{a}_p} g_0^\intercal c_{\mathfrak{a}_p}^{-1} Z_p - \frac{1}{2}\xi_{\mathfrak{a}_p}^2 g_0^\intercal c_{\mathfrak{a}_p}^{-1} g_0\end{aligned}$$

Maximizing $\ell(\xi_{\mathfrak{a}_p})$ over $\xi_{\mathfrak{a}_p}$ generates its MLE invoking the null, that is,

$$g_0^\intercal c_{\mathfrak{a}_p}^{-1} Z_p - \hat{\xi}_{\mathfrak{a}_p} g_0^\intercal c_{\mathfrak{a}_p}^{-1} g_0 = 0 \quad \Rightarrow \quad \hat{\xi}_{\mathfrak{a}_p} = \frac{g_0^\intercal c_{\mathfrak{a}_p}^{-1} Z_p}{g_0^\intercal c_{\mathfrak{a}_p}^{-1} g_0}.$$

Furthermore, note the density of $Z_p \mid \xi_{\mathfrak{a}_p}$ is

$$\begin{aligned}f(Z_p \mid \xi_{\mathfrak{a}_p}) &= (2\pi)^{-1}|c_{\mathfrak{a}_p}|^{-1}\exp\big[-\frac{1}{2}(Z_p - g_0\xi_{\mathfrak{a}_p})^\intercal c_{\mathfrak{a}_p}^{-1}(Z_p - g_0\xi_{\mathfrak{a}_p})\big] \\ &= (2\pi)^{-1}|c_{\mathfrak{a}_p}|^{-1}\exp\left[-\frac{1}{2}Z_p^\intercal c_{\mathfrak{a}_p}^{-1} Z_p + \left(\xi_{\mathfrak{a}_p}\frac{g_0^\intercal c_{\mathfrak{a}_p}^{-1} Z_p}{g_0^\intercal c_{\mathfrak{a}_p}^{-1} g_0} - \frac{\xi_{\mathfrak{a}_p}^2}{2}\right) g_0^\intercal c_{\mathfrak{a}_p}^{-1} g_0\right]\end{aligned}$$

By the factorization theorem (Corollary 2.6.1 of Lehmann and Romano (2005)), $\hat{\xi}_{\mathfrak{a}_p}$ is sufficient for $\xi_{\mathfrak{a}_p}$.

(b) $\mathbb{E}(S_p) = h_0^\intercal g_0 \xi_{\mathfrak{a}_p} = (-\beta_0, 1)(1, \beta_0)^\intercal \xi_{\mathfrak{a}_p} = 0$ and $\mathbb{E}(\hat{\xi}_{\mathfrak{a}_p}) = g_0^\intercal c_{\mathfrak{a}_p}^{-1} g_0 \xi_{\mathfrak{a}_p}/(g_0^\intercal c_{\mathfrak{a}_p}^{-1} g_0) = \xi_{\mathfrak{a}_p}$, variances $\mathrm{Var}(S_p)$ and $\mathrm{Var}(\hat{\xi}_{\mathfrak{a}_p})$ are easily computed, and

$$\mathrm{Cov}(S_p, \hat{\xi}_{\mathfrak{a}_p}) = \mathrm{Cov}\left(h_0^\intercal Z_p,\ \frac{g_0^\intercal c_{\mathfrak{a}_p}^{-1} Z_p}{g_0^\intercal c_{\mathfrak{a}_p}^{-1} g_0}\right) = \frac{h_0^\intercal c_{\mathfrak{a}_p} c_{\mathfrak{a}_p}^{-1} g_0}{g_0^\intercal c_{\mathfrak{a}_p}^{-1} g_0} = \frac{h_0^\intercal g_0}{g_0^\intercal c_{\mathfrak{a}_p}^{-1} g_0} = 0,$$

since $h_0^\intercal g_0 = (-\beta_0) \cdot 1 + 1 \cdot \beta_0 = 0$ and that zero correlation implies independence in the joint normality.

Proof of Theorem 3

For $U_p = (S_p^*, T_p^*)^\intercal$, its log-likelihood is

$$\ell\left(\delta, \{\xi_{\mathfrak{a}_p}^*\}_{p \in \mathcal{P}_{\mathfrak{a}_B}}\right) = -\frac{1}{2}\sum_{p \in \mathcal{P}_{\mathfrak{a}_B}}\left[(S_p^* - \delta\lambda_p\xi_{\mathfrak{a}_p}^*)^2 + \{T_p^* - (1 + \delta\theta_p)\xi_{\mathfrak{a}_p}^*\}^2\right] - \frac{|\mathcal{P}_{\mathfrak{a}_B}|}{2}\log(2\pi).$$

The score with respect to δ is

$$\frac{\partial}{\partial \delta}\ell\left(\delta, \{\xi^*_{\mathfrak{a}_p}\}_{p\in\mathcal{P}_{\mathfrak{a}_B}}\right) = \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \left[\lambda_p \xi^*_{\mathfrak{a}_p}(S^*_p - \delta\lambda_p\xi^*_{\mathfrak{a}_p}) + \theta_p\xi^*_{\mathfrak{a}_p}\{T^*_p - (1+\delta\theta_p)\xi^*_{\mathfrak{a}_p}\}\right].$$

Under the restriction $\delta = 0$, the likelihood separates as

$$\ell\left(0, \{\xi^*_{\mathfrak{a}_p}\}_{p\in\mathcal{P}_{\mathfrak{a}_B}}\right) = -\frac{1}{2}\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \left[(S^*_p)^2 + (T^*_p - \xi^*_{\mathfrak{a}_p})^2\right] - \frac{|\mathcal{P}_{\mathfrak{a}_B}|}{2}\log(2\pi),$$

so the restricted MLE is $\tilde{\xi}^*_{\mathfrak{a}_p} = T^*_p$. Let $\hat{\xi}^*_{\mathfrak{a}_p}(\delta)$ denote the profile MLE of $\xi^*_{\mathfrak{a}_p}$ for a fixed δ, and define the profiled log-likelihood

$$\ell_{\text{prof}}(\delta) = \ell\left(\delta, \{\hat{\xi}^*_{\mathfrak{a}_p}(\delta)\}_{p\in\mathcal{P}_{\mathfrak{a}_B}}\right).$$

By the chain rule, the profile score restricted at the null is

$$\ell'_{\text{prof}}(0) = \frac{\partial}{\partial \delta}\ell\left(\delta, \{\xi^*_{\mathfrak{a}_p}\}_{p\in\mathcal{P}_{\mathfrak{a}_B}}\right)\Bigg|_{\delta=0,\, \xi^*_{\mathfrak{a}_p}=\tilde{\xi}^*_{\mathfrak{a}_p}} + \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \frac{\partial}{\partial \xi^*_{\mathfrak{a}_p}}\ell\left(0, \{\xi^*_{\mathfrak{a}_p}\}_{p\in\mathcal{P}_{\mathfrak{a}_B}}\right)\Bigg|_{\xi^*_{\mathfrak{a}_p}=\tilde{\xi}^*_{\mathfrak{a}_p}} \cdot \frac{\partial\hat{\xi}^*_{\mathfrak{a}_p}(0)}{\partial\delta}.$$

The second term is zero by the restricted likelihood first-order condition. Hence, substituting $\tilde{\xi}^*_{\mathfrak{a}_p} = T^*_p$ into $\ell'_{\text{prof}}(0)$ gives

$$\begin{aligned}\ell'_{\text{prof}}(0) &= \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \left[\lambda_p T^*_p S^*_p + \theta_p T^*_p(T^*_p - T^*_p)\right] \\ &= \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \lambda_p T^*_p S^*_p.\end{aligned}$$

Under H_0, conditional on $(T^*_p)_{p\in\mathcal{P}_{\mathfrak{a}_B}}$, the conditional variance of the profile score is

$$\text{Var}\left(\ell'_{\text{prof}}(0) \mid (T^*_p)_{p\in\mathcal{P}_{\mathfrak{a}_B}}\right) = \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \lambda^2_p (T^*_p)^2.$$

The Lagrange multiplier statistic is the squared profile score divided by its null conditional variance, the counter-component in the Fisher information matrix (Engle (1984)):

$$LM = \frac{\ell'_{\text{prof}}(0)^2}{\text{Var}\left(\ell'_{\text{prof}}(0) \mid (T^*_p)_{p\in\mathcal{P}_{\mathfrak{a}_B}}\right)} = \frac{\left(\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \lambda_p S^*_p T^*_p\right)^2}{\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \lambda^2_p (T^*_p)^2},$$

Under H_0, $(S^*_p)_{p\in\mathcal{P}_{\mathfrak{a}_B}} \sim \mathcal{N}(0, I_{|\mathcal{P}_{\mathfrak{a}_B}|})$ and $S^*_p \perp\!\!\!\perp T^*_p$ for all p. Conditional on $T^* = (T^*_p)_{p\in\mathcal{P}_{\mathfrak{a}_B}}$, the numerator in (27) is normal with mean zero and variance $\sum_p \lambda^2_p(T^*_p)^2$, and therefore the conditional statistic $LM \mid T^* \sim \chi^2_1$. Since the denominator is positive with probability one, the unconditional null distribution is also χ^2_1. And the homogeneous case is immediate.

Proof of Theorem 4

We first show a lemma.

Lemma 2. Let $g, t, \lambda, \theta \in \mathbb{R}^P$ where P is a finite integer and $\lambda > 0$, then for the function

$$L(g; t, \lambda, \theta) = \sup_{\delta \in \mathbb{R}} \left\{ \sum_{p=1}^{P} \frac{[\delta \lambda_p g_p + (1 + \delta \theta_p) t_p]^2}{\delta^2 \lambda_p^2 + (1 + \delta \theta_p)^2} - \|t\|^2 \right\}$$

we have $0 \le L(g; t, \lambda, \theta) \le \|g\|^2$.

Proof of Lemma 2. For each p and $\delta \in \mathbb{R}$, define

$$a_p(\delta) = \big(\delta \lambda_p,\, 1 + \delta \theta_p\big)^\intercal \quad \text{and} \quad z_p = (g_p, t_p)^\intercal.$$

Under the maintained positive-definiteness condition,

$$v_{S,p} = h_0^\intercal c_{\mathfrak{a}_p} h_0 > 0 \quad \text{and} \quad v_{T,p} = (g_0^\intercal c_{\mathfrak{a}_p}^{-1} g_0)^{-1} > 0.$$

Hence $\lambda_p = (v_{T,p}/v_{S,p})^{1/2} > 0$. If $a_p(\delta) = 0$, its first coordinate would imply $\delta \lambda_p = 0$, and therefore $\delta = 0$; but its second coordinate would then equal $1 + \delta \theta_p = 1$, a contradiction. Thus $a_p(\delta) \neq 0$ for every $\delta \in \mathbb{R}$, and in particular

$$\|a_p(\delta)\|^2 = \delta^2 \lambda_p^2 + (1 + \delta \theta_p)^2 > 0.$$

By the Cauchy–Schwarz inequality,

$$0 \le \frac{\big[a_p(\delta)^\intercal z_p\big]^2}{\|a_p(\delta)\|^2} \le \|z_p\|^2 = g_p^2 + t_p^2.$$

Consequently, for every $\delta \in \mathbb{R}$,

$$\sum_{p=1}^{P} \frac{[\delta \lambda_p g_p + (1 + \delta \theta_p) t_p]^2}{\delta^2 \lambda_p^2 + (1 + \delta \theta_p)^2} - \|t\|^2 \le \sum_{p=1}^{P} (g_p^2 + t_p^2) - \|t\|^2 = \|g\|^2.$$

Taking the supremum over δ establishes the upper bound. For the lower bound, evaluating the expression at $\delta = 0$ gives $\sum_{p=1}^{P} t_p^2 - \|t\|^2 = 0$. Hence

$$0 \le L(g; t, \lambda, \theta) \le \|g\|^2,$$

as claimed. $\square$

Given the truth of jump beta β, we have the limiting experiment $Z_p = (X_p, Y_p)^\intercal \sim \mathcal{N}(\xi_{\mathfrak{a}_p} g, c_{\mathfrak{a}_p})$ with $g = (1, \beta)^\intercal$. Define $\delta = \beta - \beta_0$, so $g = g_0 + \delta e_2$ with $e_2 = (0,1)^\intercal$. Note $\mathbb{E}(S_p^*) = \mathbb{E}(h_0^\intercal \xi_{\mathfrak{a}_p}(g_0 + \delta e_2)/\sqrt{v_{S,p}}) = \delta \xi_{\mathfrak{a}_p}/\sqrt{v_{S,p}}$ and $\mathbb{E}(T_p^*) = \sqrt{v_{T,p}} g_0^\intercal c_{\mathfrak{a}_p}^{-1}(g_0 + \delta e_2)\xi_{\mathfrak{a}_p} = \xi_{\mathfrak{a}_p}(1+\delta\theta_p)/\sqrt{v_{T,p}}$ by defining $\theta_p = v_{T,p} g_0^\intercal c_{\mathfrak{a}_p}^{-1} e_2$. Define $\lambda_p \equiv \sqrt{v_{T,p}/v_{S,p}}$ and $\xi_{\mathfrak{a}_p}^* \equiv \xi_{\mathfrak{a}_p}/\sqrt{v_{T,p}}$, so we have

$$U_p \equiv (S_p^*, T_p^*)^\intercal = A_p Z_p \sim \mathcal{N}\left(\xi_{\mathfrak{a}_p}^* \begin{pmatrix} \delta\lambda_p \\ 1+\delta\theta_p \end{pmatrix}, I_2\right), \text{ where } A_p = \begin{pmatrix} h_0^\intercal/\sqrt{v_{S,p}} \\ \sqrt{v_{T,p}} g_0^\intercal c_{\mathfrak{a}_p}^{-1} \end{pmatrix}.$$

Therefore, the Gaussian limit experiment Z_p for parameters $(\beta, \xi_{\mathfrak{a}_p})$ can be reparametrized as another normal limit experiment $U_p = (S_p^*, T_p^*)^\intercal$ with parameters $(\delta, \xi_{\mathfrak{a}_p}^*)$ via variance normalization and U_p's are independent across p. Let $m_p(\delta) = (\delta\lambda_p, 1+\delta\theta_p)^\intercal$, then the log-likelihood of the $\{U_p\}_{p=1}^P$ is

$$\ell(\delta, \{\xi_{\mathfrak{a}_p}^*\}) = -\frac{1}{2}\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \|U_p - \xi_{\mathfrak{a}_p}^* m_p(\delta)\|_2^2 - \frac{|\mathcal{P}_{\mathfrak{a}_B}|}{2}\log(2\pi). \tag{S5}$$

For any δ, profiling out $\xi_{\mathfrak{a}_p}^*$ gives

$$\hat{\xi}_{\mathfrak{a}_p}^*(\delta) = \frac{m_p^\intercal(\delta) U_p}{m_p^\intercal(\delta) m_p(\delta)}$$

Hence we have the concentrated log-likelihood,

$$\ell_c(\delta) = -\frac{1}{2}\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \|U_p\|_2^2 + \frac{1}{2}\sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \frac{[m_p^\intercal(\delta)U_p]^2}{m_p^\intercal(\delta)m_p(\delta)} - \frac{|\mathcal{P}_{\mathfrak{a}_B}|}{2}\log(2\pi). \tag{S6}$$

Under the null $\delta = 0$, so $m_p(0) = e_2$ and

$$\frac{[m_p^\intercal(0)U_p]^2}{m_p^\intercal(0)m_p(0)} = (T_p^*)^2.$$

Thus

$$\begin{aligned} LR_{\text{het}} &= \sup_{\delta\in\mathbb{R}} 2\{\ell_c(\delta) - \ell_c(0)\} = \sup_{\delta\in\mathbb{R}} \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \frac{[m_p^\intercal(\delta)U_p]^2}{m_p^\intercal(\delta)m_p(\delta)} - Q_T \\ &= \sup_{\delta\in\mathbb{R}} \sum_{p\in\mathcal{P}_{\mathfrak{a}_B}} \frac{\left\{\delta\lambda_p S_p^* + (1+\delta\theta_p)T_p^*\right\}^2}{\delta^2\lambda_p^2 + (1+\delta\theta_p)^2} - Q_T. \end{aligned}$$

If $c_{\mathfrak{a}_p}$'s are equal for all p, write $\lambda_p = \lambda$ and $\theta_p = \theta_h$ for all p, where $\theta_h \in \mathbb{R}$. Then $m_p(\delta) = (\delta\lambda, 1+\delta\theta_h)^\intercal$, and by Rayleigh quotient ratio (Theorem 4.2.2 of Horn and Johnson (2012)), LR_{het} simplifies to

$$LR_{\text{homo}} = \sup_{\delta\in\mathbb{R}} \frac{m^\intercal(\delta)Qm(\delta)}{m^\intercal(\delta)m(\delta)} - Q_T = \lambda_{\max}(Q) - Q_T$$

$$= \frac{1}{2}\left[(Q_S - Q_T) + \sqrt{(Q_S + Q_T)^2 - 4(Q_S Q_T - Q_{ST}^2)}\right]$$

Proof of Corollary 1

Let $m = \mathbb{E}(S^*) = \delta\Lambda\xi^*$, so $S^* \sim \mathcal{N}(m, I_{|\mathcal{P}_{\mathfrak{a}_B}|})$. The transformed Gaussian experiment is can be expressed as

$$\begin{pmatrix} S^* \\ T^* \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} m \\ (I_{|\mathcal{P}_{\mathfrak{a}_B}|} + \delta\Theta)\xi^* \end{pmatrix}, I_{2|\mathcal{P}_{\mathfrak{a}_B}|}\right).$$

And the testing problem becomes

$$H_0 : m = 0 \qquad \text{against} \qquad H_1 : m \in \mathbb{R}^{|\mathcal{P}_{\mathfrak{a}_B}|} \setminus \{0\}$$

Consequently, for every fixed t,

$$S^* \mid T^* = t \sim \mathcal{N}_P(m, I_P), \qquad m = \delta\Lambda\xi^*.$$

Since Λ is nonsingular and $\xi^* \in \Xi = \mathbb{R}^P \setminus \{0\}$,

$$\{\delta\Lambda\xi^* : \delta \neq 0,\ \xi^* \in \Xi\} = \mathbb{R}^P \setminus \{0\}.$$

Thus, conditional on $T^* = t$, the weak-jump experiment is precisely the Gaussian mean problem stated in the corollary.

We next establish the geometry of the two acceptance regions. For $t \neq 0$, define

$$K_h(s,t) \equiv \frac{(t^\intercal s)^2}{t^\intercal t}, \qquad J_h(s,t) \equiv s^\intercal s - K_h(s,t), \qquad q \equiv t^\intercal t.$$

Let $c_h(t) > 0$ denote the critical value used by the fixed-t section of $\varphi^{\alpha}_{CLR^h}$. Direct algebra applied to (29) gives

$$LR^h(s,t) \leq c_h(t) \quad \Longleftrightarrow \quad \frac{K_h(s,t)}{c_h(t)} + \frac{J_h(s,t)}{c_h(t) + q} \leq 1.$$

Hence the fixed-t acceptance region of $\varphi^{\alpha}_{CLR^h}$ is a closed, bounded ellipsoid with nonempty interior. If $t = 0$, then $LR^h(s,0) = s^\intercal s$, so its acceptance region is a closed ball. In either case the acceptance region is convex and its boundary is Lebesgue-null.

For the heterogeneous CLR, compactify the profile parameter as $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$ and define

$$F_\eta(s,t) \equiv \sum_{p=1}^{P} \frac{\{\eta\lambda_p s_p + (1+\eta\theta_p)t_p\}^2}{\eta^2\lambda_p^2 + (1+\eta\theta_p)^2}, \qquad \eta \in \mathbb{R},$$

$$F_{\pm\infty}(s,t) \equiv \sum_{p=1}^{P} \frac{(\lambda_p s_p + \theta_p t_p)^2}{\lambda_p^2 + \theta_p^2}.$$

If $c_{CLR}(t) > 0$ is the critical value used by the fixed-t section of φ^α_{CLR}, its acceptance region is

$$\mathcal{A}_{CLR}(t) = \bigcap_{\eta \in \overline{\mathbb{R}}} \{s : F_\eta(s,t) \le t^\intercal t + c_{CLR}(t)\}.$$

Each member of this intersection is a convex quadratic sublevel set; therefore $\mathcal{A}_{CLR}(t)$ is closed and convex. The $\eta = \pm\infty$ member is coercive in s, because its quadratic coefficient is

$$\operatorname{diag}\left(\left(\frac{\lambda_p^2}{\lambda_p^2 + \theta_p^2}\right)_{p=1}^{P}\right) \succ 0,$$

so $\mathcal{A}_{CLR}(t)$ is bounded. Moreover, $F_\eta(0,t) \le t^\intercal t$ for every η, with equality at $\eta = 0$, and hence $LR(0,t) = 0$. The map $F_\eta(s,t)$ is jointly continuous in (s,η) on $\mathbb{R}^P \times \overline{\mathbb{R}}$. Compactness of $\overline{\mathbb{R}}$ then implies that $\sup_\eta F_\eta(s,t)$ is continuous in s. It follows that $\mathcal{A}_{CLR}(t)$ has nonempty interior, and its boundary is Lebesgue-null.

It remains to invoke the complete-class result. In the Gaussian exponential-family representation, $t_0(s) = -\|s\|^2/2 - (P/2)\log(2\pi)$ is continuous, which verifies Birnbaum's Assumption 2. His Assumption 3 also holds in every finite dimension P. To see this directly, let

$$H = \{s : a^\intercal s + a_0 \le 0\}, \qquad \|a\| = 1,$$

be any half-space, and let $s_0 \notin H$. Choose a bounded neighborhood U of s_0 such that

$$\inf_{s \in U}(a^\intercal s + a_0) = \epsilon > 0.$$

For $r > 0$, set $m_r = ra$. Under $\mathcal{N}_P(m_r, I_P)$, $\Pr_{m_r}(H) = \Phi(-r - a_0)$. For all sufficiently large r, the Mills bound $\Phi(-x) \le \phi(x)/x$ therefore yields, uniformly over $s \in U$,

$$\frac{\phi_P(s - m_r)}{\Pr_{m_r}(H)} \ge (2\pi)^{-(P-1)/2}(r + a_0)\exp\left\{-\frac{\|s\|^2}{2} + \frac{a_0^2}{2} + r(a^\intercal s + a_0)\right\}$$

$$\geq C_U(r+a_0)e^{r\epsilon} \longrightarrow \infty,$$

where $C_U > 0$ is independent of r. This verifies Birnbaum's Assumption 3.

By Birnbaum (1955, Theorem 4, pp. 29–31), tests that are almost-everywhere equivalent to tests with open convex acceptance regions form a minimal complete class. Each acceptance region established above differs from its open interior only on a Lebesgue-null boundary and thus has identical risk under every Gaussian mean. Both conditional tests therefore belong to this minimal complete class. If either test were inadmissible, a test dominating it, followed by completeness, would permit that member to be deleted while preserving completeness, contradicting minimality. Thus φ^{α}_{CLR} and $\varphi^{\alpha}_{CLR^h}$ are admissible in every fixed-t conditional experiment.

Proof of Theorem 5

(a) In the limit experiment, we have $S^* \perp\!\!\!\perp D$ because $D = \Lambda T^*$, $S^* \perp\!\!\!\perp T^*$ due to Lemma 1, and $T^* \mapsto D$ is a bijective mapping. It follows that $S^* \mid D \sim \mathcal{N}(m, I_{|\mathcal{P}_{\mathfrak{a}_B}|})$ for which $m = \delta\Lambda\xi^*$. On the $\mathbb{P}$-a.s. event $\{D \neq 0\}$, let $P_D \equiv D(D^\intercal D)^{-1}D^\intercal$ and $M_D \equiv I - P_D$, then by definition $LM = S^{*\intercal}P_dS^*$ and $J = S^{*\intercal}M_dS^*$. The matrices P_d and M_d are $|\mathcal{P}_{\mathfrak{a}_B}| \times |\mathcal{P}_{\mathfrak{a}_B}|$ symmetric matrices with ranks 1 and $|\mathcal{P}_{\mathfrak{a}_B}|-1$, respectively. Cochran's theorem (see Theorem 1.5 of Shao (2003)) therefore gives

$$LM \mid D \sim \chi^2_1(m^\intercal P_D m), \qquad J \mid D \sim \chi^2_{|\mathcal{P}_{\mathfrak{a}_B}|-1}(m^\intercal(I-P_D)m), \qquad LM \perp\!\!\!\perp J \mid D.$$

Note under the null $\delta = 0$ implies $m = 0$, thus

$$LM + a(D)\cdot J \mid D \overset{\mathcal{L},H_0}{=} \chi^2_1 + a(D)\cdot\chi^2_{|\mathcal{P}_{\mathfrak{a}_B}|-1},$$

where the two central chi-squares in the above are independent. Therefore $\mathbb{E}_{H_0,\mu_D}(\varphi^{\alpha}_{CLC,a(D)} \mid D) = \alpha$ outside the $\mathbb{P}$-null set $\{D = 0\}$ and the size control is free of μ_D.

(b) Consider following alternative to include strong and semi-strong cases

$$\xi^*_n = s_n u_n, \qquad s_n \longrightarrow \infty, \qquad u_n \longrightarrow u \neq 0, \qquad \delta_n = \frac{b}{s_n}, \tag{S7}$$

The Gaussian experiment along (S7) admits the representation

$$S^*_n = b\Lambda u_n + \varepsilon_{S,n}, \qquad\qquad T^*_n = s_n u_n + b\Theta u_n + \varepsilon_{T,n},$$

where $\varepsilon_{S,n}$ and $\varepsilon_{T,n}$ are independent $\mathcal{N}(0, I_P)$ vectors. Hence

$$\mu_{D,n} = s_n \Lambda u_n + b\Lambda\Theta u_n, \qquad \nu_{D,n} = \Lambda u_n + \frac{b}{s_n}\Lambda\Theta u_n \longrightarrow \nu_{D,\infty}.$$

It follows that

$$\frac{D_n}{s_n} \xrightarrow{\mathbb{P}} \nu_{D,\infty}, \qquad \frac{Q_{T,n}}{s_n^2} \xrightarrow{\mathbb{P}} \|u\|^2, \qquad P_{D_n} = P_{D_n/s_n} \xrightarrow{\mathbb{P}} P_{\nu_{D,\infty}} \equiv \frac{\nu_{D,\infty}\nu_{D,\infty}^{\intercal}}{\nu_{D,\infty}^{\intercal}\nu_{D,\infty}}.$$

The mean of S_n^* converges to $b\nu_{D,\infty}$, which belongs to the range of $P_{\nu_{D,\infty}}$ and has zero projection on its orthogonal complement. Since $\varepsilon_{S,n}$ is independent of D_n, Slutsky's theorem and the orthogonal Gaussian decomposition induced by $P_{\nu_{D,\infty}}$ give

$$(LM_n, J_n) \xrightarrow{\mathcal{L},H_{\mathfrak{a}}^{s_n}} (LM_\infty, J_\infty), \quad LM_\infty \sim \chi_1^2\left(b^2\|\nu_{D,\infty}\|^2\right), \quad J_\infty \sim \chi^2_{|\mathcal{P}_{\mathfrak{a}_B}|-1}, \quad LM_\infty \perp\!\!\!\perp J_\infty.$$

For $s > 0$, define the compatible mean set in the normalized experiment by

$$\mathcal{M}_s(v) \equiv \mathcal{M}_D(sv).$$

Fix arbitrary deterministic sequences $s_k \to \infty$ and $\nu_k \to \nu_{D,\infty}$ as $k \to \infty$, and let $B_C \equiv \{m \in \mathbb{R}^{|\mathcal{P}_{\mathfrak{a}_B}|} : \|m\| \le C\}$ for a constant C. If $m_k \in \mathcal{M}_{s_k}(\nu_k) \cap B_C$, take a compatible pair $(\tilde{\delta}_k, \tilde{\xi}_k^*)$ and choose p_0-announment such that $(\nu_{D,\infty})_{p_0} \neq 0$ where $(\nu_{D,\infty})_{p_0}$ is the p_0-th coordinate of $\nu_{D,\infty}$. Recall definition of $\mathcal{M}_D(\mu_D)$, we have $(m_k)_{p_0} = \tilde{\delta}_k \lambda_{p_0} (\tilde{\xi}_k^*)_{p_0}$ and $(s_k \nu_k)_{p_0} = \lambda_{p_0}(1 + \theta_{p_0}\tilde{\delta}_k)(\tilde{\xi}_k^*)_{p_0}$, taking ratio yields

$$\frac{\tilde{\delta}_k}{1 + \theta_{p_0}\tilde{\delta}_k} = \frac{(m_k)_{p_0}}{(s_k\nu_k)_{p_0}} \equiv x_k \overset{(i)}{=} O(s_k^{-1}).$$

where equality (i) is due to $m_k \in B_C$ implying $|(m_k)_{p_0}| \le C$ and $(\nu_k)_{p_0} \to (\nu_{D,\infty})_{p_0} \neq 0$. So

$$\tilde{\delta}_k = \frac{x_k}{1 - \theta_{p_0} x_k} = O(s_k^{-1}).$$

Thus $h_k \equiv s_k \tilde{\delta}_k = O(1)$, using that Λ and Θ are diagonal,

$$\begin{aligned}
\mu_{D,k} &\equiv s_k \nu_k = \Lambda\left(I + \tilde{\delta}_k \Theta\right)\tilde{\xi}_k^*, \\
\tilde{\xi}_k^* &= \left(I + \tilde{\delta}_k\Theta\right)^{-1}\Lambda^{-1} s_k \nu_k, \\
m_k &= \tilde{\delta}_k \Lambda \tilde{\xi}_k^* \\
&= s_k \tilde{\delta}_k \Lambda\left(I + \tilde{\delta}_k\Theta\right)^{-1}\Lambda^{-1}\nu_k
\end{aligned}$$

$$= s_k\tilde{\delta}_k \left(I + \tilde{\delta}_k\Theta\right)^{-1} \nu_k$$

$$= h_k \left(I + \frac{h_k}{s_k}\Theta\right)^{-1} \nu_k = h_k\nu_{D,\infty} + o(1). \tag{S8}$$

Conversely, for any bounded h, take

$$\tilde{\delta}_k = \frac{h}{s_k}, \qquad \tilde{\xi}^*_k = s_k \left(I + \frac{h}{s_k}\Theta\right)^{-1} \Lambda^{-1}\nu_k.$$

The unrestricted parameter spaces imply that this pair is eventually in the parameter space, and its induced mean converges to $h\nu_{D,\infty}$. The preceding two inclusions are uniform over bounded h; points on the boundary of B_C are obtained by an inward sequence. Therefore, for every $C > 0$,

$$d_H(\mathcal{M}_{s_k}(\nu_k) \cap B_C,\, \{h\nu_{D,\infty} : h \in \mathbb{R}\} \cap B_C) \longrightarrow 0. \tag{S9}$$

where $d_H(A_1, A_2)$ is the Hausdorff distance betwee sets A_1 and A_2, defined as

$$d_H(A_1, A_2) \equiv \max\{\sup_{x_1\in A_1} \inf_{x_2\in A_2} \|x_1 - x_2\|, \sup_{x_2\in A_2} \inf_{x_1\in A_1} \|x_1 - x_2\|\}.$$

We now make the correspondence with Andrews (2016, Theorem 5) explicit. For $s > 0$, define the normalized statistic and its mean by

$$\bar{D}_s \equiv \frac{D}{s}, \qquad \bar{\mu}_D \equiv \frac{\mu_D}{s}.$$

Since $S^* \perp\!\!\!\perp D$, their joint distribution is

$$\begin{pmatrix} S^* \\ \bar{D}_s \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} m \\ \bar{\mu}_D \end{pmatrix}, \begin{pmatrix} I_P & 0 \\ 0 & \Lambda^2/s^2 \end{pmatrix}\right). \tag{S10}$$

In Andrews' notation, take his moment vector to be $g^A \equiv S^*$ and his derivative vector to be $\widetilde{g}^A_s \equiv \bar{D}_s$; hence his dimensions are $k = P$ and $p = 1$. Let γ^A_s denote the covariance configuration of this normalized experiment. The two covariance blocks appearing in Andrews' theorem are therefore, explicitly,

$$\Sigma^A_{\widetilde{g}g}(\gamma^A_s) \equiv \mathrm{Cov}(\widetilde{g}^A_s, g^A) = 0, \qquad \Sigma^A_{\widetilde{g}\widetilde{g}}(\gamma^A_s) \equiv \mathrm{Var}(\widetilde{g}^A_s) = \frac{\Lambda^2}{s^2}.$$

Consequently, Andrews' orthogonalized derivative statistic is

$$D^A_s \equiv \widetilde{g}^A_s - \Sigma^A_{\widetilde{g}g}(\gamma^A_s) g^A = \bar{D}_s, \qquad \mathbb{E}[D^A_s] = \bar{\mu}_D.$$

Thus, for finite s, Andrews' correspondence between the mean of D_s^A and the compatible values of the mean m of g^A is, in our notation,

$$\mathcal{M}_D^A(\bar{\mu}_D, \gamma_s^A) \equiv \mathcal{M}_D(s\bar{\mu}_D). \tag{S11}$$

Let γ_∞^A denote the limiting covariance configuration at which

$$\Sigma_{\tilde{g}g}^A(\gamma_\infty^A) = \Sigma_{\tilde{g}\tilde{g}}^A(\gamma_\infty^A) = 0,$$

and take

$$\Gamma^A \equiv \{\gamma_s^A : s > 0\} \cup \{\gamma_\infty^A\}, \qquad \mathcal{M}_D^A(\bar{\mu}_D, \gamma_\infty^A) \equiv \{h\bar{\mu}_D : h \in \mathbb{R}\}.$$

Because $\Lambda^2/s^2 \to 0$, $\gamma_s^A \to \gamma_\infty^A$ as $s \to \infty$. We apply Andrews' theorem at $(\nu_{D,\infty}, \gamma_\infty^A)$. The condition $\nu_{D,\infty} \neq 0$ means that this $P \times 1$ mean matrix has full column rank.

It remains to check the "for every sequence" requirement in Andrews' theorem. Consider any $(\bar{\mu}_{D,k}, \gamma_k^A) \to (\nu_{D,\infty}, \gamma_\infty^A)$ in $\mathbb{R}^P \times \Gamma^A$. Along terms for which $\gamma_k^A = \gamma_{s_k}^A$ with finite s_k, necessarily $s_k \to \infty$, and

$$\mathcal{M}_D^A(\bar{\mu}_{D,k}, \gamma_{s_k}^A) = \mathcal{M}_D(s_k\bar{\mu}_{D,k}) = \mathcal{M}_{s_k}(\bar{\mu}_{D,k}).$$

Hence (S9) gives Andrews' required Hausdorff convergence. Along terms for which $\gamma_k^A = \gamma_\infty^A$, the compatible set is exactly $\{h\bar{\mu}_{D,k} : h \in \mathbb{R}\}$, whose intersection with B_C converges in Hausdorff distance to $\{h\nu_{D,\infty} : h \in \mathbb{R}\} \cap B_C$ because $\bar{\mu}_{D,k} \to \nu_{D,\infty} \neq 0$. Splitting an arbitrary sequence into these two types of terms proves the universal sequence condition. Thus all hypotheses of Andrews (2016, Theorem 5) are satisfied.

Multiplying D/s by s does not change its direction and therefore does not change LM, J, the power of any constant-weight CLC rule, the power envelope, or the maximum-regret objective. Consequently, the MMRU weight in the normalized problem with mean ν_k is exactly $a_{\mathrm{MMRU}}(s_k\nu_k)$. Andrews' theorem therefore implies

$$a_{\mathrm{MMRU}}(s_k\nu_k) \longrightarrow 0 \qquad \text{for every deterministic } s_k \longrightarrow \infty \text{ and } \nu_k \longrightarrow \nu_{D,\infty}. \tag{S12}$$

Taking $s_k = s_n$ and $\nu_k = \nu_{D,n} = \mu_{D,n}/s_n$ proves (35).

It remains to establish feasibility. Write

$$\mu_{T,n} \equiv (I_P + \delta_n\Theta)\xi_n^* = s_n u_n + b\Theta u_n.$$

Then

$$\mathfrak{c}_{D,n} = \|\mu_{T,n}\|^2 = s_n^2\|u\|^2 + o(s_n^2).$$

Since $T_n^* = \mu_{T,n} + \varepsilon_{T,n}$,

$$Q_{T,n} = \mathfrak{c}_{D,n} + 2\mu_{T,n}^{\intercal}\varepsilon_{T,n} + \|\varepsilon_{T,n}\|^2, \qquad \frac{Q_{T,n}}{\mathfrak{c}_{D,n}} \xrightarrow{\mathbb{P}} 1.$$

In particular, $Q_{T,n} \xrightarrow{\mathbb{P}} \infty$. For a generic realization $q \to \infty$, the three estimators satisfy

$$\begin{aligned}
&\widehat{\mathfrak{c}}_{D,\mathrm{MLE}}(q) = q - (P-1) + o(1), \qquad\qquad \widehat{\mathfrak{c}}_{D,\mathrm{PP}}(q) = q - P \quad \text{eventually},\\
&\widehat{\mathfrak{c}}_{D,\mathrm{KRS}}(q) = q - P + o(1).
\end{aligned}$$

The MLE expansion follows from its noncentral-chi-square score equation and $I_{\ell+1}(z)/I_\ell(z) = 1-(2\ell+1)/(2z)+O(z^{-2})$, where $\ell = P/2-1$; the PP expansion is immediate from $(q-P)_+$, and the KRS expansion follows directly from its defining series. Hence

$$\frac{\widehat{\mathfrak{c}}_{D,i}(q)}{q} \longrightarrow 1, \qquad i \in \{\mathrm{MLE}, \mathrm{PP}, \mathrm{KRS}\}.$$

For any estimator satisfying,

$$\frac{\widehat{\mu}_{D,n}}{s_n} = \frac{D_n}{s_n}\sqrt{\frac{\widehat{\mathfrak{c}}_{D,n}}{Q_{T,n}}} \xrightarrow{\mathbb{P}} \nu_{D,\infty}.$$

Because (S12) holds for every deterministic sequence converging to $\nu_{D,\infty}$, for every $\epsilon > 0$ there exist $\eta > 0$ and $N < \infty$ such that

$$\sup_{\|v-\nu_{D,\infty}\|\le\eta} a_{\mathrm{MMRU}}(s_n v) \le \epsilon \qquad \text{for every } n \ge N.$$

Otherwise, one could select a deterministic countersequence $v_n \to \nu_{D,\infty}$, contradicting (S12). Applying this bound to $v = \widehat{\mu}_{D,n}/s_n$ gives

$$\widehat{a}_n^{\mathrm{PI}} = a_{\mathrm{MMRU}}(\widehat{\mu}_{D,n}) \xrightarrow{\mathbb{P}} 0,$$

Moreover, $Q_{T,n} = D_n^{\intercal}\Lambda^{-2}D_n$, so $\widehat{a}_n^{\mathrm{PI}}$ is measurable with respect to D_n, and part (a) applies.

Finally, the joint limit of (LM_n, J_n) implies $J_n = O_{\mathbb{P}}(1)$. Together with $\widehat{a}_n^{\mathrm{PI}} \to_{\mathbb{P}} 0$, this gives $\widehat{a}_n^{\mathrm{PI}} J_n = o_{\mathbb{P}}(1)$. The distribution of $\chi_1^2 + a\chi_{P-1}^2$ is continuous in a, so its $(1-\alpha)$ quantile converges to $\chi^2_{1,1-\alpha}$ as $a \downarrow 0$. The limiting noncentral chi-square distribution has no atom at this critical value. Hence the PI–CLC and LM rejection indicators differ with probability tending to zero.

S2. Additional Propositions

Proposition S1 (Strong-identification equivalence of LM and LTT–GLS)**.** Let $P = |\mathcal{P}_{\mathfrak{a}_B}| < \infty$ and suppose Assumptions 2, 3, and 4 hold. Work conditionally on $\mathcal{F}$ and on sample paths for which the announcement-level spot covariance matrices $\boldsymbol{c}_{\mathfrak{a}_p} \equiv \boldsymbol{c}_p$ are positive definite and the strong-jump information is nondegenerate. More precisely, let $J_p \equiv \Delta X_{\mathfrak{a}_p}$ be the pth X-jump and suppose that, under the local alternatives

$$\beta_n = \beta_0 + d\Delta_n^{1/2}, \qquad d \in \mathbb{R}, \tag{S13}$$

the event increments admit the joint strong-jump expansion

$$\Delta_{i(p)}^n Z = J_p g_n + \Delta_n^{1/2} \mathcal{E}_{n,p} + o_{\mathbb{P}}(\Delta_n^{1/2}), \qquad g_n = (1, \beta_n)^\intercal, \tag{S14}$$

where, stably in law,

$$(\mathcal{E}_{n,p})_{p=1}^P \longrightarrow (\mathcal{E}_p)_{p=1}^P, \qquad \mathcal{E}_p \mid \mathcal{F} \sim \mathcal{N}(0, \boldsymbol{c}_p),$$

and the $\mathcal{E}_p$'s are conditionally independent. For fixed announcement times, (S14) is understood to include asymptotically disjoint event cells containing no unrelated jump with probability approaching one. Define

$$v_{S,p} \equiv h_0^\intercal \boldsymbol{c}_p h_0, \qquad v_{T,p} \equiv (g_0^\intercal \boldsymbol{c}_p^{-1} g_0)^{-1}, \qquad \mathcal{I}^* \equiv \sum_{p=1}^P \frac{J_p^2}{v_{S,p}} > 0. \tag{S15}$$

Let $\widehat{\boldsymbol{c}}_{n,p} \equiv (\hat{\boldsymbol{c}}_{n,i(p)-} + \hat{\boldsymbol{c}}_{n,i(p)+})/2$ be consistent for $\boldsymbol{c}_p$, and construct LM_n by replacing $(Z_p, \boldsymbol{c}_p)$ in (22)–(27) with $(\Delta_n^{-1/2}\Delta_{i(p)}^n Z, \widehat{\boldsymbol{c}}_{n,p})$. Let $\hat{\beta}_n^{\mathrm{GLS}} \equiv \hat{\beta}_n(\mathfrak{a}_B, w^*)$ be the LTT–GLS estimator in (13), based on a preliminary estimator satisfying $\widetilde{\beta}_n - \beta_n = o_{\mathbb{P}}(1)$ along each sequence in (S13), and set

$$\hat{v}_{S,n,p}^{\mathrm{GLS}} \equiv (-\widetilde{\beta}_n, 1)\widehat{\boldsymbol{c}}_{n,p}(-\widetilde{\beta}_n, 1)^\intercal, \qquad \widehat{\mathcal{I}}_n \equiv \sum_{p=1}^P \frac{(\Delta_{i(p)}^n X)^2}{\hat{v}_{S,n,p}^{\mathrm{GLS}}}, \tag{S16}$$

and define the squared LTT–GLS Wald statistic

$$W_{\text{LTT-GLS},n} \equiv \frac{\widehat{\mathcal{I}}_n}{\Delta_n} (\hat{\beta}_n^{\mathrm{GLS}} - \beta_0)^2. \tag{S17}$$

Then, for every fixed $d \in \mathbb{R}$, the following statements hold:

(a) The signed square roots of LM_n and $W_{\text{LTT-GLS},n}$ have the common efficient-score expansion

$$K_{LM,n} = K_{\text{LTT-GLS},n} + o_{\mathbb{P}}(1) = \frac{1}{\sqrt{\mathcal{I}^*}} \sum_{p=1}^{P} \frac{J_p}{v_{S,p}} \frac{\Delta_{i(p)}^n Y - \beta_0 \Delta_{i(p)}^n X}{\sqrt{\Delta_n}} + o_{\mathbb{P}}(1), \tag{S18}$$

where $LM_n = K_{LM,n}^2$ and $W_{\text{LTT-GLS},n} = K_{\text{LTT-GLS},n}^2$. Consequently,

$$LM_n - W_{\text{LTT-GLS},n} = o_{\mathbb{P}}(1). \tag{S19}$$

(b) Conditionally on $\mathcal{F}$,

$$(LM_n, W_{\text{LTT-GLS},n}) \xrightarrow{\mathcal{L}\text{-}s} (Q_d, Q_d), \qquad Q_d \mid \mathcal{F} \sim \chi_1^2(d^2\mathcal{I}^*). \tag{S20}$$

Hence, with $c_\alpha = \chi^2_{1,1-\alpha}$,

$$\mathbb{P}_{\beta_n}(1\{LM_n > c_\alpha\} \neq 1\{W_{\text{LTT-GLS},n} > c_\alpha\}) \longrightarrow 0.$$

(c) If, in addition, LTT's Assumptions 4 and 5 hold so that the LAMN argument in Theorem 3 of Li, Todorov, and Tauchen (2017) applies to the selected event experiment (or its fixed-arrival conditional-Gaussian analogue applies), then the LTT semiparametric efficiency bound is

$$\Sigma^* = (\mathcal{I}^*)^{-1}. \tag{S21}$$

Thus the two tests have the same local noncentrality parameter $d^2/\Sigma^* = d^2\mathcal{I}^*$ and are generated by the same semiparametrically efficient central sequence.

Proof of Proposition S1. Put

$$\overline{Z}_{n,p} \equiv \Delta_n^{-1/2} \Delta_{i(p)}^n Z, \qquad A_{n,p} \equiv h_0^\intercal \overline{Z}_{n,p} = \frac{\Delta_{i(p)}^n Y - \beta_0 \Delta_{i(p)}^n X}{\sqrt{\Delta_n}}.$$

Since $h_0^\intercal g_0 = 0$, $h_0^\intercal e_2 = 1$, and $g_n = g_0 + d\sqrt{\Delta_n} e_2$, expansion (S14) gives

$$A_{n,p} = dJ_p + h_0^\intercal \mathcal{E}_{n,p} + o_{\mathbb{P}}(1). \tag{S22}$$

In particular, $(A_{n,p})_{p=1}^P = O_{\mathbb{P}}(1)$ for every fixed d.

We first rewrite the LM statistic. Denote its feasible restricted profile estimator and normalizations by

$$\tilde{\xi}^0_{n,p} \equiv \frac{g_0^\intercal \widehat{\boldsymbol{c}}_{n,p}^{-1} \overline{Z}_{n,p}}{g_0^\intercal \widehat{\boldsymbol{c}}_{n,p}^{-1} g_0}, \qquad \hat{v}^0_{S,n,p} \equiv h_0^\intercal \widehat{\boldsymbol{c}}_{n,p} h_0,$$

$$\hat{v}_{T,n,p} \equiv (g_0^\intercal \widehat{\boldsymbol{c}}_{n,p}^{-1} g_0)^{-1}, \qquad \hat{\lambda}_{n,p} \equiv \sqrt{\hat{v}_{T,n,p}/\hat{v}^0_{S,n,p}}.$$

If $\hat{S}^*_{n,p} = A_{n,p}/\sqrt{\hat{v}^0_{S,n,p}}$ and $\hat{T}^*_{n,p} = \tilde{\xi}^0_{n,p}/\sqrt{\hat{v}_{T,n,p}}$, then the identities

$$\hat{\lambda}_{n,p}\hat{S}^*_{n,p}\hat{T}^*_{n,p} = \frac{\tilde{\xi}^0_{n,p} A_{n,p}}{\hat{v}^0_{S,n,p}}, \qquad \hat{\lambda}^2_{n,p}(\hat{T}^*_{n,p})^2 = \frac{(\tilde{\xi}^0_{n,p})^2}{\hat{v}^0_{S,n,p}}$$

hold exactly. Define the restricted jump estimator on the original scale by $\hat{J}^0_{n,p} \equiv \sqrt{\Delta_n}\tilde{\xi}^0_{n,p}$. Using (S14),

$$\begin{aligned}
\hat{J}^0_{n,p} &= \frac{g_0^\intercal \widehat{\boldsymbol{c}}_{n,p}^{-1} \Delta^n_{i(p)} Z}{g_0^\intercal \widehat{\boldsymbol{c}}_{n,p}^{-1} g_0} \\
&= J_p + \sqrt{\Delta_n}\frac{g_0^\intercal \widehat{\boldsymbol{c}}_{n,p}^{-1}(dJ_p e_2 + \mathcal{E}_{n,p} + o_{\mathbb{P}}(1))}{g_0^\intercal \widehat{\boldsymbol{c}}_{n,p}^{-1} g_0} = J_p + O_{\mathbb{P}}(\sqrt{\Delta_n}).
\end{aligned}$$

Notice that covariance-estimation error does not multiply the leading jump term: for every nonsingular matrix C, $g_0^\intercal C^{-1}(J_p g_0)/(g_0^\intercal C^{-1} g_0) = J_p$. Consistency of $\widehat{\boldsymbol{c}}_{n,p}$ therefore yields $\hat{v}^0_{S,n,p} \xrightarrow{\mathbb{P}} v_{S,p}$. Choosing the sign of the LM square root to agree with its score numerator, we obtain

$$K_{LM,n} \equiv \frac{\sum_{p=1}^{P} \hat{J}^0_{n,p} A_{n,p}/\hat{v}^0_{S,n,p}}{\left\{\sum_{p=1}^{P} (\hat{J}^0_{n,p})^2/\hat{v}^0_{S,n,p}\right\}^{1/2}} = \frac{1}{\sqrt{\mathcal{I}^*}} \sum_{p=1}^{P} \frac{J_p A_{n,p}}{v_{S,p}} + o_{\mathbb{P}}(1). \tag{S23}$$

We next expand the LTT–GLS statistic directly from its original-sample normal equation. Because $\widehat{\boldsymbol{c}}_{n,p} \xrightarrow{\mathbb{P}} \boldsymbol{c}_p$ and $\widetilde{\beta}_n \xrightarrow{\mathbb{P}} \beta_0$ under (S13), LTT's optimal weight in their equation (4.5) satisfies

$$w^*(\hat{\boldsymbol{c}}_{n,i(p)-}, \hat{\boldsymbol{c}}_{n,i(p)+}, \widetilde{\beta}_n) = \frac{1}{\hat{v}^{\mathrm{GLS}}_{S,n,p}} \xrightarrow{\mathbb{P}} \frac{1}{v_{S,p}}.$$

Moreover, (S14) gives $\Delta^n_{i(p)} X = J_p + O_{\mathbb{P}}(\sqrt{\Delta_n})$ and hence $\widehat{\mathcal{I}}_n \xrightarrow{\mathbb{P}} \mathcal{I}^*$. The WLS identity in (13) now gives, without approximation,

$$K_{\text{LTT-GLS},n} \equiv \sqrt{\frac{\widehat{\mathcal{I}}_n}{\Delta_n}}(\hat{\beta}^{\mathrm{GLS}}_n - \beta_0) \tag{S24}$$

$$= \frac{\sum_{p=1}^{P} (\Delta_{i(p)}^{n} X) A_{n,p} / \hat{v}_{S,n,p}^{\mathrm{GLS}}}{\sqrt{\widehat{\mathcal{I}}_n}} = \frac{1}{\sqrt{\mathcal{I}^*}} \sum_{p=1}^{P} \frac{J_p A_{n,p}}{v_{S,p}} + o_{\mathbb{P}}(1). \tag{S25}$$

Equations (S23) and (S25) prove (S18). Their common leading term is $O_{\mathbb{P}}(1)$ by (S22), so squaring proves (S19).

Conditionally on $\mathcal{F}$, (S22) implies

$$\frac{1}{\sqrt{\mathcal{I}^*}} \sum_{p=1}^{P} \frac{J_p A_{n,p}}{v_{S,p}} \xrightarrow{\mathcal{L}\text{-}s} \mathcal{N}(d\sqrt{\mathcal{I}^*}, 1),$$

because the conditional variance of $\sum_p J_p h_0^\intercal \mathcal{E}_p / v_{S,p}$ is $\sum_p J_p^2 / v_{S,p} = \mathcal{I}^*$. Part (b) follows by squaring. Since the limiting distribution is continuous at c_α, the difference between the two rejection indicators converges to zero in probability.

Finally, LTT's optimal asymptotic variance in their equation (4.8) is

$$\Sigma^* = \left\{ \sum_{p=1}^{P} \frac{2J_p^2}{h_0^\intercal (\boldsymbol{c}_{\mathfrak{a}_p -} + \boldsymbol{c}_{\mathfrak{a}_p}) h_0} \right\}^{-1} = \left(\sum_{p=1}^{P} \frac{J_p^2}{v_{S,p}} \right)^{-1} = (\mathcal{I}^*)^{-1},$$

where the second equality uses Assumption 4. Under the LAMN restrictions of LTT's semi-parametric efficiency analysis, their Theorem 3 shows that this bound is sharp. Equivalently, profiling the unrestricted nuisance jump size J_p from the pth conditional Gaussian likelihood gives the Schur-complement information

$$J_p^2 \left\{ e_2^\intercal \boldsymbol{c}_p^{-1} e_2 - \frac{(e_2^\intercal \boldsymbol{c}_p^{-1} g_0)^2}{g_0^\intercal \boldsymbol{c}_p^{-1} g_0} \right\} = \frac{J_p^2}{h_0^\intercal \boldsymbol{c}_p h_0} = \frac{J_p^2}{v_{S,p}}.$$

Summing over announcements yields $\mathcal{I}^*$, which proves part (c). □

Remark 1 (Scope of Proposition S1)**.** The proposition concerns the heterogeneous LM statistic in (27); $LM^h = Q_{ST}^2 / Q_T$ shares its strong-ID limit only if λ_p is constant on the support of $(J_p)_p$. Equation (S19) is local, not a finite-sample or fixed-alternative identity: under fixed alternatives, the restricted jump proxy contains $1 + (\beta - \beta_0)\theta_p$, which may cause cross-event score cancellation. Price–volatility cojumps are also excluded because LTT's interpretation of (S21) as the sharp semiparametric bound and the Gaussian independence in (27) no longer apply.

Proposition S2. In the Gaussian limit experiment (21) with $P = |\mathcal{P}_{\mathfrak{a}_B}|$, and let $\lambda_p > 0$, θ_p, $\xi^*_{\mathfrak{a}_p} \equiv \xi_{\mathfrak{a}_p}/\sqrt{v_{T,p}}$ and $\delta = \beta - \beta_0$ be as in Theorem 4, $\Lambda = \mathrm{diag}((\lambda_p)_p)$, and $\pi(\tau) \equiv \mathbb{P}\{\chi^2_1(\tau) > \chi^2_{1,1-\alpha}\}$. Consider strong or semi-strong identification drifting parameters and local alternative

$$\xi^*_{\mathfrak{a}_p} = r_n u_p, \qquad r_n \to \infty, \qquad \delta_n = d/r_n, \qquad d \in \mathbb{R}, \tag{S26}$$

for fixed $u = (u_1, \ldots, u_P)^\intercal \in \mathbb{R}^P_*$, and define

$$w_p \equiv \frac{u_p^2}{\sum_q u_q^2}, \qquad \mathcal{R}(w,\lambda) \equiv \frac{\big(\sum_p w_p\lambda_p\big)^2}{\sum_p w_p \lambda_p^2}, \qquad \tau \equiv d^2 \sum_p \lambda_p^2 u_p^2. \tag{S27}$$

(a) Along the local alternative (S26), the tests ϕ_{LM} and ϕ_{CLR} with critical value computed conditionally on T^* are asymptotically uniformly most powerful unbiased, and the local power functions satisfy $\lim_n \mathbb{E}_{\delta_n,\xi^*}[\phi_{LM}] = \lim_n \mathbb{E}_{\delta_n,\xi^*}[\phi_{CLR}] = \pi(\tau)$.

(b) The statistics $LM^h = Q^2_{ST}/Q_T$ and LR^h of Theorem 4(b), each with its exact conditional critical value, satisfy $\lim_n \mathbb{E}_{\delta_n,\xi^*}[\phi^h] = \pi\big(\mathcal{R}(w,\lambda)\cdot\tau\big)$.

(c) With $\lambda_{\min} \equiv \min_p \lambda_p$, $\lambda_{\max} \equiv \max_p \lambda_p$ (the extrema may be restricted to $\{p : w_p > 0\}$),

$$1 \;\ge\; \mathcal{R}(w,\lambda) \;\ge\; \mathcal{R}_\lambda \equiv \frac{4\lambda_{\min}\lambda_{\max}}{(\lambda_{\min}+\lambda_{\max})^2}, \tag{S28}$$

If $\lambda_{\min} < \lambda_{\max}$, equality on the left holds iff λ_p is constant on $\{p : w_p > 0\}$, and equality on the right holds iff w puts mass $\lambda_{\max}/(\lambda_{\min}+\lambda_{\max})$ on $\{p : \lambda_p = \lambda_{\min}\}$ and the remainder on $\{p : \lambda_p = \lambda_{\max}\}$. If $\lambda_{\min} = \lambda_{\max}$, both inequalities are equalities. Thus $\mathcal{R}_\lambda$ is sharp uniformly over the unknown jump-size configuration.

(d) Since π is strictly increasing,

$$0 \;\le\; \lim_n \mathbb{E}[\phi_{CLR}] - \lim_n \mathbb{E}[\phi_{CLR^h}] = \pi(\tau) - \pi\big(\mathcal{R}(w,\lambda)\tau\big) \;\le\; G(\mathcal{R}_\lambda) \;\le\; C_\alpha \log(1/\mathcal{R}_\lambda), \tag{S29}$$

where $G : (0,1] \to [0, 1-\alpha)$ is defined by $G(\kappa) \equiv \sup_{t\ge 0}\{\pi(t) - \pi(\kappa t)\}$ and is strictly decreasing, with $G(1) = 0$, and $C_\alpha \equiv \sup_{t\ge0} t\,\pi'(t)$; at $\alpha = 0.05$, $C_\alpha \approx 0.435$ and, e.g., $G(0.99) \approx 0.004$, $G(0.9) \approx 0.046$, $G(0.75) \approx 0.124$, $G(0.5) \approx 0.292$, $G(0.19) \approx 0.609$.

Proof of Proposition S2. (a) Write the generic term of (28) as $f_p(\delta') = \{\delta'\lambda_p S_p^* + (1 + \delta'\theta_p)T_p^*\}^2/\{\delta'^2\lambda_p^2 + (1+\delta'\theta_p)^2\}$, so $LR = \sup_{\delta'\in\overline{\mathbb{R}}}\sum_p f_p(\delta') - Q_T$. Two elementary facts drive the argument. First, for every δ' (including $\delta' = \pm\infty$), put $a = \delta'\lambda_p$, $b = 1 + \delta'\theta_p$, and $D = a^2 + b^2$. Then $g_p(\delta') = a^2/D$, $b^2/D = 1 - g_p(\delta')$, and $|ab|/D = \{g_p(\delta')(1 - g_p(\delta'))\}^{1/2}$. Expanding the square gives

$$\begin{aligned} f_p(\delta') - (T_p^*)^2 &= \frac{(aS_p^* + bT_p^*)^2 - (a^2+b^2)(T_p^*)^2}{D} = \frac{a^2[(S_p^*)^2 - (T_p^*)^2] + 2abS_p^*T_p^*}{D} \\ &= g_p(\delta')\{(S_p^*)^2 - (T_p^*)^2\} + 2\frac{ab}{D}S_p^*T_p^*. \end{aligned}$$

The desired bound is just a control of the cross term: indeed,

$$\begin{aligned} &2(S_p^*)^2 - \tfrac{1}{2}g_p(\delta')(T_p^*)^2 - \{f_p(\delta') - (T_p^*)^2\} \\ &\quad = \{2 - g_p(\delta')\}(S_p^*)^2 + \tfrac{1}{2}g_p(\delta')(T_p^*)^2 - 2\frac{ab}{D}S_p^*T_p^* \;\geq\; 0, \end{aligned}$$

because the discriminant of the quadratic polynomial is non-positive. Hence

$$f_p(\delta') - (T_p^*)^2 \;\leq\; 2(S_p^*)^2 - \tfrac{1}{2}\, g_p(\delta')(T_p^*)^2, \qquad g_p(\delta') \equiv \frac{\delta'^2\lambda_p^2}{\delta'^2\lambda_p^2 + (1+\delta'\theta_p)^2}, \tag{S30}$$

where g_p is continuous on $\overline{\mathbb{R}}$ and vanishes only at $\delta' = 0$. To see why this concentrates the profiled maximizer near zero, sum (S30) over p:

$$\sum_p f_p(\delta') - Q_T \leq 2\sum_p (S_p^*)^2 - \frac{1}{2}\sum_p g_p(\delta')(T_p^*)^2.$$

The scale λ_p in the first display below comes only from rewriting the mean of S_p^* in terms of the standardized jump size:

$$\mathbb{E}_{\delta,\xi^*}[S_p^*] = \frac{\delta\xi_{\mathfrak{a}_p}}{\sqrt{\upsilon_{S,p}}} = \delta\sqrt{\frac{\upsilon_{T,p}}{\upsilon_{S,p}}}\xi^*_{\mathfrak{a}_p} = \delta\lambda_p\xi^*_{\mathfrak{a}_p}, \qquad \mathbb{E}_{\delta,\xi^*}[T_p^*] = (1+\delta\theta_p)\xi^*_{\mathfrak{a}_p}.$$

Thus, along (S26),

$$S_p^* = d\lambda_p u_p + O_{\mathbb{P}}(1), \qquad T_p^* = r_n u_p + d\theta_p u_p + O_{\mathbb{P}}(1) = r_n u_p + O_{\mathbb{P}}(1),$$

so $\sum_p (S_p^*)^2 = O_{\mathbb{P}}(1)$. The negative term, by contrast, is large once $|\delta'|$ is larger than the local scale $1/r_n$. Indeed, on the local annulus $M/r_n \leq |\delta'| \leq \varepsilon$, for ε small enough, $g_p(\delta') \geq c\,\delta'^2\lambda_p^2$ uniformly in p for some $c > 0$, and therefore

$$\sum_p g_p(\delta')(T_p^*)^2 \geq c\, r_n^2\delta'^2\sum_p \lambda_p^2 u_p^2 + o_{\mathbb{P}}(r_n^2\delta'^2) \geq cM^2\sum_p \lambda_p^2 u_p^2 + o_{\mathbb{P}}(1).$$

Outside any fixed neighborhood of zero, continuity of g_p and $u \neq 0$ make $\sum_p g_p(\delta')u_p^2$ bounded away from zero, so the same negative term is of order r_n^2. Hence, after choosing M large, with probability approaching one,

$$\sup_{|\delta'| \geq M/r_n} \left\{ \sum_p f_p(\delta') - Q_T \right\} < 0.$$

Since the objective equals zero at $\delta' = 0$, the profiled supremum is attained, up to an $o_{\mathbb{P}}(1)$ error, inside $|\delta'| \leq M/r_n$. Second, the exact identity

$$f_p(\delta') - (T_p^*)^2 = \frac{\delta' \lambda_p \big\{ 2(1 + \delta'\theta_p) S_p^* T_p^* + \delta' \lambda_p \big((S_p^*)^2 - (T_p^*)^2 \big) \big\}}{\delta'^2 \lambda_p^2 + (1 + \delta'\theta_p)^2},$$

Evaluating at $\delta' = h/r_n$, uniformly over $|h| \leq M$, the denominator $\delta'^2 \lambda_p^2 + (1 + \delta'\theta_p)^2 = 1 + 2h\theta_p/r_n + h^2(\theta_p^2 + \lambda_p^2)/r_n^2 = 1 + o(1)$ and

$$\begin{aligned} \frac{h\lambda_p}{r_n} 2\Big(1 + \frac{h\theta_p}{r_n}\Big) S_p^* T_p^* &= 2h\lambda_p u_p S_p^* + o_{\mathbb{P}}(1), \\ \frac{h^2 \lambda_p^2}{r_n^2} \big((S_p^*)^2 - (T_p^*)^2 \big) &= -h^2 \lambda_p^2 u_p^2 + o_{\mathbb{P}}(1), \end{aligned}$$

because $T_p^*/r_n \xrightarrow{\mathbb{P}} u_p$ and $(S_p^*)^2/r_n^2 \xrightarrow{\mathbb{P}} 0$. Summing over p yields

$$\sum_p f_p(h/r_n) - Q_T = 2h\langle \Lambda u, S^* \rangle - h^2 \|\Lambda u\|^2 + o_{\mathbb{P}}(1).$$

Maximizing the limiting quadratic in h gives

$$LR = \frac{\langle \Lambda u, S^* \rangle^2}{\|\Lambda u\|^2} + o_{\mathbb{P}}(1).$$

And note

$$LM = \frac{\big(\sum_p \lambda_p S_p^* T_p^* \big)^2}{\sum_p \lambda_p^2 (T_p^*)^2} = \frac{\big(r_n \sum_p \lambda_p u_p S_p^* + o_{\mathbb{P}}(r_n) \big)^2}{r_n^2 \sum_p \lambda_p^2 u_p^2 + o_{\mathbb{P}}(r_n^2)} = \frac{\langle \Lambda u, S^* \rangle^2}{\|\Lambda u\|^2} + o_{\mathbb{P}}(1).$$

Thus $LR = LM + o_{\mathbb{P}}(1)$. Since $S^* \sim \mathcal{N}(d\Lambda u, I)$ along (S26), $LM \xrightarrow{\mathcal{L}} \chi_1^2(\tau)$ with noncentrality parameter $\tau = d^2 \|\Lambda u\|^2$. The same expansion under H_0 conditionally on $T^* = t_n^*$ with $t_n^*/r_n \to u$ yields $LR \mid T^* \xrightarrow{\mathcal{L}} \chi_1^2$ and hence $cv_{CLR}^{\alpha}(T^*) \xrightarrow{\mathbb{P}} \chi_{1,1-\alpha}^2$; the LM test needs no argument since $LM \mid T^* \sim \chi_1^2$ exactly. It remains to spell out the efficiency claim. Along (S26), conditioning on T^* asymptotically fixes the nuisance direction, because $T^*/r_n \xrightarrow{\mathbb{P}} u$. The relevant limit experiment for the local parameter d is therefore the Gaussian shift

$$S^* = d\,\Lambda u + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I_P).$$

All information about d is contained in the scalar projection

$$Z_u \equiv \frac{\langle \Lambda u, S^* \rangle}{\|\Lambda u\|}, \qquad Z_u \sim \mathcal{N}(d\|\Lambda u\|, 1),$$

since the component of S^* orthogonal to Λu has a distribution that does not depend on d. Equivalently, the likelihood ratio in the limit experiment is

$$\log \frac{\mathrm{d}\mathbb{P}_d}{\mathrm{d}\mathbb{P}_0}(S^*) = d\langle \Lambda u, S^* \rangle - \frac{1}{2} d^2 \|\Lambda u\|^2.$$

Thus the efficient information is $\|\Lambda u\|^2$ and the local noncentrality is $d^2\|\Lambda u\|^2 = \tau = \lim_n \delta_n^2 \sum_p \xi_{\mathfrak{a}_p}^2 / v_{S,p}$. For the two-sided normal mean problem testing $H_0 : \eta = 0$ against $H_1 : \eta \neq 0$ given $Z_u \sim \mathcal{N}(\eta, 1)$, we check the two conditions of Theorem 4.4.1 of Lehmann and Romano (2005),

$$\mathbb{P}_0(Z_u < z_{\alpha/2} \text{ or } Z_u > z_{1-\alpha/2}) = \alpha$$
$$\mathbb{E}_{H_0}[Z_u 1\{Z_u < z_{\alpha/2} \text{ or } Z_u > z_{1-\alpha/2}\}] = 0$$

where $z_{\alpha/2}$ and $z_{1-\alpha/2}$ are the lower and upper $\alpha/2$-quantiles of the standard normal distribution, respectively. Therefore, the equivalent chi-square test $1\{Z_u^2 > \chi^2_{1,1-\alpha}\}$ is the uniformly most powerful unbiased (UMPU)x level-α test with the local power function $\mathbb{P}\{\chi_1^2(\tau) > \chi^2_{1,1-\alpha}\} = \pi(\tau)$.

(b) In the semi-strong/strong identification cases, we have $Q_T \to \infty$. And $LR^h = LM^h + o_{\mathbb{P}}(1)$ and $cv^{\alpha}_{CLR^h}(Q_T) \to \chi^2_{1,1-\alpha}$ from (30); the homogeneous statistics project S^* onto $T^*/\|T^*\| \to u/\|u\|$ rather than $\Lambda u/\|\Lambda u\|$, with noncentrality $d^2\langle u, \Lambda u\rangle^2/\|u\|^2 = \mathcal{R}(w,\lambda)\tau$ by (S27).

(c) With $\lambda_{\min} \equiv \min_p \lambda_p$ and $\lambda_{\max} \equiv \max_p \lambda_p$, we have

$$\underline{\mathcal{R}}_\lambda \equiv \frac{4\lambda_{\min}\lambda_{\max}}{(\lambda_{\min}+\lambda_{\max})^2} \overset{(i)}{\leq} \mathcal{R}(w,\lambda) = \frac{(\sum_p w_p \lambda_p)^2}{\sum_p w_p \lambda_p^2} \overset{(ii)}{\leq} 1. \tag{S31}$$

To prove inequality (i), let $\bar{\lambda} \equiv \sum_p w_p \lambda_p$. Since $\lambda_{\min} > 0$, $\sum_p w_p = 1$, and $\lambda_{\min} \leq \lambda_p \leq \lambda_{\max}$ on the support of w, we have

$$0 \leq (\lambda_p - \lambda_{\min})(\lambda_{\max} - \lambda_p) \quad \Longrightarrow \quad \lambda_p^2 \leq (\lambda_{\min}+\lambda_{\max})\lambda_p - \lambda_{\min}\lambda_{\max}.$$

Multiplying by w_p and summing over p therefore yields

$$\sum_p w_p\lambda_p^2 \le (\lambda_{\min}+\lambda_{\max})\bar{\lambda} - \lambda_{\min}\lambda_{\max}.$$

Consequently,

$$\begin{aligned}
\mathcal{R}(w,\lambda) \ge \frac{\bar{\lambda}^2}{(\lambda_{\min}+\lambda_{\max})\bar{\lambda}-\lambda_{\min}\lambda_{\max}}&, \\
\frac{\bar{\lambda}^2}{(\lambda_{\min}+\lambda_{\max})\bar{\lambda}-\lambda_{\min}\lambda_{\max}} - \frac{4\lambda_{\min}\lambda_{\max}}{(\lambda_{\min}+\lambda_{\max})^2}& \\
= \frac{\big((\lambda_{\min}+\lambda_{\max})\bar{\lambda}-2\lambda_{\min}\lambda_{\max}\big)^2}{(\lambda_{\min}+\lambda_{\max})^2\big((\lambda_{\min}+\lambda_{\max})\bar{\lambda}-\lambda_{\min}\lambda_{\max}\big)}& \ge 0.
\end{aligned}$$

The denominators are positive because $\lambda_{\min} > 0$ and $\bar{\lambda} \in [\lambda_{\min}, \lambda_{\max}]$. This proves inequality (i). If $\lambda_{\min} < \lambda_{\max}$, equality holds iff $\lambda_p \in \{\lambda_{\min}, \lambda_{\max}\}$ on the support of w and

$$\bar{\lambda} = \frac{2\lambda_{\min}\lambda_{\max}}{\lambda_{\min}+\lambda_{\max}},$$

or, equivalently, the total weights assigned to $\lambda_{\min}$ and $\lambda_{\max}$ are $\lambda_{\max}/(\lambda_{\min}+\lambda_{\max})$ and $\lambda_{\min}/(\lambda_{\min}+\lambda_{\max})$, respectively. If $\lambda_{\min} = \lambda_{\max}$, the bound holds with equality trivially. Finally, weighted Cauchy–Schwarz gives

$$\left(\sum_p w_p\lambda_p\right)^2 \le \left(\sum_p w_p\right)\left(\sum_p w_p\lambda_p^2\right) = \sum_p w_p\lambda_p^2,$$

which proves $\mathcal{R}(w,\lambda) \le 1$. Equality in Cauchy–Schwarz holds iff λ_p is constant on the support of w.

(d) π is strictly increasing because $\chi_1^2(\tau)$ is stochastically increasing in τ. For the log bound, $\pi(t) - \pi(\kappa t) = \int_\kappa^1 t\,\pi'(ts)\,ds \le C_\alpha \log(1/\kappa)$, using $\pi'(\tau) = \frac{1}{2}\{\mathbb{P}(\chi_3^2(\tau) > \chi_{1,1-\alpha}^2) - \mathbb{P}(\chi_1^2(\tau) > \chi_{1,1-\alpha}^2)\}$. □

S3. Additional Simulation Results

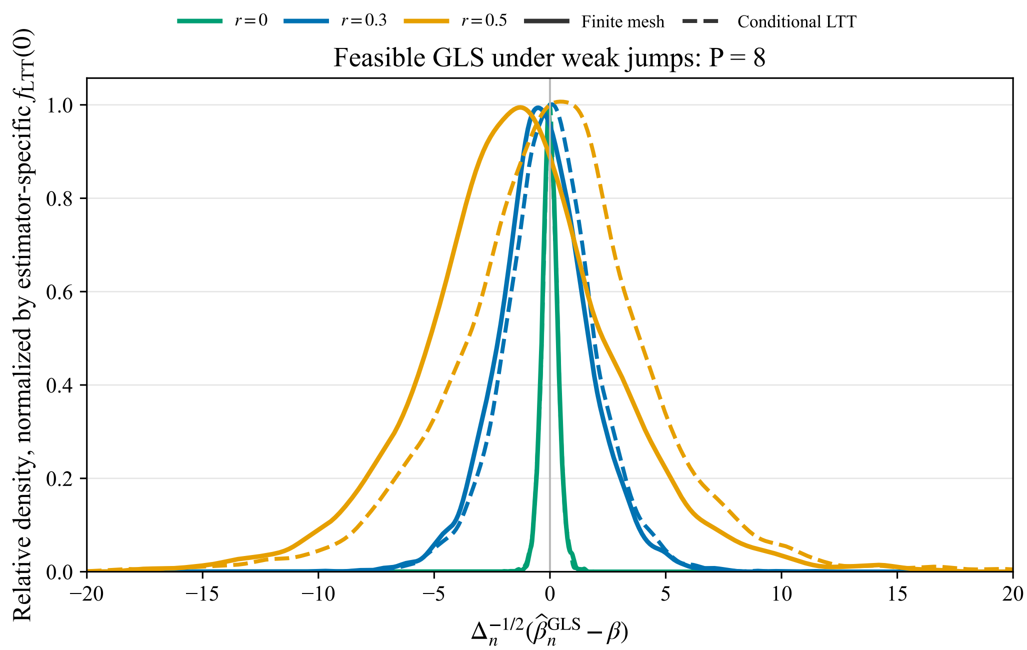
Feasible GLS with Weak Jumps

Figure S1 uses the density calibration, rather than the separate main-simulation table: one year of 250 trading days with $\Delta_n = 1/390$ and eight scheduled announcements. Every X

jump is $s_p(1+0.5u_p)\Delta_n^r$, where s_p is a random sign, $u_p \sim N(0,1)$, and $r \in \{0, 0.3, 0.5\}$. The remaining parameters are $\beta = 1$, $\beta^c = 0.89$, $\rho = -0.7$, idiosyncratic-Y jump intensity 15, and exponential announcement-time volatility jumps with mean 0.5.

For each rate, the figure uses 5,000 paths, truncated pre/post large-k covariance estimates with $k = 3\lceil \Delta_n^{-1/2} \rceil = 60$, preliminary OLS, and feasible weights $2/(\widehat{v}_{S,p}^{-}+\widehat{v}_{S,p}^{+})$. Each path uses 8,000 conditional LTT draws; the displayed pooled density stores one conditional draw from each outer replication. The four OLS/GLS finite-mesh/conditional-LTT samples share one rate-specific bandwidth, and each estimator pair is divided by that estimator's conditional-LTT density at zero. The comparison remains calibration-specific and does not imply a universal ordering of OLS and GLS robustness.

Figure S1: Feasible GLS under weak jumps with eight announcements



Notes: The figure compares finite-mesh draws of $\Delta_n^{-1/2}(\widehat{\beta}_n^{\mathrm{GLS}} - \beta)$ with pooled sample-specific conditional LTT reference draws for $r = 0, 0.3, 0.5$. Solid curves are feasible-GLS finite-mesh distributions and dashed curves are the corresponding conditional LTT reference laws. Within each rate, the four OLS/GLS finite-mesh and conditional-LTT samples are Gaussian-kernel smoothed using one common rate-specific bandwidth. Each GLS pair is divided by its conditional-LTT density at zero. All eight scheduled announcements shrink at the displayed rate; feasible GLS uses preliminary OLS and same-replication large-k covariance estimates. The comparison is calibration-specific and does not imply a universal ordering of OLS and GLS robustness.

Comparing Aggregations of AR Tests

Figure S2 compares four natural ways to aggregate the per-announcement Anderson–Rubin residuals under the finite-sample high-frequency data-generating process used in the simulation study. Unlike the limit-experiment calculations in the main text, the figure simulates the full intraday paths, including stochastic volatility, announcement-time volatility jumps, announcement return jumps, and idiosyncratic jumps in Y. For each candidate β_0, we form

$$S_p(\beta_0) = \frac{\Delta Y_{\tau_p} - \beta_0 \Delta X_{\tau_p}}{\{\Delta_n h(\beta_0)' \widehat{C}_p h(\beta_0)\}^{1/2}}, \qquad h(\beta_0) = (-\beta_0, 1)',$$

$$S_1 \equiv \sum_{p \in \mathcal{P}_{\mathfrak{a}_B}} \left| \frac{S_{n,p}(\beta_0)}{\sqrt{\hat{\nu}_p}} \right|, \quad S_2 \equiv \sum_{p \in \mathcal{P}_{\mathfrak{a}_B}} \left[\frac{S_{n,p}(\beta_0)}{\sqrt{\hat{\nu}_p}} \right]^2, \quad S_{\sup} \equiv \max_{p \in \mathcal{P}_{\mathfrak{a}_B}} \left| \frac{S_{n,p}(\beta_0)}{\sqrt{\hat{\nu}_p}} \right| \tag{S32}$$

where $\widehat{C}_p$ is the average of the large-K pre- and post-announcement spot covariance estimates. The sign-invariant AR tests aggregate the components as

$$Q_1^{\mathrm{AR}}(\beta_0) = \sum_p |S_p(\beta_0)|, \quad Q_2^{\mathrm{AR}}(\beta_0) = Q_S = \sum_p S_p(\beta_0)^2, \quad Q_\infty^{\mathrm{AR}}(\beta_0) = \max_p |S_p(\beta_0)|.$$

We also include a signed self-normalized t-ratio. Let

$$e_p(\beta_0) = \Delta_n^{-1/2}\{\Delta Y_{\tau_p} - \beta_0 \Delta X_{\tau_p}\}, \qquad \bar{e}(\beta_0) = P^{-1} \sum_{p=1}^{P} e_p(\beta_0).$$

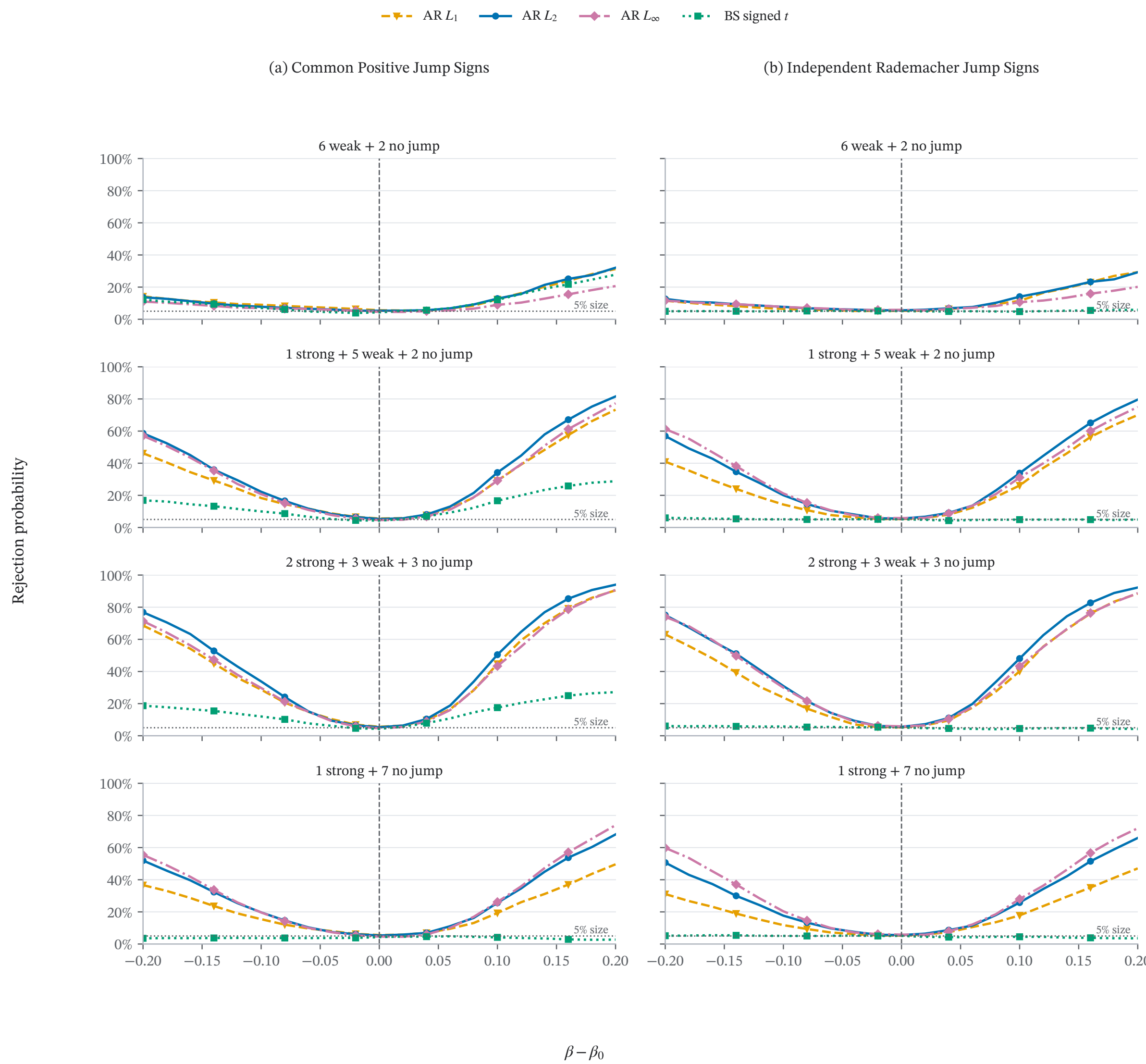
The Bakirov–Szekely statistic is

$$T_{\mathrm{BS}}(\beta_0) = \frac{\sqrt{P}\, \bar{e}(\beta_0)}{\left\{(P-1)^{-1} \sum_{p=1}^{P} [e_p(\beta_0) - \bar{e}(\beta_0)]^2\right\}^{1/2}}.$$

This is a natural benchmark: under H_0, conditional on the local volatility path and jump-timing variables, the $e_p(\beta_0)$ are independent Gaussian variables with common mean zero and heterogeneous scales. The robustness result of Bakirov and Szekely (2005) therefore gives size control for the ordinary Student t critical value in this Gaussian-scale-mixture setting. The statistic is therefore mean-targeted rather than omnibus: sign cancellation is exactly what a test of the average residual should exploit, but it also means that the statistic answers a different question from an AR test of the full vector of announcement residuals.

The eight panels separate two distinct dimensions of the alternative. The rows retain the four jump-strength designs used in the simulation section: six weak plus two no-jump announcements; one strong plus five weak plus two no-jump announcements; two strong plus three weak plus three no-jump announcements; and one strong plus seven no-jump announcements. The columns then vary only the sign geometry, comparing common positive signs with independent Rademacher signs. This comparison shows why the baseline procedure uses the L_2 AR aggregation. The signed t-ratio is informative as a diagnostic for common signed drift or news-day factors, and it gains power when the average residual shift is nonzero. When signs cancel, its near-size behavior is appropriate for a mean test, but it is weak for rejecting false β_0 values that generate offsetting announcement-level residuals. The L_∞ statistic is attractive for sparse alternatives, whereas L_2 provides the most stable compromise across the original four DGPs and the two sign designs.

Figure S2: Power of AR Aggregations under Alternative Jump-Sign Designs



Notes: Rejection probabilities use the full one-minute DGP with $P = 8$, $r = 0.5$, $\beta = 1$, and $\beta_c = 0.89$. Active price jumps coincide with positive announcement-time volatility jumps, whereas no-jump announcements retain only the volatility jump. Weak jumps have scale $5\Delta_n^r$, and strong jumps are four times larger. Large-K pre/post covariance estimates normalize the L_q-AR statistics. Each subfigure uses 2,000 null and 2,000 alternative replications; the horizontal reference marks 5% size.

Simulated intraday trajectories

Figure S3: The intraday trajectory of X and Y processes at announcement days when $r = 0$.

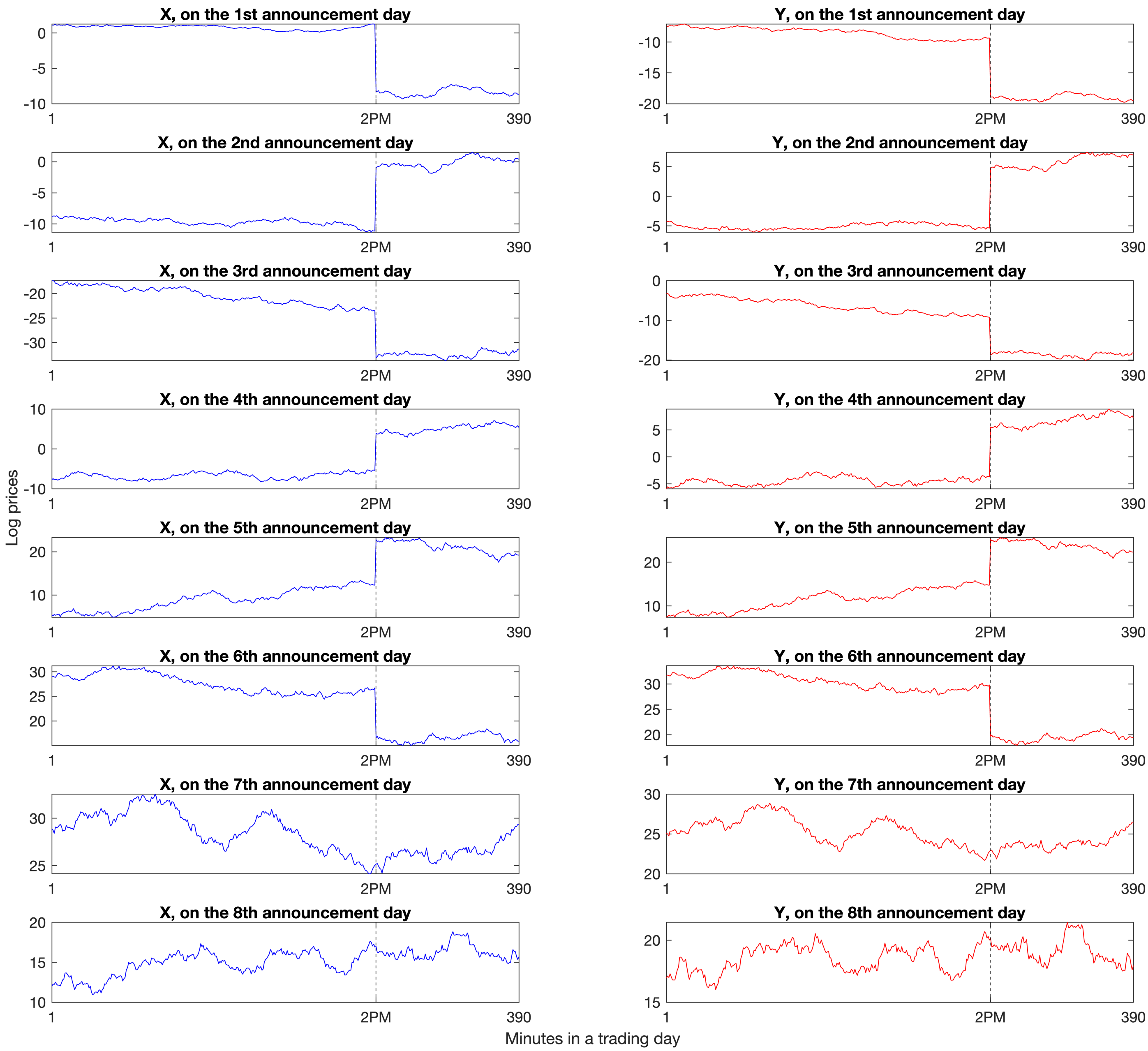
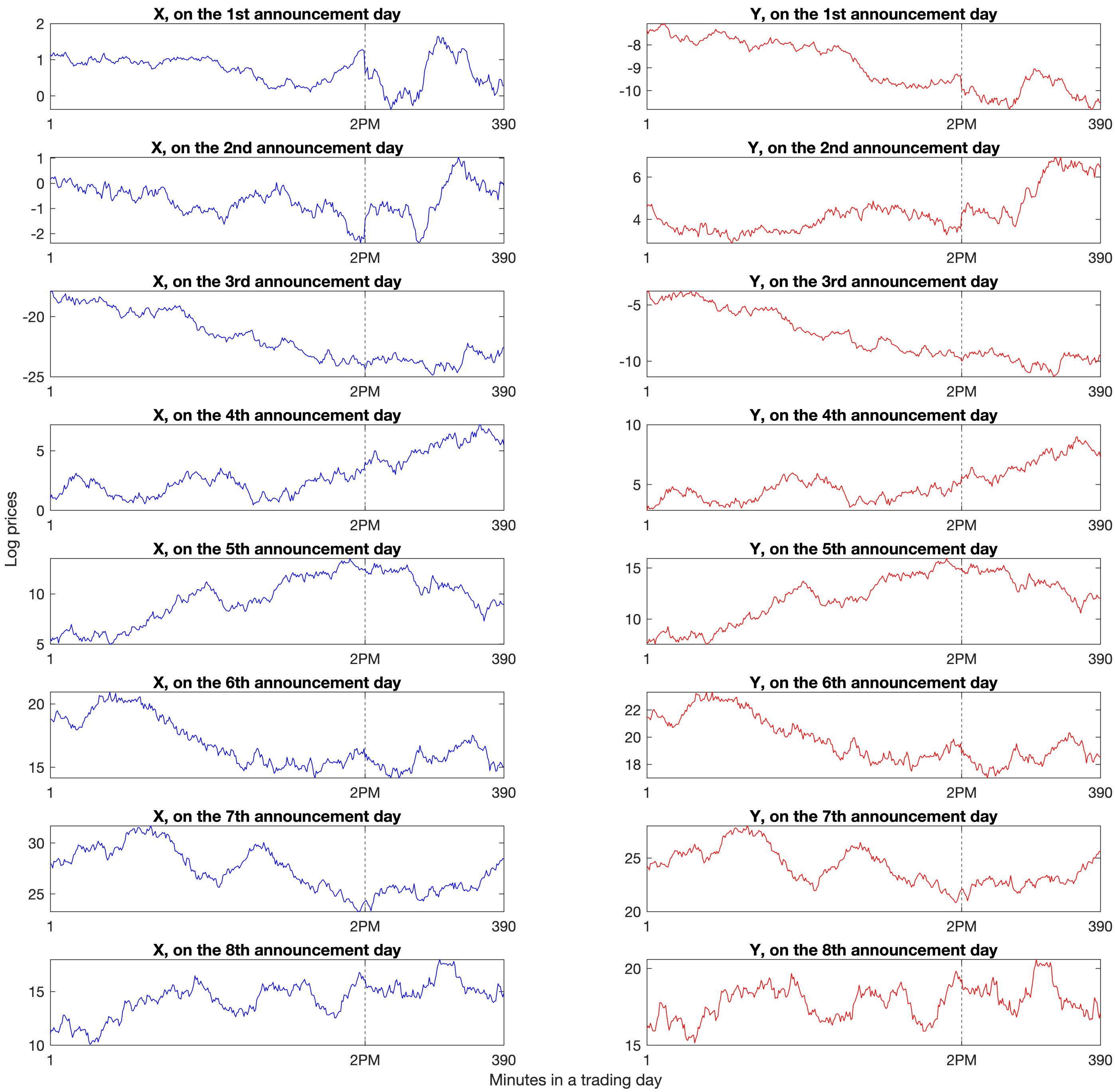


Figure S4: The intraday trajectory of X and Y processes at announcement days when $r = 0.5$.



Notes. Sampling is at the one-minute frequency, with 390 minutes per trading day. Figure S3 features jumps on the first six announcement days and none on the last two. Figure S4 features news-based jumps shrinking at rate $n^{-1/2}$. Each row plots the X and Y processes for a common announcement day.

S4. Additional Empirical Results

IRCI using Drift-Robust Volatility Estimators

Grouping FOMC Announcements by the Target-rate Decision

As a robustness check on calendar-time grouping, we group FOMC announcements by the realized target-rate decision,

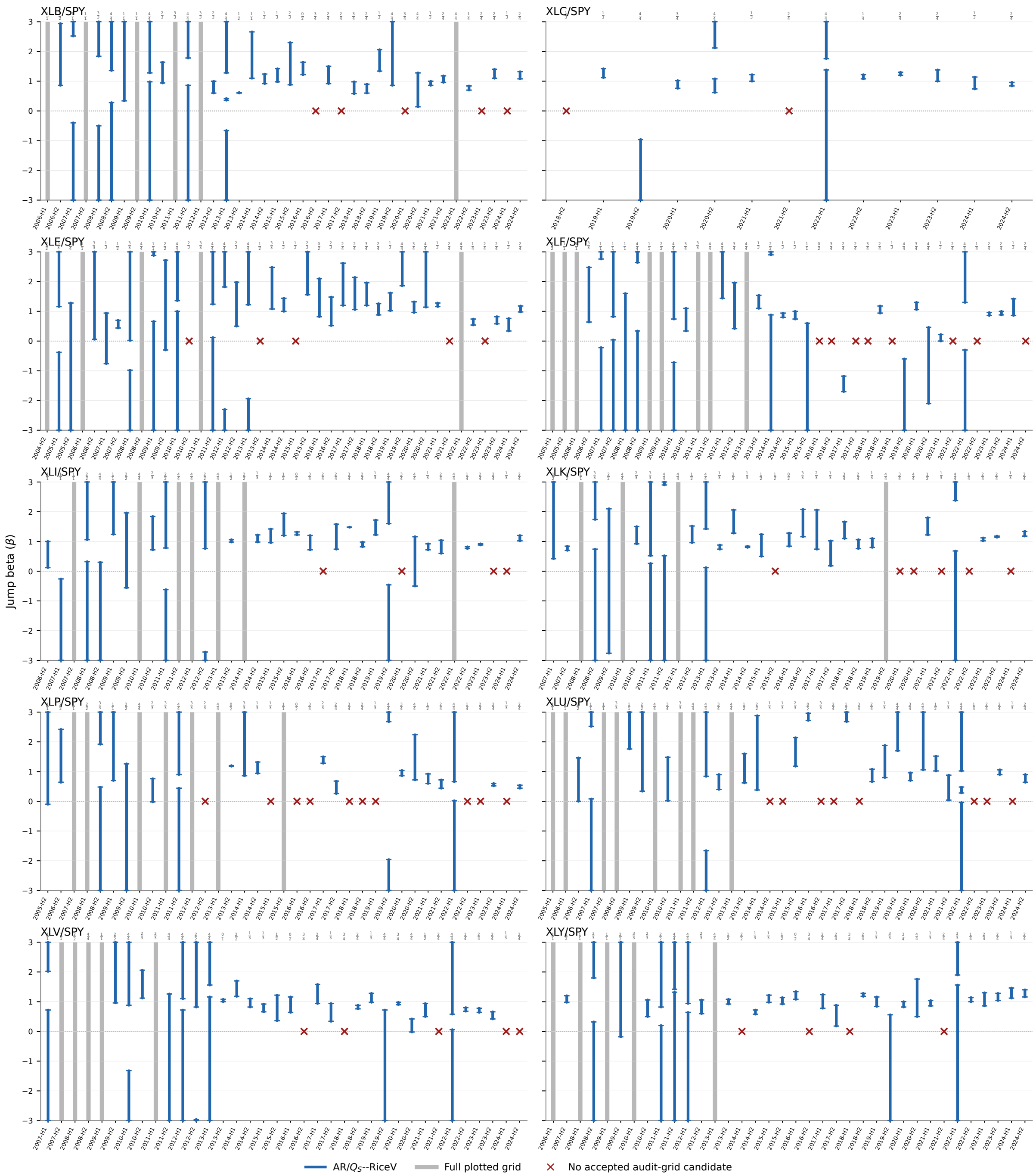
$$\Delta\mathrm{FFR}_p = \mathrm{Actual}_p^{\mathrm{upper}} - \mathrm{Prior}_p^{\mathrm{upper}},$$

measured in basis points, and invert the AR–RiceV test separately within each decision-size bin. Figure S6 reports the 95% confidence sets. The two empirically large bins, unchanged target rates (0bp) and standard hikes (+25bp), continue to produce many empty confidence sets: at the asset-by-bin level, 9 out of 10 sector ETFs are empty in the 0bp bin and 10 out of 10 are empty in the +25bp bin. Hence the empty-set pattern is not only an artifact of mixing rate cuts, holds, and hikes in a calendar window.

Figure S7 repeats the same exercise at the 90% confidence level. The intervals shrink, as expected, but the qualitative pattern is unchanged: the 0bp bin remains empty for 9 out of 10 sector ETFs, and the +25bp bin remains empty for all 10 sector ETFs.

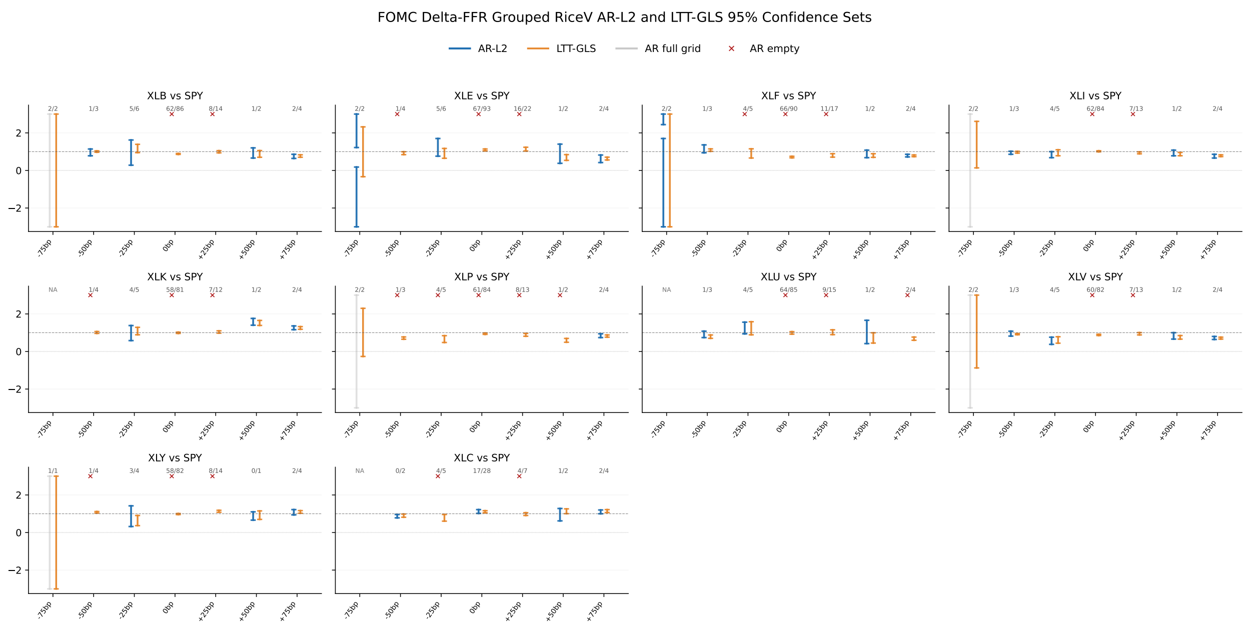
B.3. Additional FOMC and macro-announcement figures

Figure S5: Half-year FOMC jump-beta confidence sets: AR/Q_S–RiceV



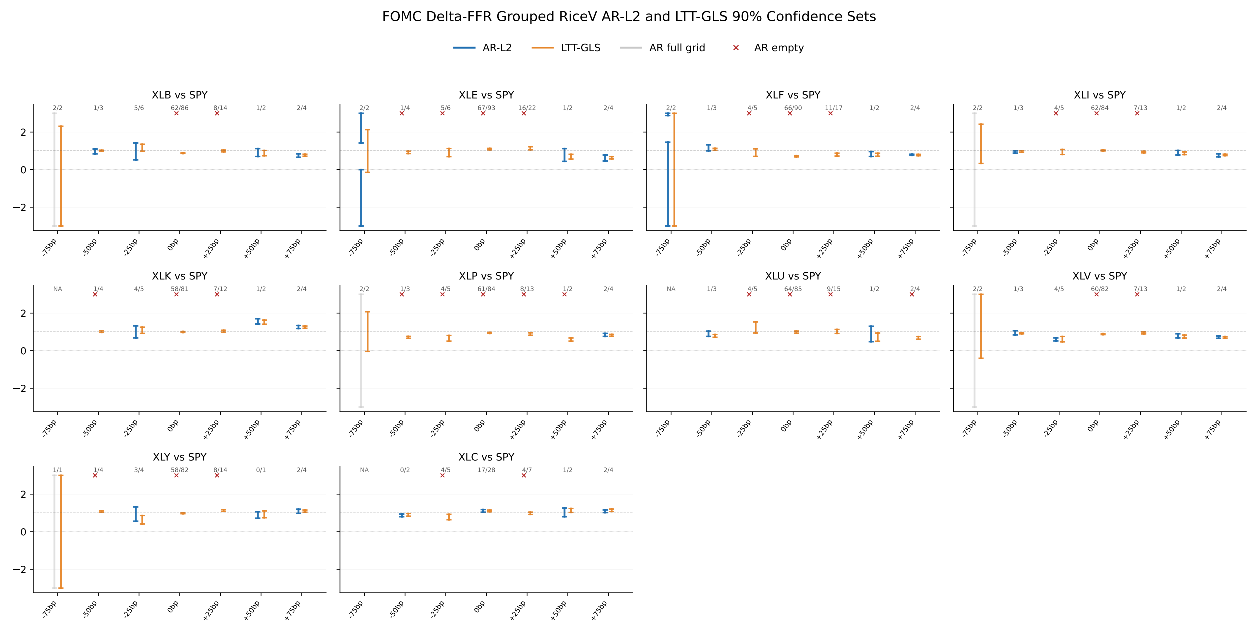
Notes: Blue: 95% AR/Q_S confidence sets using RiceV volatility estimators. Gray: the full display grid is accepted. Red: no candidate on the audit grid is accepted. Fractions report weak/valid FOMC announcements. Boundary arrows indicate off-scale acceptance.

Figure S6: AR-RiceV confidence sets around FOMC announcements grouped by ΔFFR



Notes: The figure reports 95% AR–RiceV confidence sets for sector ETF jump betas relative to SPY around FOMC announcements, grouping events by the realized change in the FOMC target-rate upper bound, $\Delta\text{FFR} = \text{Actual}^{\text{upper}} - \text{Prior}^{\text{upper}}$, measured in basis points. Blue line segments denote nonempty confidence sets, gray line segments indicate that the entire plotted grid $[-3, 3]$ is accepted, and red crosses indicate empty confidence sets. The numbers above each bin report the number of weak first-stage SPY jumps over the number of valid announcements in that bin. Critical values are simulated from the heterogeneous pre/post mixed-normal null law with 10,000 Monte Carlo draws. This figure uses the legacy unstandardized aggregation, is retained for context, and is not directly comparable to the normalized Q_S baseline.

Figure S7: AR-RiceV 90% confidence sets around FOMC announcements grouped by ΔFFR



Notes: The figure repeats Figure S6 at the 90% confidence level. All implementation choices are otherwise unchanged: AR–RiceV statistic, β grid $[-3, 3]$ with step size 0.02, weak/valid labels above each bin, and mixed-normal critical values based on 10,000 Monte Carlo draws. This figure uses the legacy unstandardized aggregation, is retained for context, and is not directly comparable to the normalized Q_S baseline.

Table S1: Binding single-announcement AR-RiceV sets for empty half-year cells. Each row gives a minimal subset of single-announcement AR-RiceV accepted sets whose intersection is already empty; remaining announcements in the same half-year are omitted.

ETF	Half-year	W/P	Binding AR-RiceV sets
XLB	2013-H2	1/2	2013-09-18: $[0.58, 0.62]$; 2013-10-30*: $[-10.00, 0.46] \cup [0.94, 10.00]$
XLB	2016-H2	3/4	2016-11-02*: $[-0.16, 1.12]$; 2016-12-14*: $[1.38, 3.24]$
XLB	2023-H1	2/4	2023-02-01*: $[1.18, 10.00]$; 2023-03-22: $[0.90, 1.14]$
XLB	2024-H1	1/3	2024-01-31: $[0.90, 1.42]$; 2024-06-12: $[1.98, 3.40]$
XLE	2013-H2	1/2	2013-09-18: $[1.04, 1.26]$; 2013-10-30*: $[-10.00, -0.32] \cup [2.56, 10.00]$
XLE	2015-H1	1/3	2015-01-28: $[-0.24, 1.08]$; 2015-04-29*: $[1.72, 8.34]$
XLE	2021-H2	2/4	2021-11-03: $[1.40, 3.06]$; 2021-12-15: $[0.50, 0.96]$
XLE	2023-H1	2/4	2023-02-01*: $[-10.00, 0.84]$; 2023-03-22: $[0.92, 1.34]$
XLF	2016-H1	0/2	2016-03-16: $[0.24, 0.60]$; 2016-06-15: $[-4.12, -0.36]$
XLF	2016-H2	3/4	2016-07-27: $[-0.86, 0.48]$; 2016-12-14*: $[-8.80, -1.66]$
XLF	2017-H2	2/4	2017-07-26: $[2.20, 4.46]$; 2017-09-20*: $[-10.00, -5.06] \cup [7.56, 10.00]$
XLF	2018-H1	3/4	2018-03-21: $[-4.04, 0.08]$; 2018-05-02*: $[0.70, 4.66]$
XLF	2021-H2	2/4	2021-07-28*: $[-10.00, -2.52]$; 2021-09-22*: $[1.22, 7.28]$
XLF	2022-H2	1/4	2022-09-21: $[0.82, 0.94]$; 2022-11-02: $[0.52, 0.72]$
XLI	2023-H2	2/4	2023-07-26*: $[-0.26, 0.60]$; 2023-09-20: $[0.88, 1.22]$
XLI	2024-H1	1/3	2024-01-31: $[0.80, 1.02]$; 2024-06-12: $[1.64, 2.64]$
XLK	2015-H2	1/2	2015-09-17*: $[1.20, 10.00]$; 2015-10-28: $[0.52, 1.00]$
XLK	2021-H2	1/3	2021-07-28*: $[-10.00, 0.28] \cup [1.94, 10.00]$; 2021-11-03: $[0.88, 1.22]$
XLK	2022-H2	1/4	2022-09-21: $[1.16, 1.32]$; 2022-12-14: $[1.42, 1.70]$
XLK	2024-H1	1/3	2024-01-31: $[0.64, 1.16]$; 2024-05-01*: $[1.22, 10.00]$
XLP	2015-H1	1/3	2015-03-18: $[1.12, 1.44]$; 2015-04-29*: $[-2.56, 0.70]$
XLP	2016-H1	0/2	2016-03-16: $[0.66, 1.12]$; 2016-06-15: $[1.16, 2.64]$
XLP	2016-H2	3/4	2016-09-21*: $[1.42, 5.14]$; 2016-11-02*: $[-0.84, 1.28]$
XLP	2019-H1	1/3	2019-01-30: $[0.90, 1.32]$; 2019-05-01*: $[-10.00, -0.30] \cup [1.60, 10.00]$
XLP	2022-H2	1/4	2022-09-21: $[0.78, 0.92]$; 2022-12-14: $[0.42, 0.64]$
XLP	2023-H1	2/4	2023-03-22: $[0.70, 1.04]$; 2023-05-03*: $[-10.00, 0.50] \cup [1.48, 10.00]$
XLP	2024-H1	1/3	2024-05-01*: $[-3.82, 0.54]$; 2024-06-12: $[0.70, 1.94]$
XLU	2015-H1	1/3	2015-01-28: $[-0.22, 0.64]$; 2015-03-18: $[1.28, 2.04]$
XLU	2016-H2	3/3	2016-09-21*: $[-2.06, 1.06]$; 2016-12-14*: $[3.80, 10.00]$
XLU	2017-H1	2/4	2017-02-01: $[-0.76, 0.80]$; 2017-03-15: $[2.74, 4.54]$
XLU	2018-H1	3/4	2018-03-21: $[-0.40, 1.62]$; 2018-06-13*: $[4.24, 10.00]$
XLU	2023-H1	2/4	2023-03-22: $[0.90, 1.46]$; 2023-05-03*: $[-10.00, 0.52] \cup [3.84, 10.00]$
XLU	2024-H1	1/3	2024-05-01*: $[-10.00, 0.58]$; 2024-06-12: $[0.88, 1.84]$
XLV	2016-H2	3/4	2016-07-27: $[0.70, 1.20]$; 2016-12-14*: $[1.46, 2.68]$

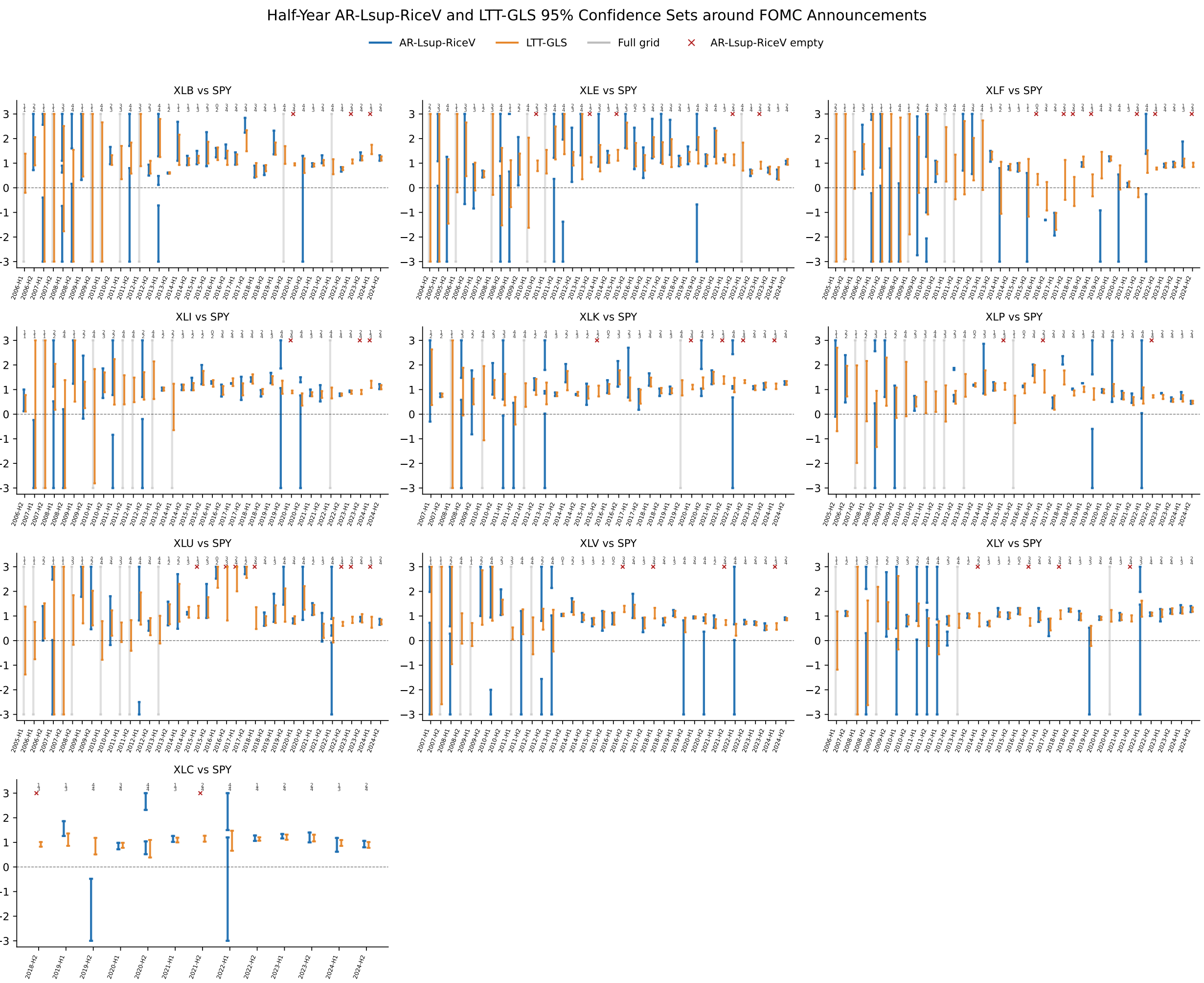
Continued on next page

Table S2: Binding single-announcement AR-RiceV sets for empty half-year cells (continued).

ETF	Half-year	W/P	Binding AR-RiceV sets
XLV	2021-H2	2/4	2021-07-28*: $[-10.00, -4.66] \cup [1.12, 10.00]$; 2021-11-03: $[0.70, 1.08]$
XLV	2024-H1	1/3	2024-01-31: $[0.68, 1.52]$; 2024-05-01*: $[-4.52, 0.44]$
XLY	2014-H2	1/3	2014-10-29: $[0.72, 1.14]$; 2014-12-17: $[0.48, 0.72]$
XLY	2016-H2	3/4	2016-07-27: $[0.70, 1.52]$; 2016-12-14*: $[-0.50, 0.56]$
XLY	2018-H1	3/4	2018-03-21: $[0.88, 1.68]$; 2018-06-13*: $[-3.20, 0.64]$

Notes: A superscript $*$ marks a weak SPY announcement return, i.e., below the truncation threshold. W/P reports the number of weak SPY returns among all announcements in the half-year. Single-announcement AR-RiceV sets are inverted on the $[-10, 10]$ grid and rounded for display. In the boundary-touch case, the rounded endpoints meet although the source half-year AR cell is classified as empty. LTT-GLS intervals are omitted because they are not part of the AR non-overlap certificate.

Figure S8: $L_{\sup}$ AR-RiceV and LTT-GLS confidence sets around FOMC announcements



Notes: The figure reports half-year 95% confidence sets for sector ETF jump betas relative to SPY around FOMC announcements. Blue line segments denote $L_{\sup}$ AR-RiceV confidence sets, orange line segments denote LTT-GLS confidence intervals, gray line segments indicate that the entire plotted grid is accepted, and red crosses indicate empty $L_{\sup}$ AR-RiceV confidence sets. The stacked numbers above each period report the number of weak first-stage jumps over the total number of valid announcements in that period.

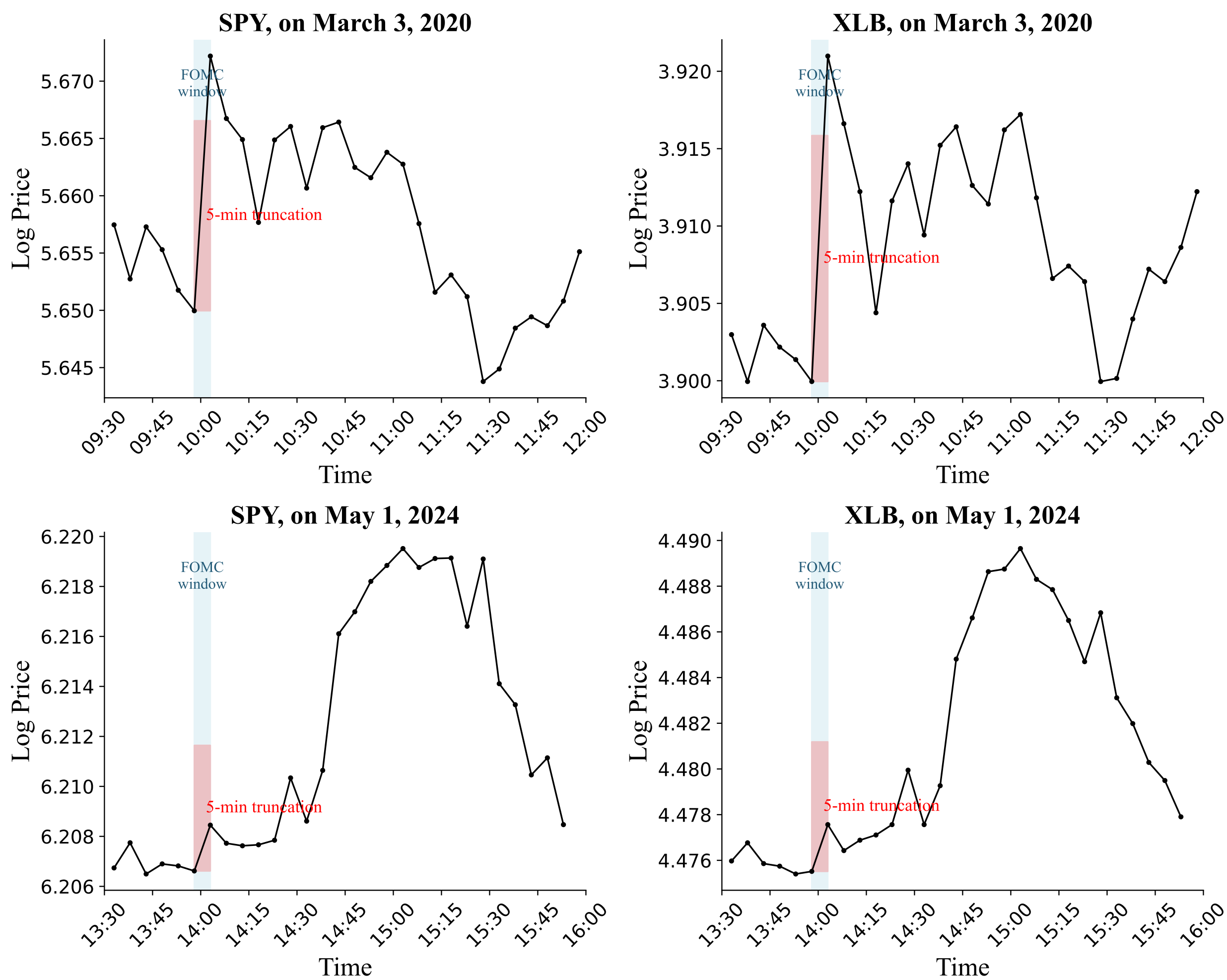


Figure S9: Five-minute-frequency log-price paths around two FOMC announcements. The figure repeats the motivating price-path illustration using the five-minute announcement window $[t-2, t+2]$. The blue shaded region marks the FOMC window and the red shaded region marks the five-minute return used for truncation comparison.

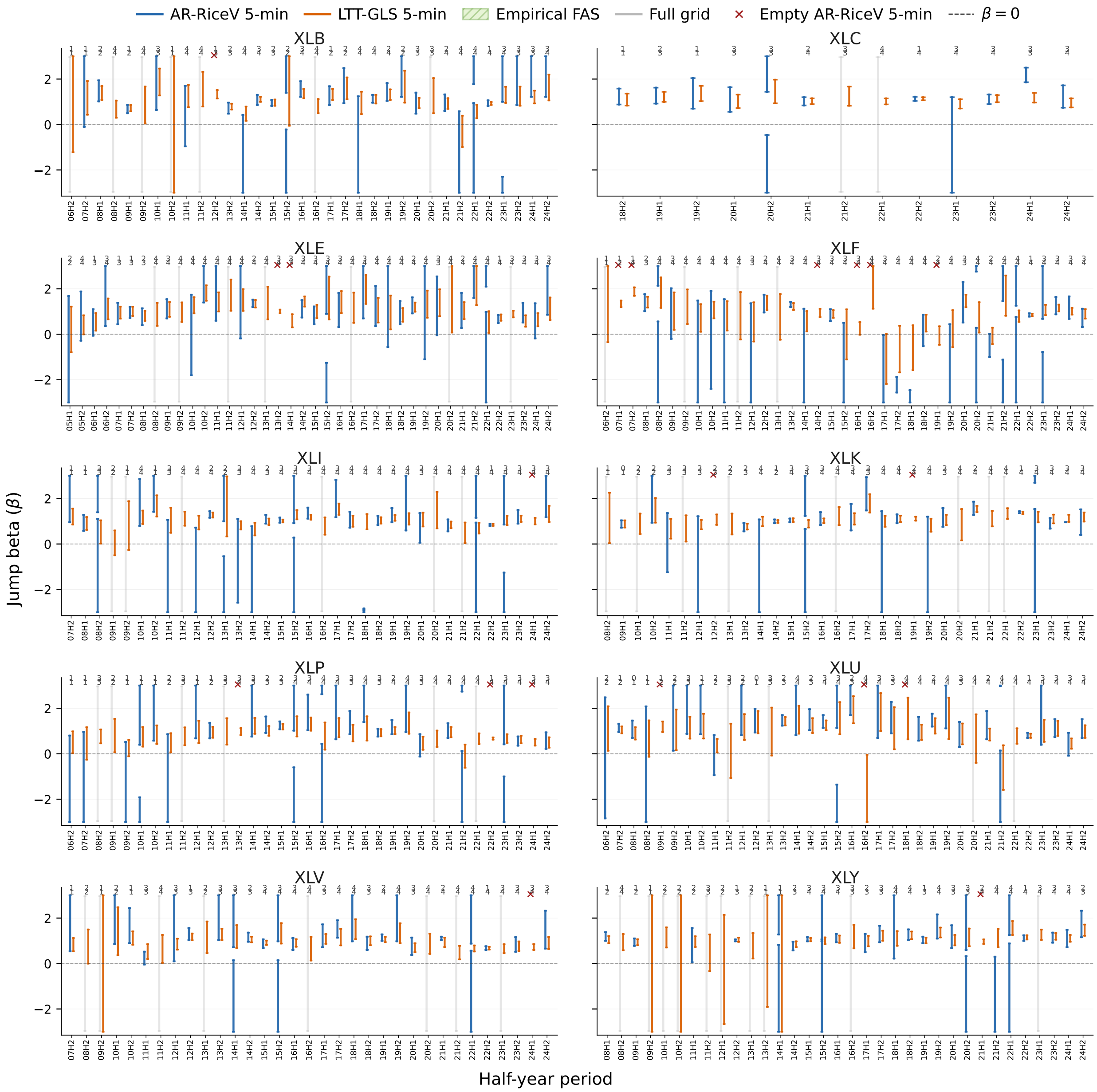


Figure S10: Five-minute-frequency AR-RiceV and LTT-GLS confidence sets around FOMC announcements. The figure reports half-year 95% confidence sets after aggregating the announcement return over $[t-2, t+2]$. Blue intervals denote AR-RiceV confidence sets, orange intervals denote LTT-GLS confidence intervals, gray intervals denote acceptance of the full plotted grid, and red crosses denote empty AR-RiceV sets. Numbers above each period report weak/valid announcements.

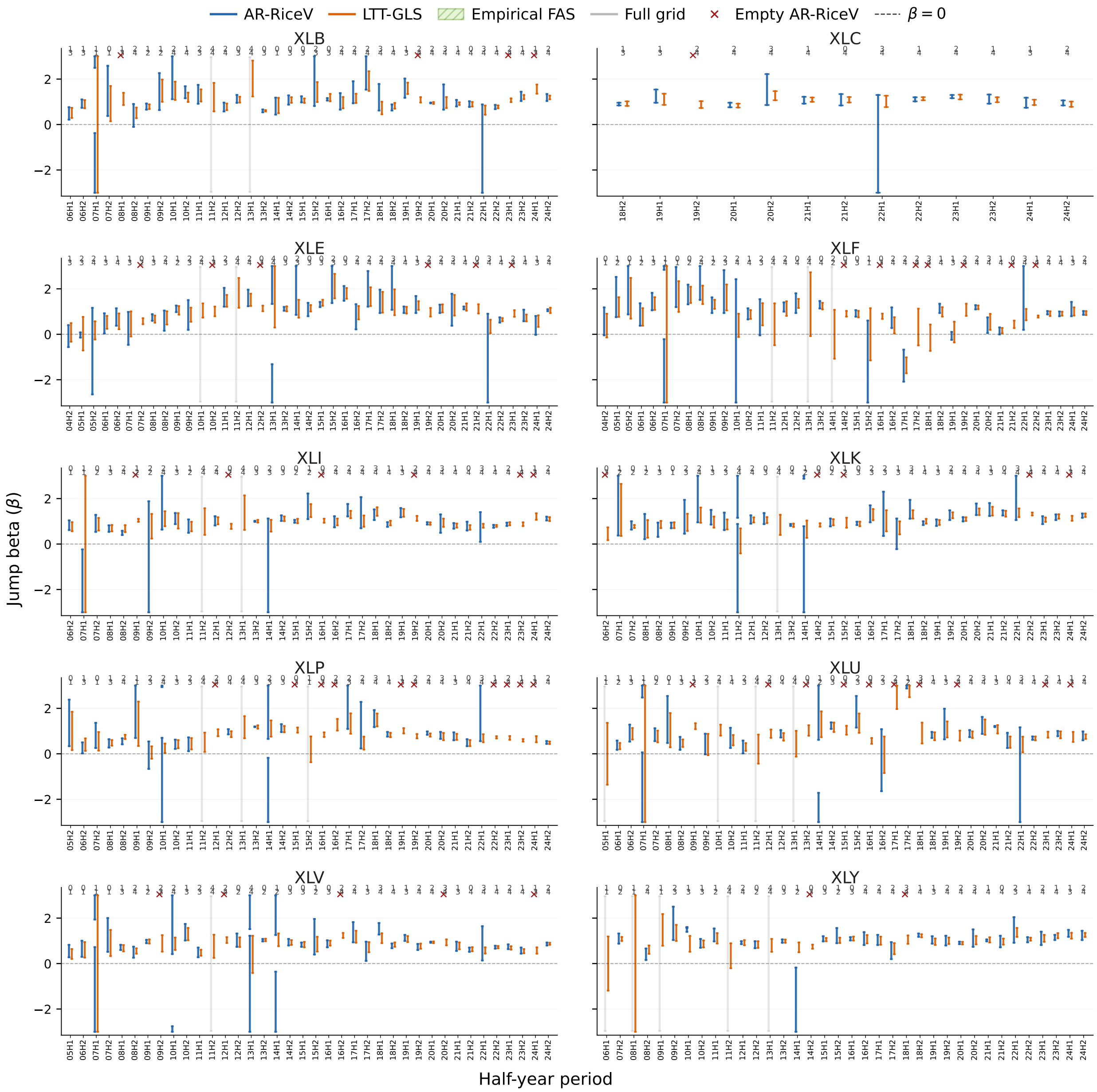
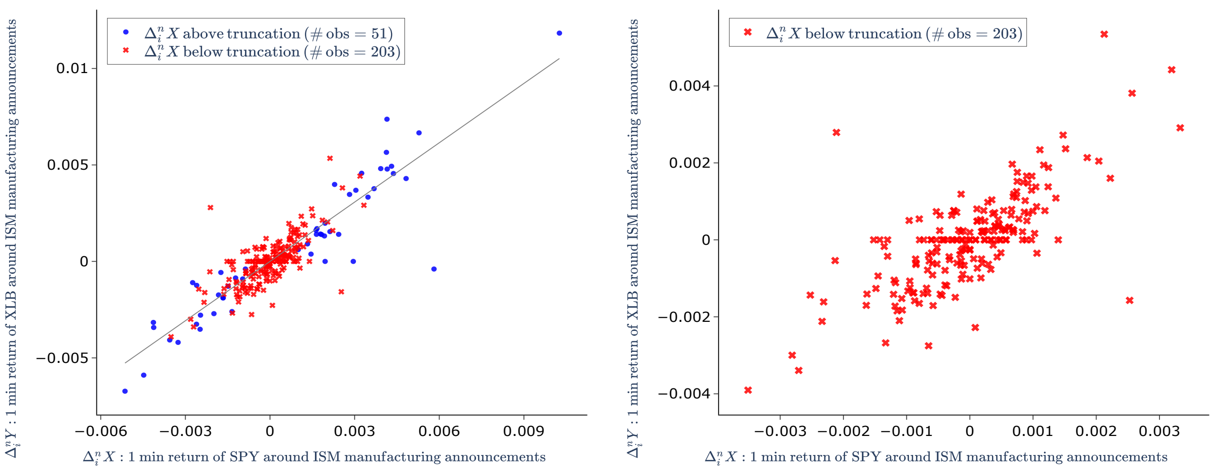


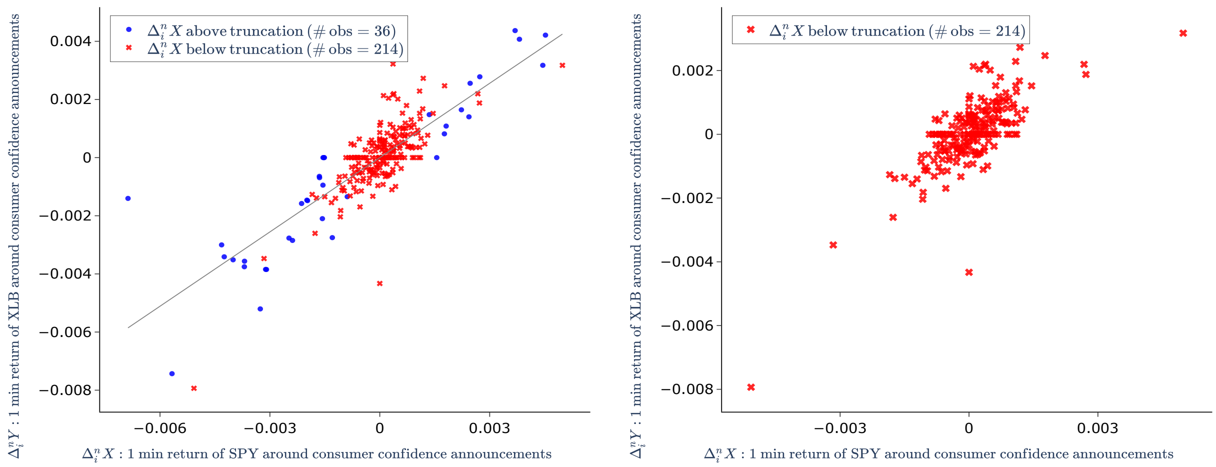
Figure S11: Reaction-time-adjusted AR-RiceV and LTT-GLS confidence sets around FOMC announcements. The figure reports half-year 95% confidence sets after allowing the announcement-time anchor to shift within the five-minute window when the observed SPY jump is outside the exact announcement minute. Blue intervals denote AR-RiceV confidence sets, orange intervals denote LTT-GLS confidence intervals, gray intervals denote acceptance of the full plotted grid, and red crosses denote empty AR-RiceV sets. Numbers above each period report weak/valid announcements.

Figure S12: The scatter plot of high frequency return pairs around ISM manufacturing announcements.



Notes: The left panel plots all one-minute high-frequency return pairs of XLB versus SPY around regular-hours ISM manufacturing announcements. Blue dots denote observations for which the X-return exceeds its truncation threshold, while red crosses denote observations for which the X-return falls below its truncation threshold. The classification is based only on the first-stage SPY return and does not further separate observations according to the Y-return. The straight line in the left panel represents the linear fit without intercept based on the blue dots, with an estimated slope of 1.02. The right panel provides a zoomed-in view of observations for which the X-return falls below its truncation threshold.

Figure S13: The scatter plot of high frequency return pairs around Conference Board consumer confidence announcements.



Notes: The left panel plots all one-minute high-frequency return pairs of XLB versus SPY around regular-hours Conference Board consumer confidence announcements. Blue dots denote observations for which the X-return exceeds its truncation threshold, while red crosses denote observations for which the X-return falls below its truncation threshold. The classification is based only on the first-stage SPY return and does not further separate observations according to the Y-return. The straight line in the left panel represents the linear fit without intercept based on the blue dots, with an estimated slope of 0.85. The right panel provides a zoomed-in view of observations for which the X-return falls below its truncation threshold.

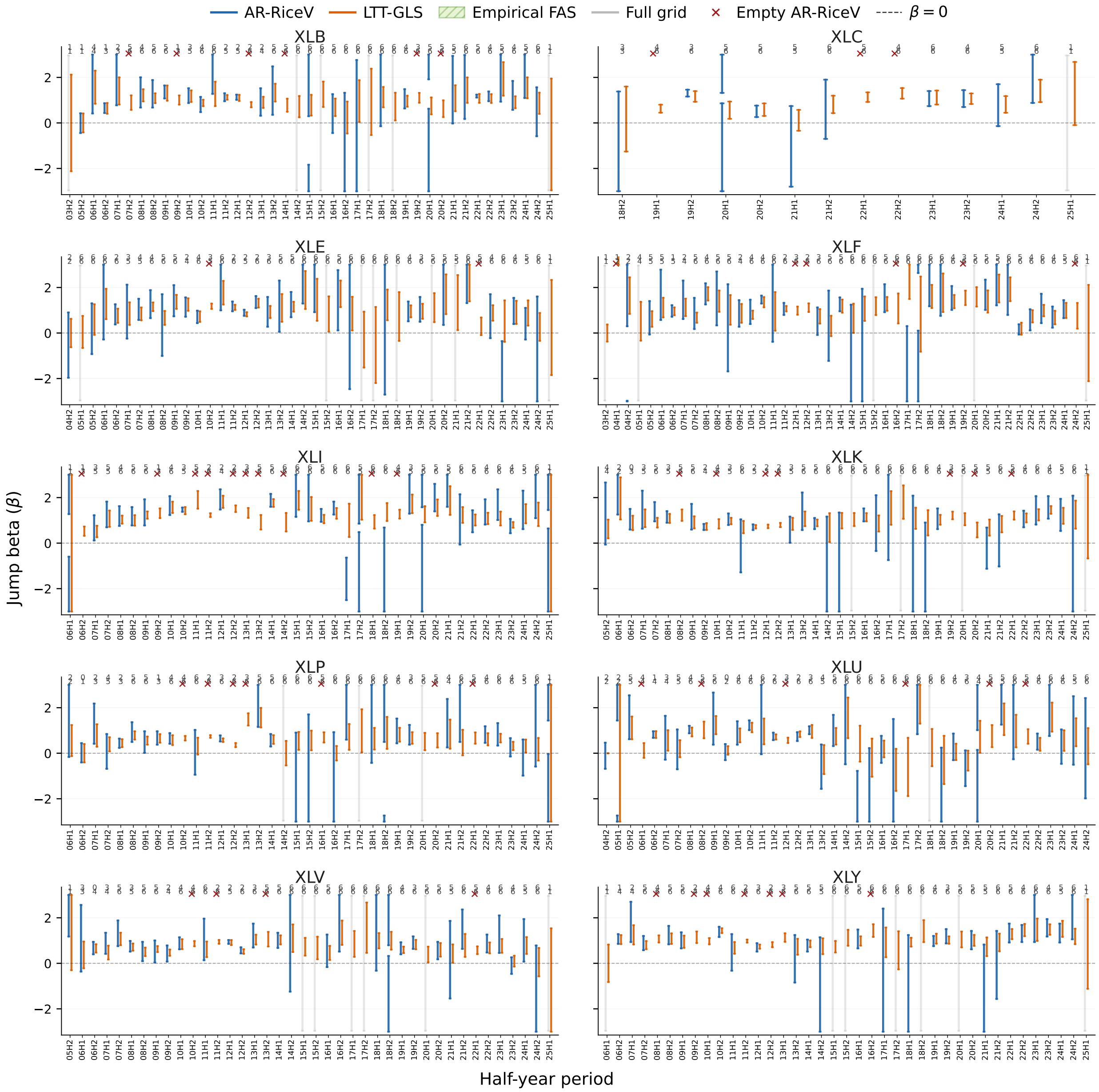


Figure S14: AR-RiceV and LTT-GLS confidence sets around ISM manufacturing announcements. The figure reports half-year 95% confidence sets for sector ETF jump betas relative to SPY around regular-hours ISM manufacturing announcements. Blue intervals denote AR-RiceV confidence sets, orange intervals denote LTT-GLS confidence intervals, gray intervals denote acceptance of the full plotted grid, and red crosses denote empty AR-RiceV sets. Numbers above each period report weak/valid announcements.

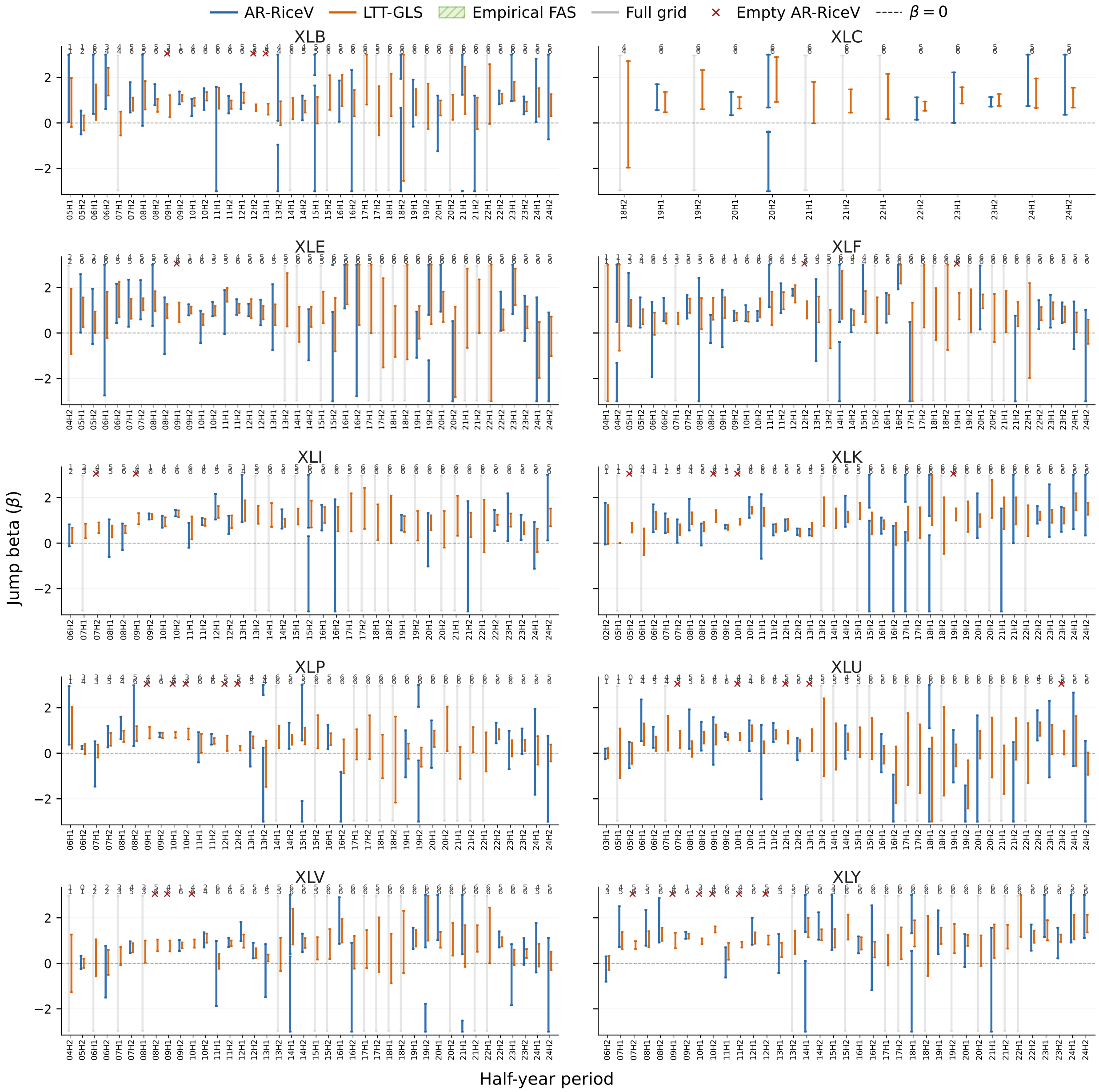


Figure S15: AR-RiceV and LTT-GLS confidence sets around Conference Board consumer confidence announcements. The figure reports half-year 95% confidence sets for sector ETF jump betas relative to SPY around regular-hours Conference Board consumer confidence announcements. Blue intervals denote AR-RiceV confidence sets, orange intervals denote LTT-GLS confidence intervals, gray intervals denote acceptance of the full plotted grid, and red crosses denote empty AR-RiceV sets. Numbers above each period report weak/valid announcements.

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